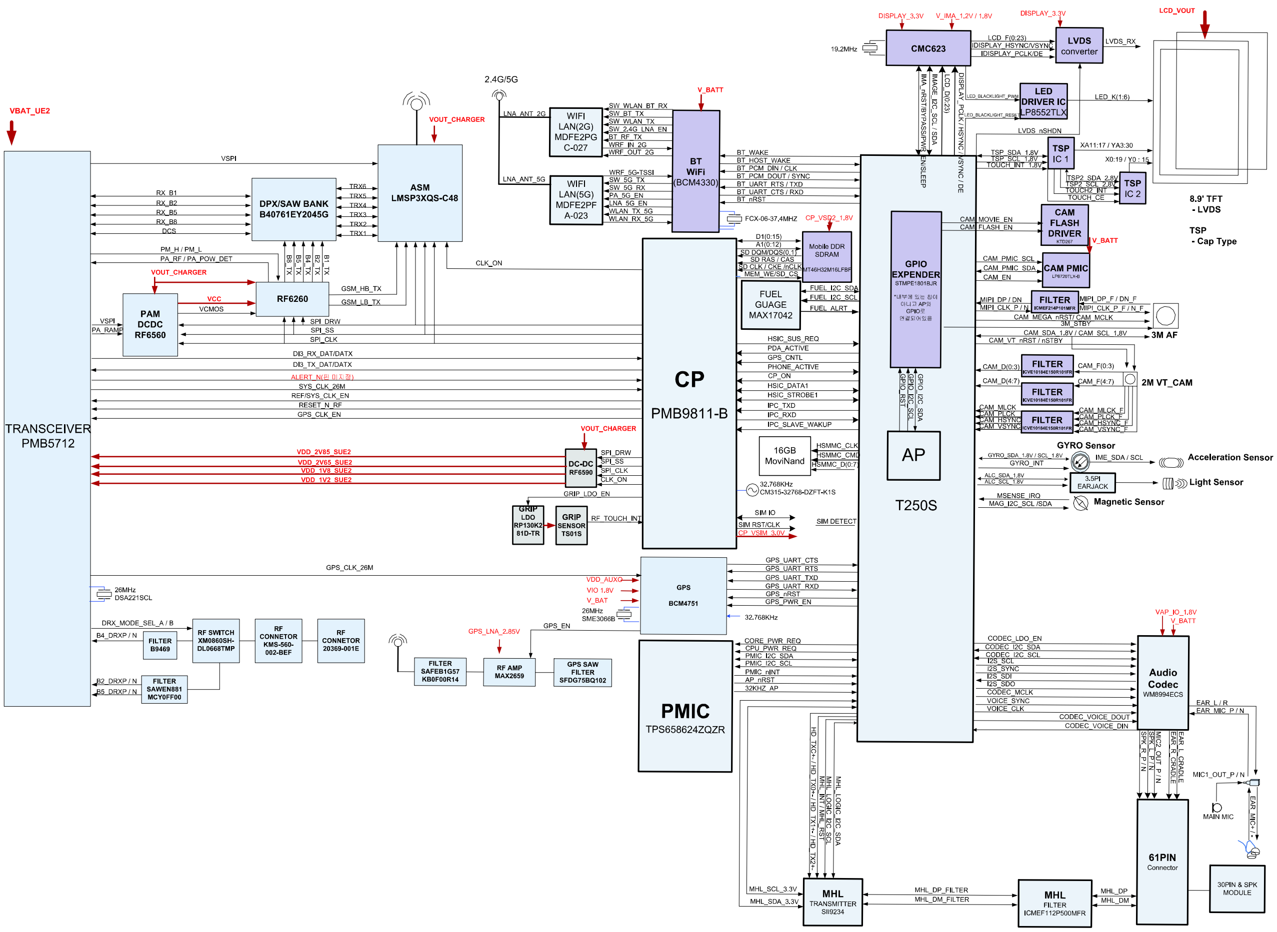
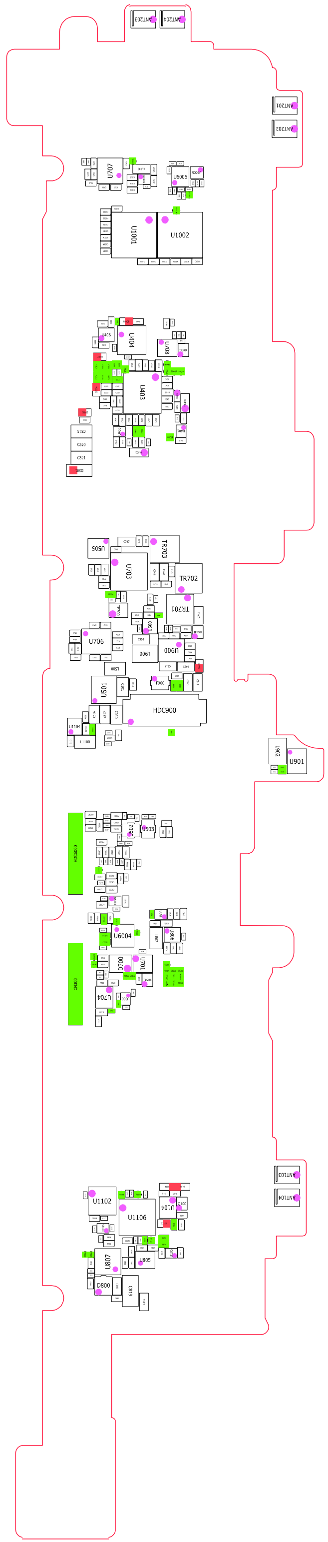
8. 3 级维修

8-1.框图

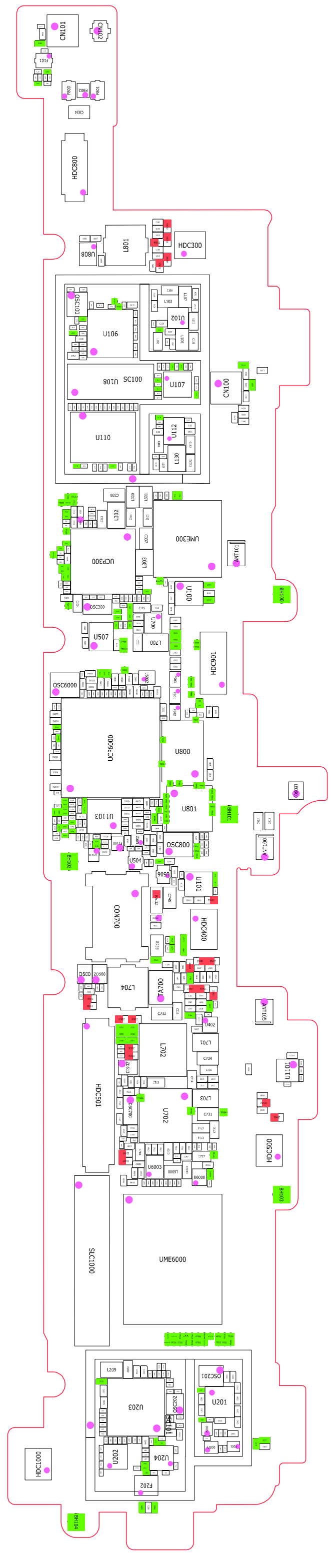


8-2.主板图

8-2-1.顶部

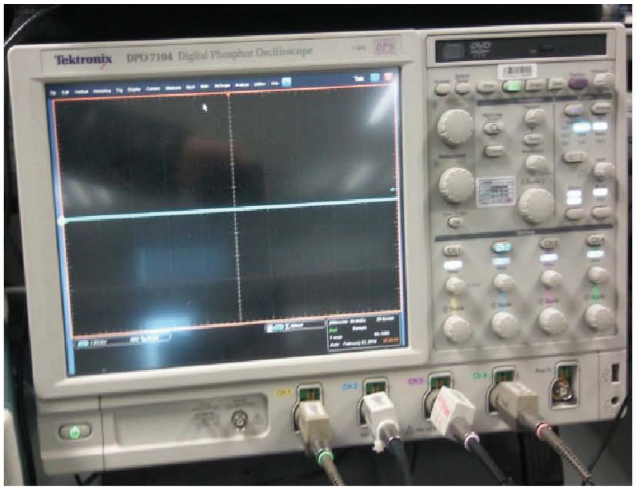


8-2-2.底部



8-3.故障排除流程图

设备

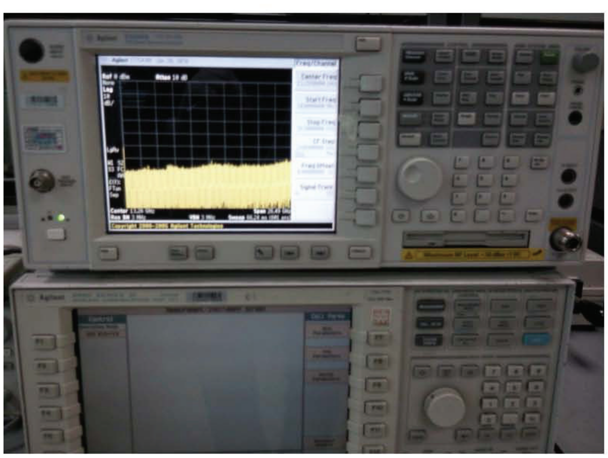


↑示波器

↑数字万用表

↑电源供电器

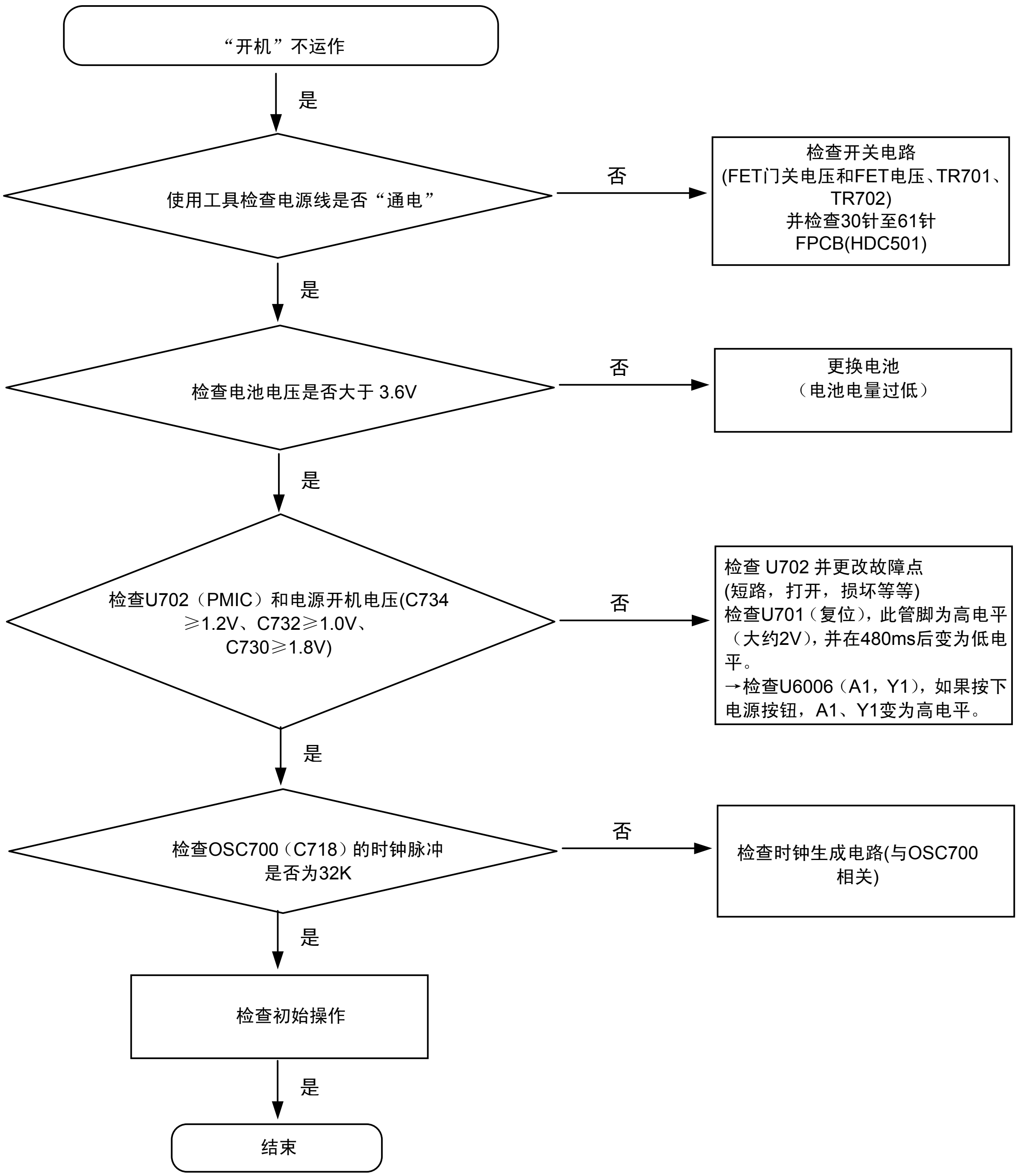
↑+字螺丝刀，小镊子

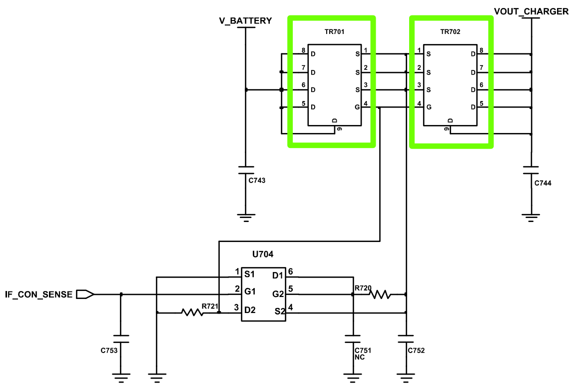


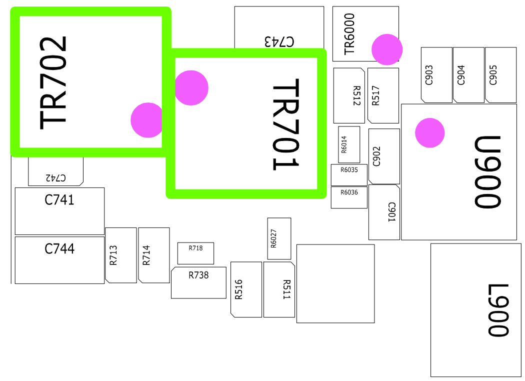
↑8960 & 光谱分析器

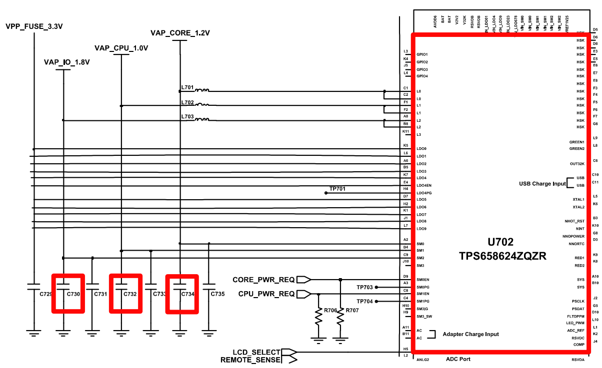
↑焊接烙铁

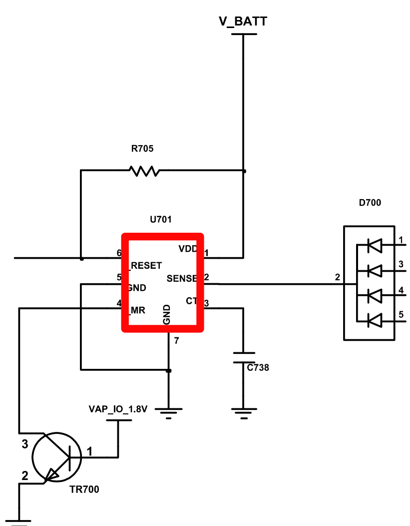
8-3-1 开机

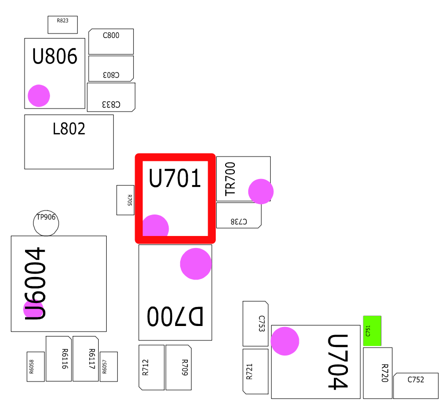

“开机”不运作
是
使用工具检查电源线是否“通电”
否
检查开关电路
(FET门关电压和FET电压、TR701、
TR702)
并检查30针至61针
FPCB(HDC501)
是
检查电池电压是否大于 3.6V
否
更换电池
（电池电量过低）
是
检查U702（PMIC）和电源开机电压(C734
≥1.2V、C732≥1.0V、
C730≥1.8V)
否
检查 U702 并更改故障点
(短路，打开，损坏等等)
检查U701（复位），此管脚为高电平
（大约2V），并在480ms后变为低电
平。
→检查U6006（A1，Y1），如果按下
电源按钮，A1、Y1变为高电平。
是
检查OSC700（C718）的时钟脉冲
是否为32K
否
检查时钟生成电路(与OSC700
相关)
是
检查初始操作
是
结束


V_BATTERY
VOUT_CHARGER
TR701
TR702
C743
C744
U704
IF_CON_SENSE
R721
R720
C753
C751
NC
C752

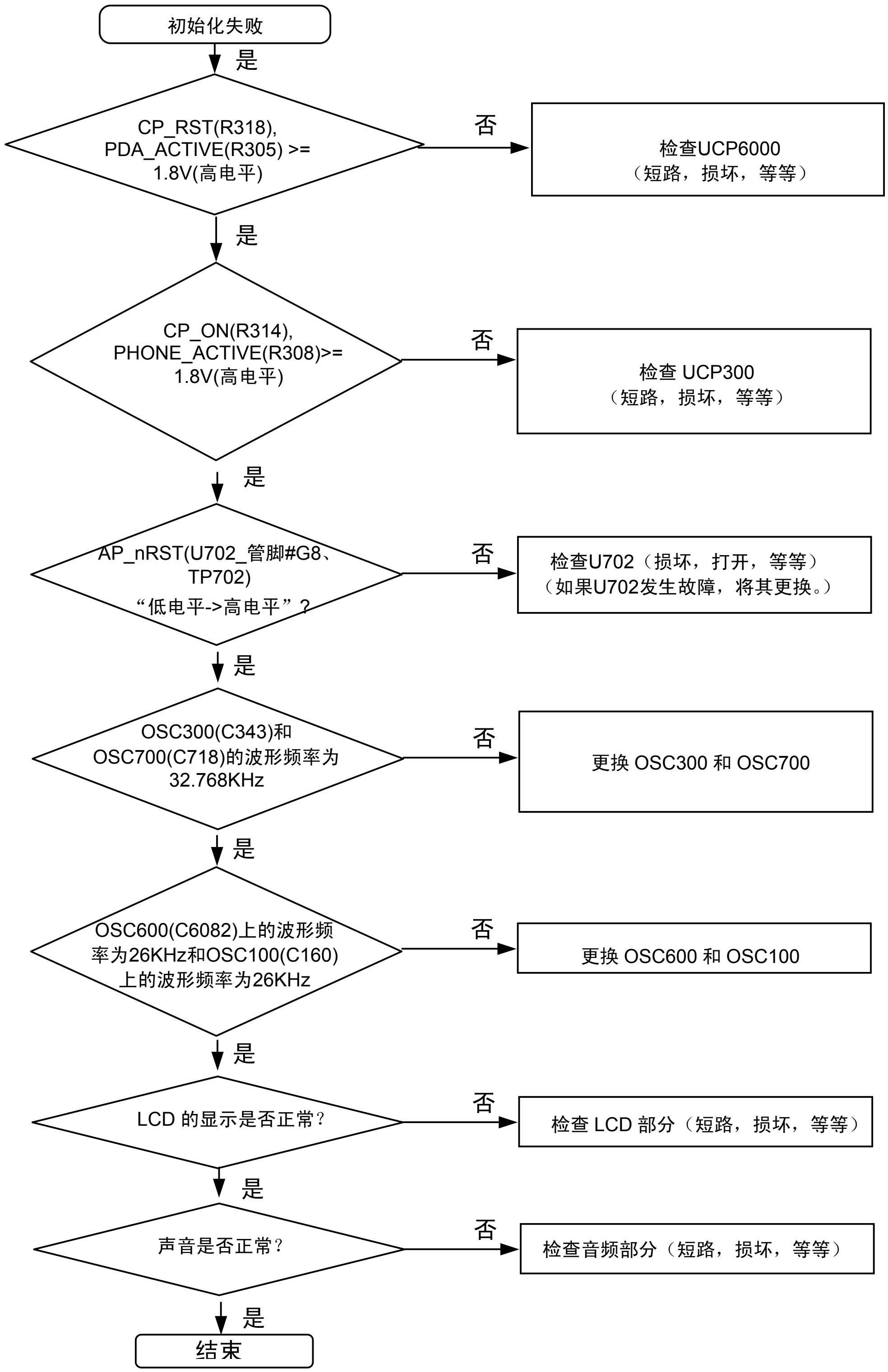

TR702
TR701
C743
TR6000
C741
C744
U900
L900

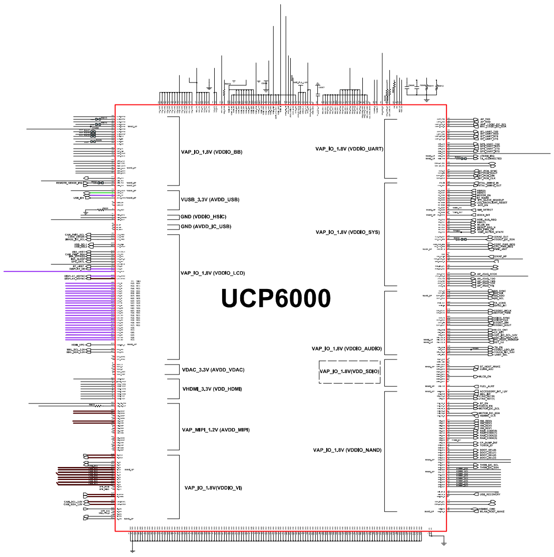

VPP_FUSE_3.3V
VAP_CORE_1.2V
VAP_CPU_1.0V
VAP_IO_1.8V
U702
TPS658624ZQZR
CORE_PWR_REQ
CPU_PWR_REQ
LCD_SELECT
REMOTE_SENSE

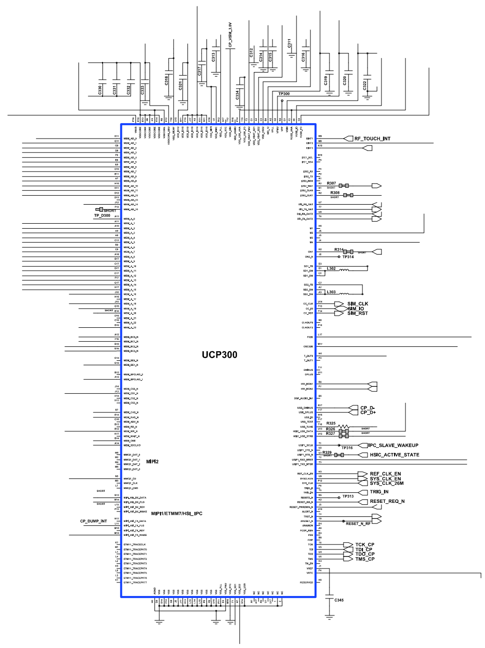

V_BATT
R705
U701
D700
C738
VAP_IO_1.8V
TR700

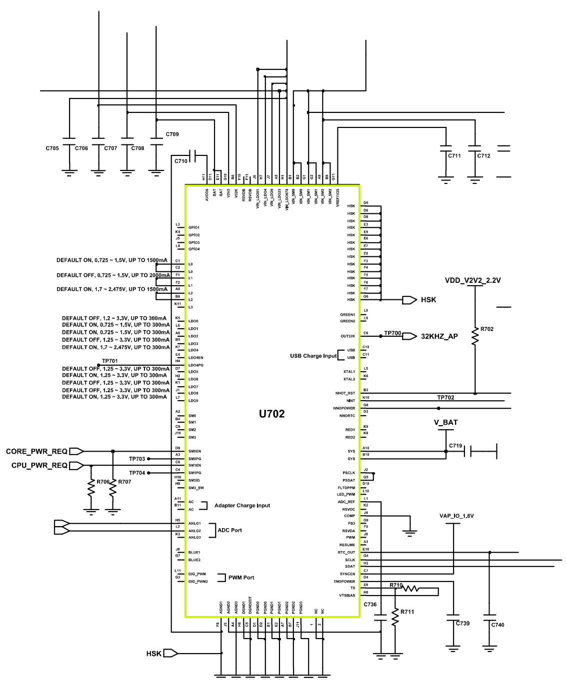

U806
L802
U701
TR700
U6004
D700
U704

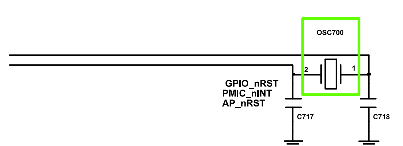
8-3-2.初始化

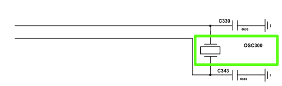

初始化失败
是
CP_RST(R318),
PDA_ACTIVE(R305) >=
1.8V(高电平)
否
检查UCP6000
（短路，损坏，等等）
是
CP_ON(R314),
PHONE_ACTIVE(R308)>=
1.8V(高电平)
否
检查 UCP300
（短路，损坏，等等）
是
AP_nRST(U702_管脚#G8、
TP702)
“低电平->高电平”？
否
检查U702（损坏，打开，等等）
（如果U702发生故障，将其更换。）
是
OSC300(C343)和
OSC700(C718)的波形频率为
32.768KHz
否
更换 OSC300 和 OSC700
是
OSC600(C6082)上的波形频
率为26KHz和OSC100(C160)
上的波形频率为26KHz
否
更换 OSC600 和 OSC100
是
LCD 的显示是否正常？
否
检查 LCD 部分（短路，损坏，等等）
是
声音是否正常？
否
检查音频部分（短路，损坏，等等）
是
结束

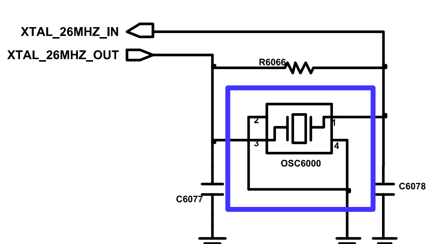

UCP6000

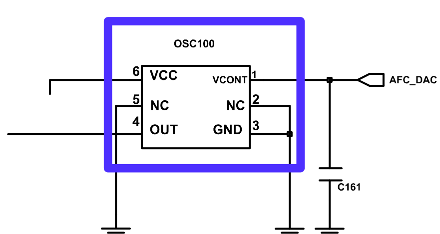

UCP300


U702
VDD_V2V2_2.2V
HSK
32KHZ_AP
V_BAT
VAP_IO_1.8V
CORE_PWR_REQ
CPU_PWR_REQ


OSC700
GPIO_nRST
PMIC_nINT
AP_nRST
C717
C718

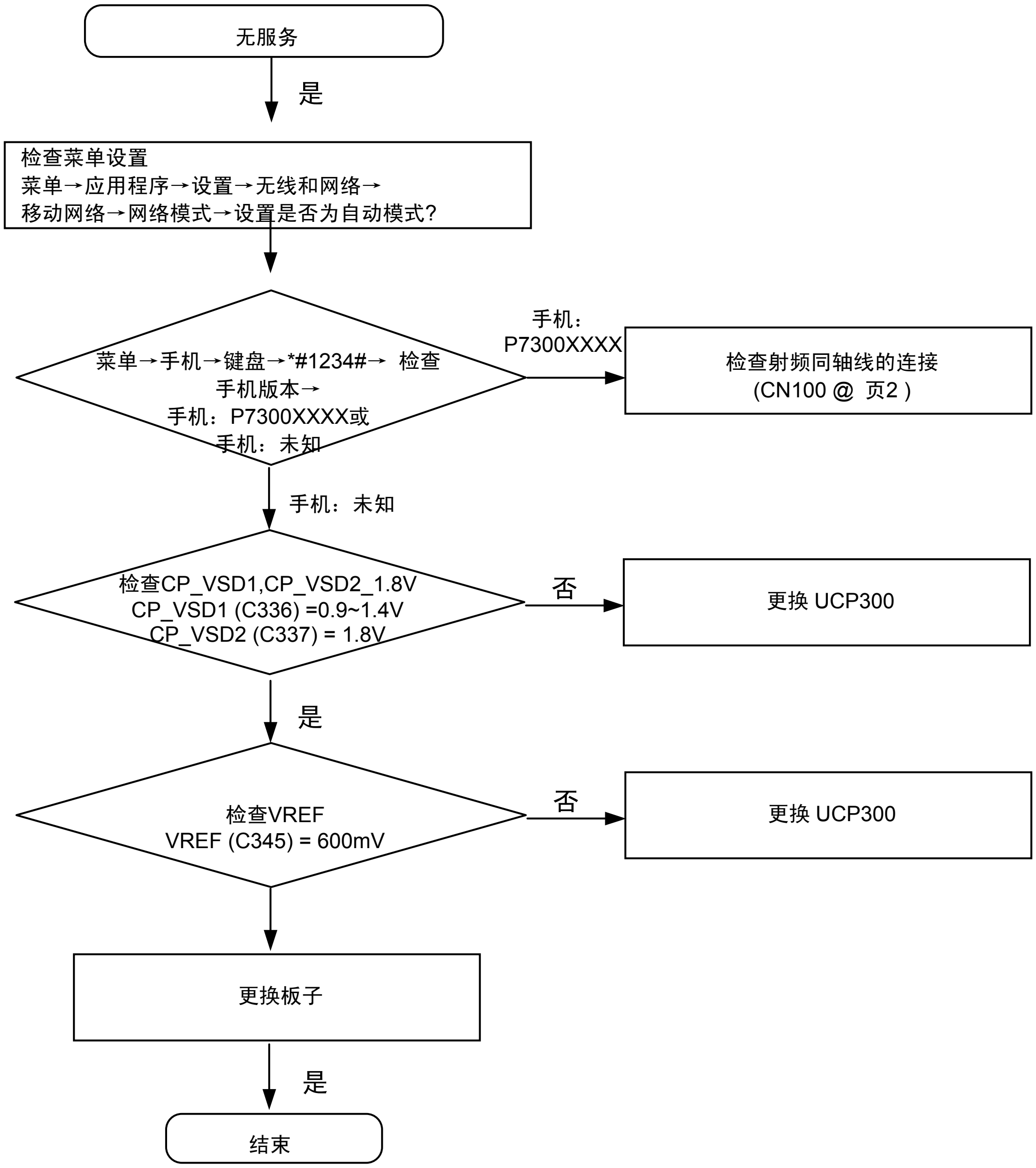

OSC300

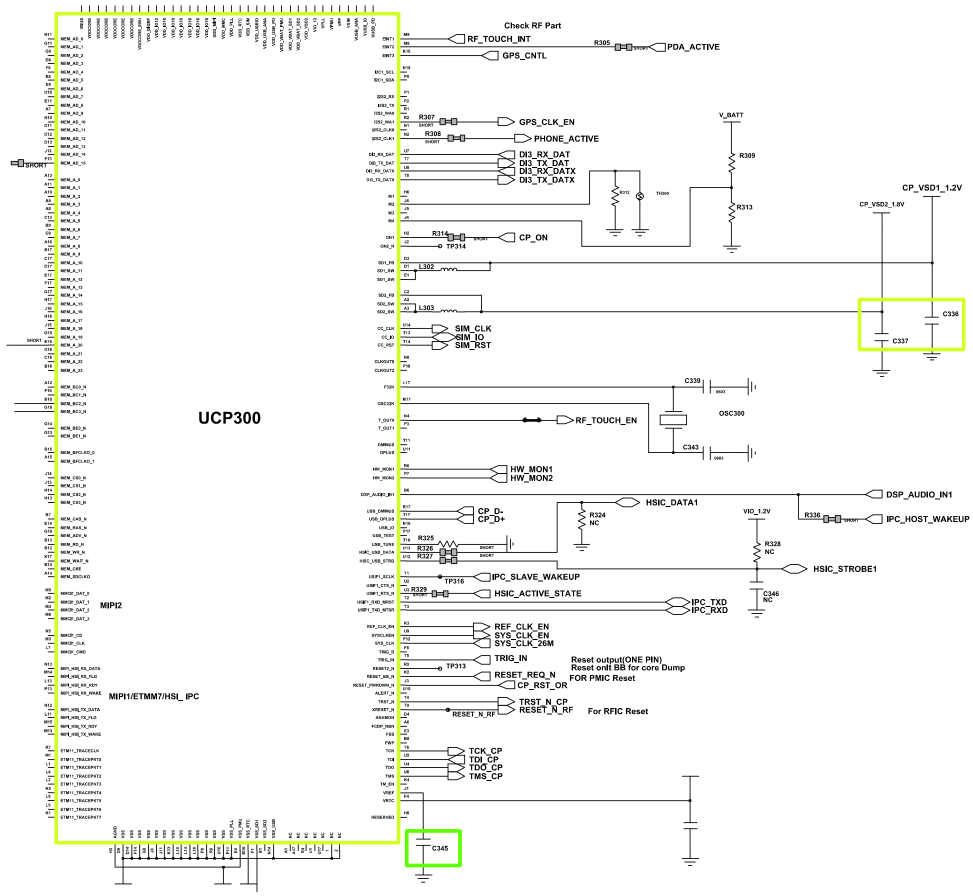

XTAL_26MHZ_IN
XTAL_26MHZ_OUT
R6066
OSC6000
C6077
C6078

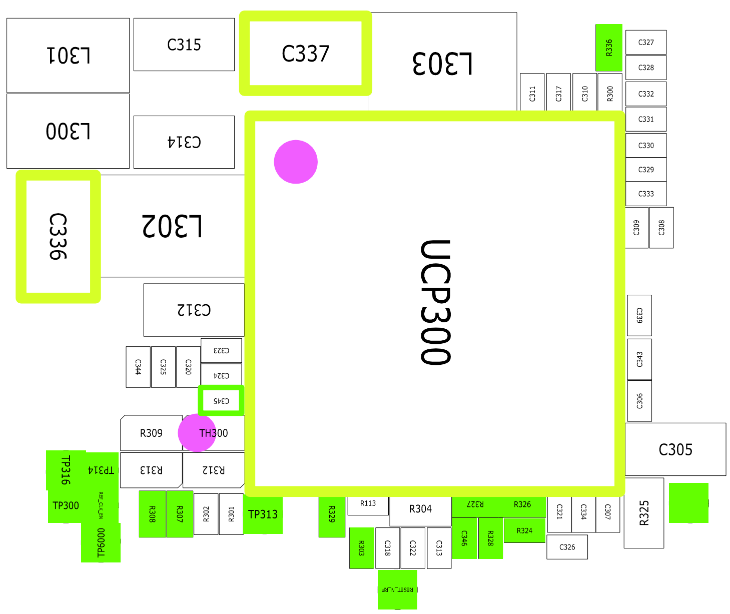

OSC100
VCC
VCONT
NC
NC
OUT
GND
AFC_DAC
C161


UME300
U100
HDC901
U800
U801
L301
C315
C337
L303
L300
C314
C336
L302
UCP300
C312
U700
L700
C715
C701
U6002
C6035
UCP6000
U1103
OSC300
U507
R530
C305
C519
C523
R309
TH300
R313
R312
R304
R325
F903
F901
F902
C6072
C6021
C6038
C6019
C6036
C6020
C6063
C6066
C6031
C6033
R6066
OSC6000
C6071
C6034
C6029
C6070
C1101
C1111
L107
U102
C121
C135
L106
L103
C136
L100
U107
L130
U112
C182
C1202
C181
L129
U106
SC100
U108
U110
OSC100
TP704
C735
C739
C707
L703
C731
C730
TP702
R6004
R6006
C726
C727
C712
C719
C737
U6000
C714
TP703
C724
C6079
U702
C711
C740
C705
L6000
C709
C713
C702
C703
U6003
C704
C706
C721
C722
C710
R6067
C6080
R6069
C6085
ZD513
OSC700
C728
C720
C729
ZD503
ZD504
C723
C725
R1008
C708
C718
C717
R702

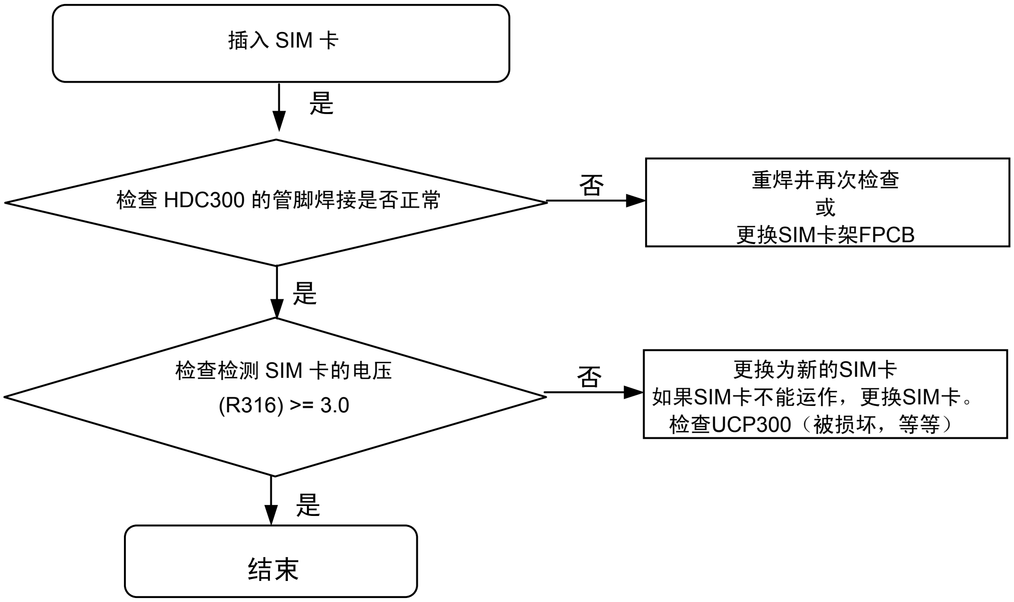
8-3-3. 无服务

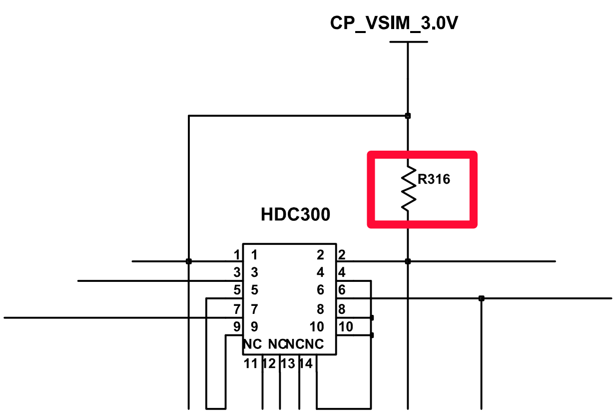

无服务
是
检查菜单设置
菜单→应用程序→设置→无线和网络→
移动网络→网络模式→设置是否为自动模式?
菜单→手机→键盘→*#1234#→ 检查
手机版本→
手机：P7300XXXX或
手机：未知
手机：
P7300XXXX
检查射频同轴线的连接
(CN100 @ 页2)
手机：未知
检查CP_VSD1,CP_VSD2_1.8V
CP_VSD1 (C336) =0.9~1.4V
CP_VSD2 (C337) = 1.8V
否
更换 UCP300
是
检查VREF
VREF (C345) = 600mV
否
更换 UCP300
更换板子
是
结束

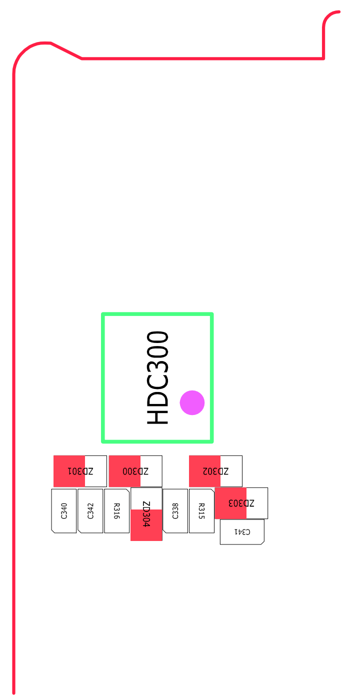

Check RF Part
RF_TOUCH_INT
GPS_CNTL
R305
PDA_ACTIVE
R307
GPS_CLK_EN
R308
PHONE_ACTIVE
V_BATT
R309
DI3_RX_DAT
DI3_TX_DAT
DI3_RX_DATX
DI3_TX_DATX
R313
CP_VSD1_1.2V
CP_VSD2_1.8V
R314
CP_ON
TP314
L302
L303
C336
C337
SIM_CLK
SIM_IO
SIM_RST
C339
OSC300
UCP300
RF_TOUCH_EN
C343
HW_MON1
HW_MON2
DSP_AUDIO_IN1
HSIC_DATA1
CP_D-
CP_D+
R324
NC
VIO_1.2V
R336
IPC_HOST_WAKEUP
R325
R326
R327
R328
NC
HSIC_STROBE1
TP316
IPC_SLAVE_WAKEUP
R329
HSIC_ACTIVE_STATE
C346
NC
MIPI2
IPC_TXD
IPC_RXD
REF_CLK_EN
SYS_CLK_EN
SYS_CLK_26M
TRIG_IN
TP313
Reset output(ONE PIN)
Reset onlt BB for core Dump
RESET_REQ_N
FOR PMIC Reset
CP_RST_OR
MIPI1/ETMM7/HSI_ IPC
TRST_N_CP
RESET_N_RF
For RFIC Reset
TCK_CP
TDI_CP
TDO_CP
TMS_CP
C345


L301
C315
C337
L303
R336
C327
C328
C311
C317
C310
R300
C332
C331
L300
C314
C330
C329
C333
C336
L302
UCP300
C309
C308
C312
C339
C323
C344
C325
C320
C324
C343
C345
C306
R309
TH300
C305
TP316
TP314
R313
R312
TP300
R308
R307
R302
R301
TP313
R329
R113
R304
R327
R326
C321
C334
C307
R325
TP6000
R303
C318
C322
C313
C346
R328
R324
C326

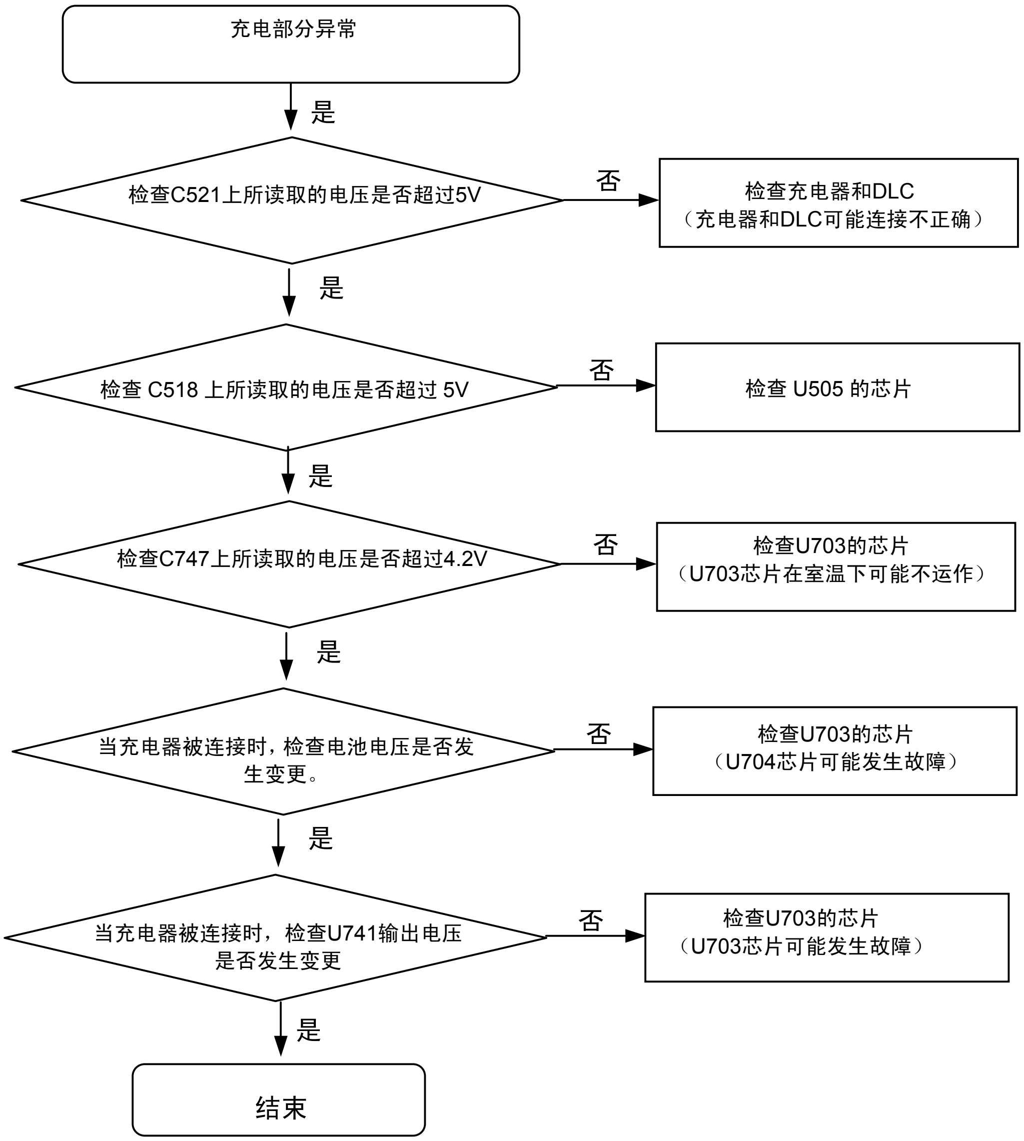
8-3-4.Sim 部分

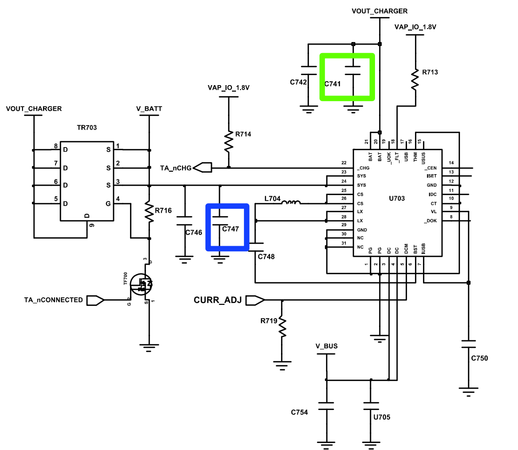

插入 SIM 卡
是
检查 HDC300 的管脚焊接是否正常
否
重焊并再次检查
或
更换SIM卡架FPCB
是
检查检测 SIM 卡的电压
(R316) >= 3.0
否
更换为新的SIM卡
如果SIM卡不能运作，更换SIM卡。
检查UCP300（被损坏，等等）
是
结束

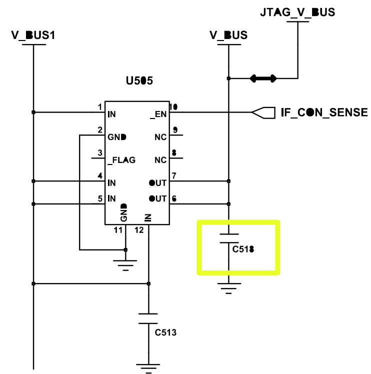

CP_VSIM_3.0V
R316
HDC300
1 1 2 2
3 3 4 4
5 5 6 6
7 7 8 8
9 9 10 10
NC NC NC NC
11 12 13 14

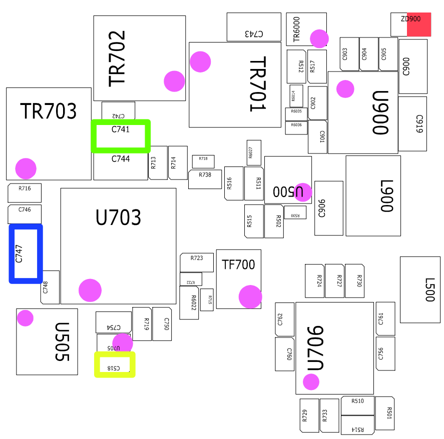

HDC300

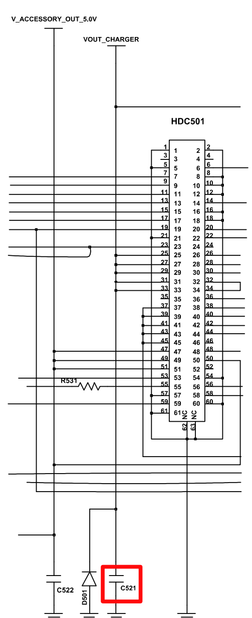
8-3-3. 充电部分

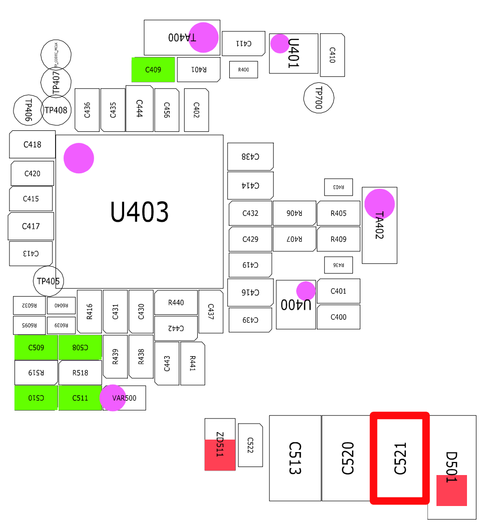

充电部分异常
是
检查C521上所读取的电压是否超过5V
否
检查充电器和DLC
（充电器和DLC可能连接不正确）
是
检查 C518 上所读取的电压是否超过 5V
否
检查 U505 的芯片
是
检查C747上所读取的电压是否超过4.2V
否
检查U703的芯片
（U703芯片在室温下可能不运作）
是
当充电器被连接时，检查电池电压是否发生变更。
否
检查U703的芯片
（U704芯片可能发生故障）
是
当充电器被连接时，检查U741输出电压是否发生变更
否
检查U703的芯片
（U703芯片可能发生故障）
是
结束



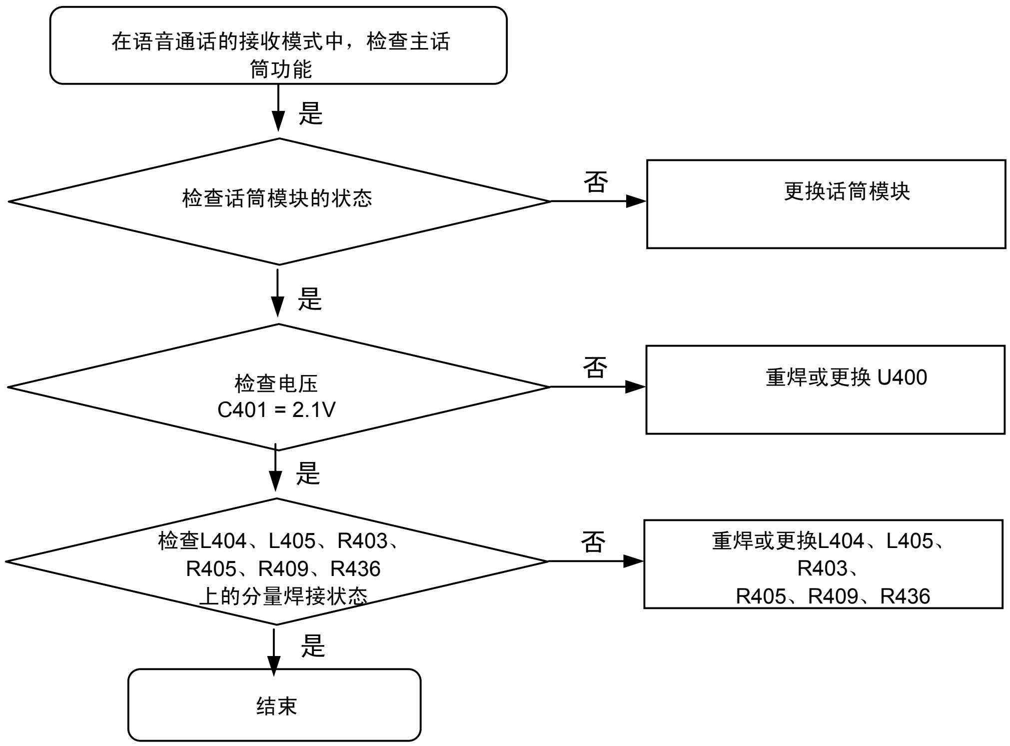




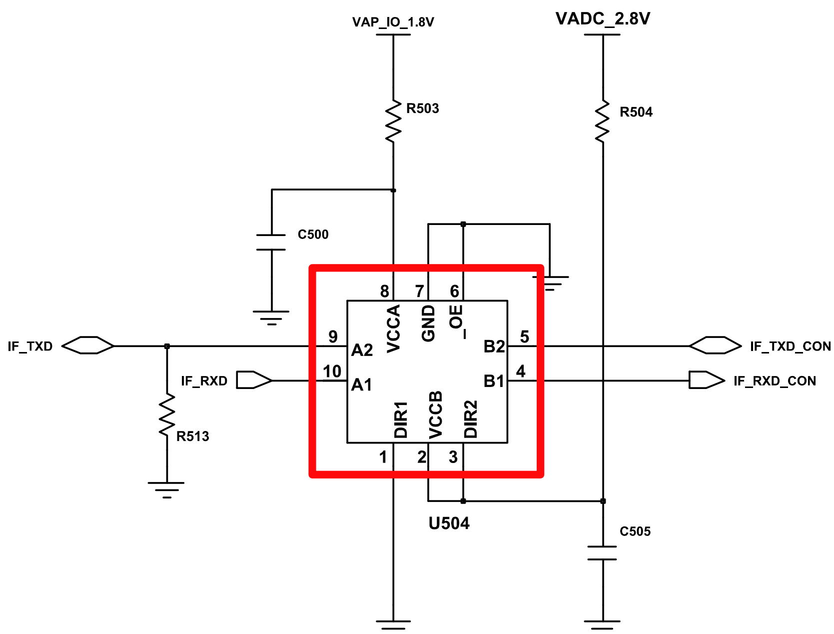




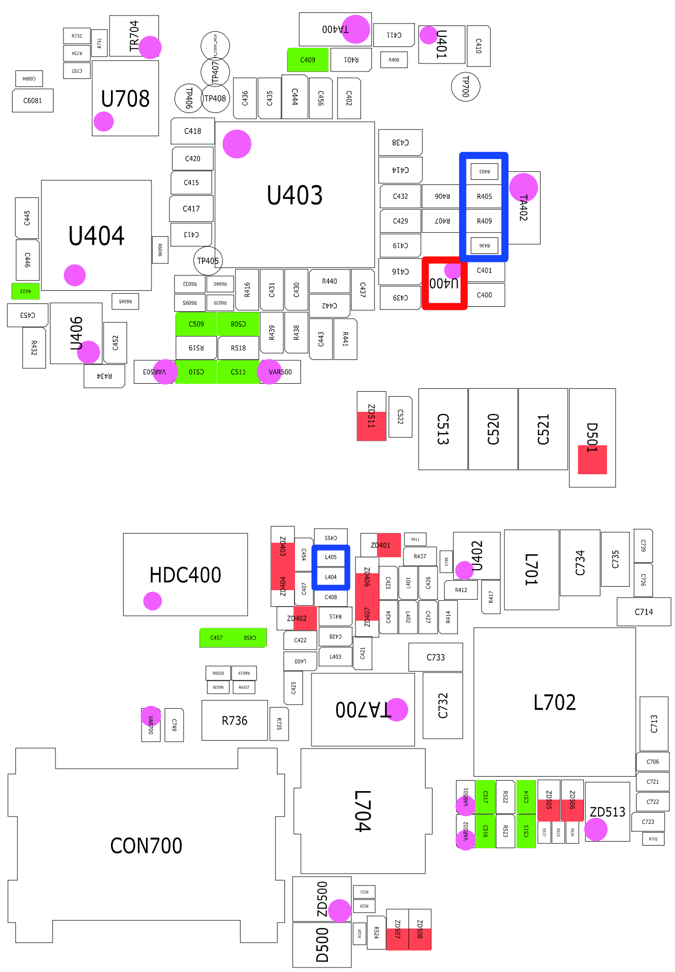
8-3-4.话筒部分



在语音通话的接收模式中，检查主话筒功能
是
检查话筒模块的状态
否
更换话筒模块
是
检查电压
C401 = 2.1V
否
重焊或更换 U400
是
检查L404、L405、R403、R405、R409、R436上的分量焊接状态
否
重焊或更换L404、L405、R403、R405、R409、R436
是
结束

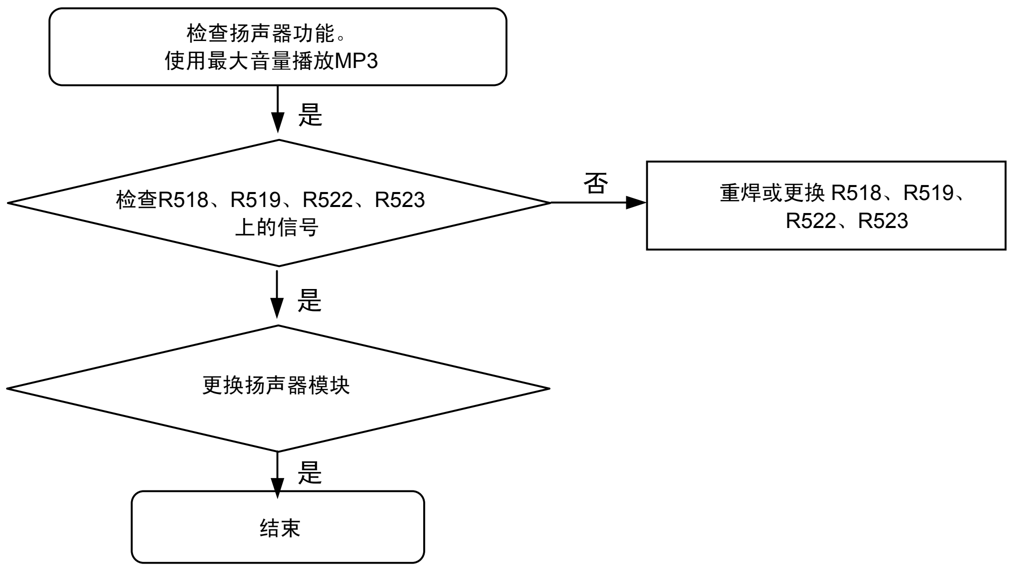


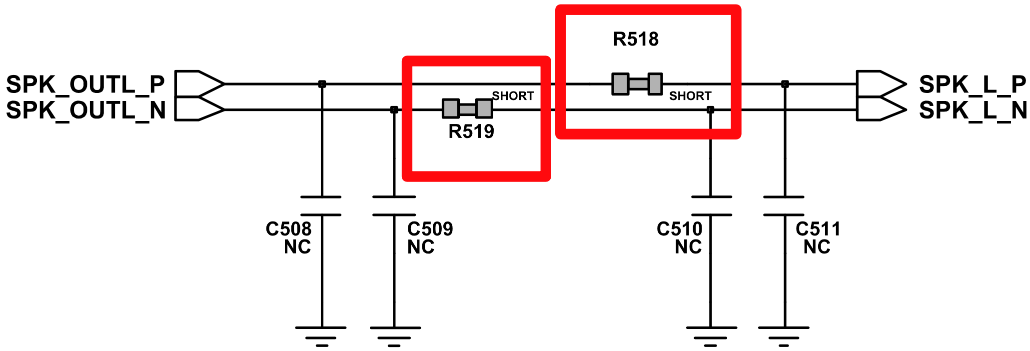
VAP_IO_1.8V
VADC_2.8V
R503
R504
C500
IF_TXD
IF_RXD
IF_TXD_CON
IF_RXD_CON
R513
VCCA
GND
_OE
A2
A1
B2
B1
DIR1
VCCB
DIR2
U504
C505

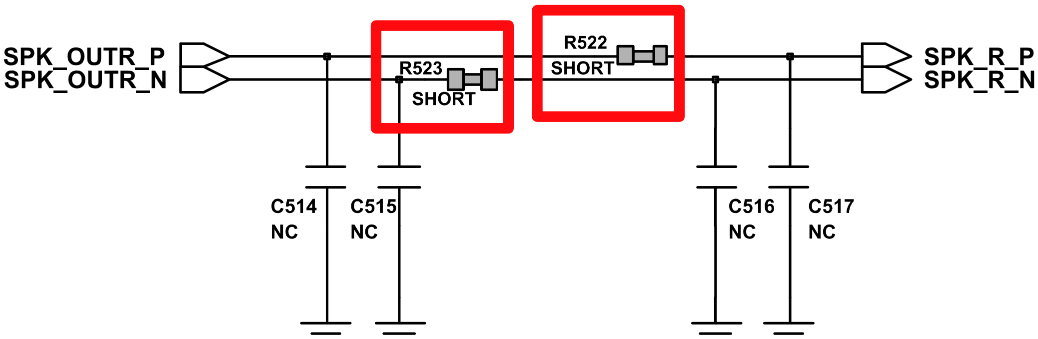


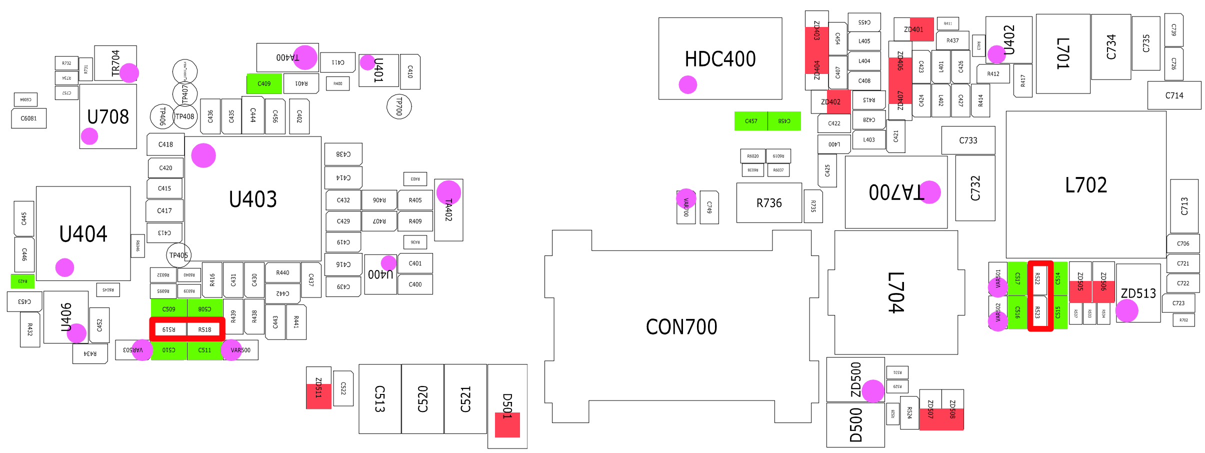
U708
U404
U403
U406
U401
U400
TA400
TA402
HDC400
CON700
TA700
L701
L702
L704
U402
C513
C520
C521
D501
ZD513
ZD500
D500
R736

8-3-5.扬声器部分

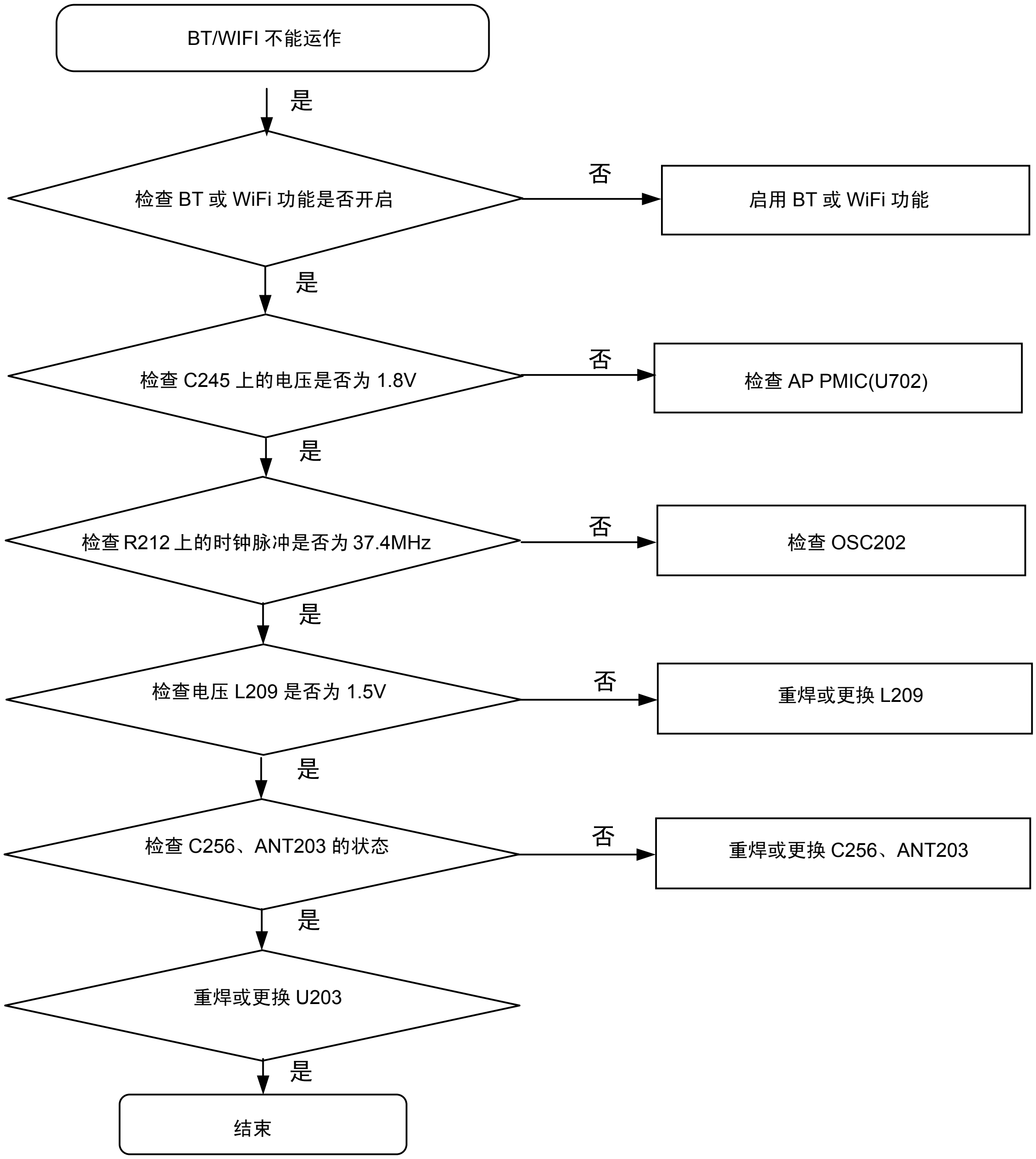

检查扬声器功能。
使用最大音量播放MP3
是
检查R518、R519、R522、R523
上的信号
否
重焊或更换 R518、R519、
R522、R523
是
更换扬声器模块
是
结束

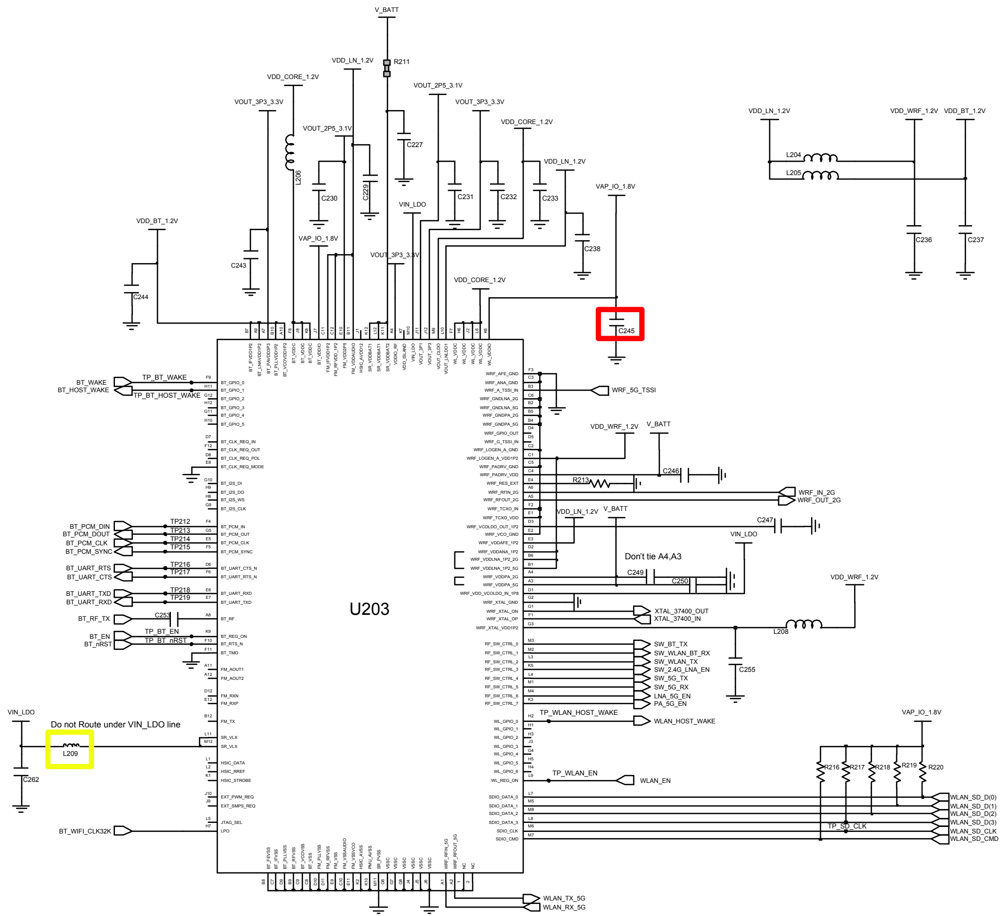

R518
SPK_OUTL_P
SPK_OUTL_N
SHORT
R519
SHORT
SPK_L_P
SPK_L_N
C508
NC
C509
NC
C510
NC
C511
NC

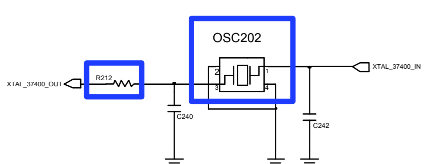

SPK_OUTR_P
SPK_OUTR_N
R523
SHORT
R522
SHORT
SPK_R_P
SPK_R_N
C514
NC
C515
NC
C516
NC
C517
NC

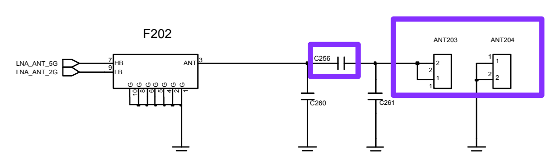

U708
U403
U404
U406
HDC400
TA700
L702
L701
L704
CON700
ZD513
C513
C520
C521
D501
D500
ZD500
R736
C734
C735
C733
C732
U402

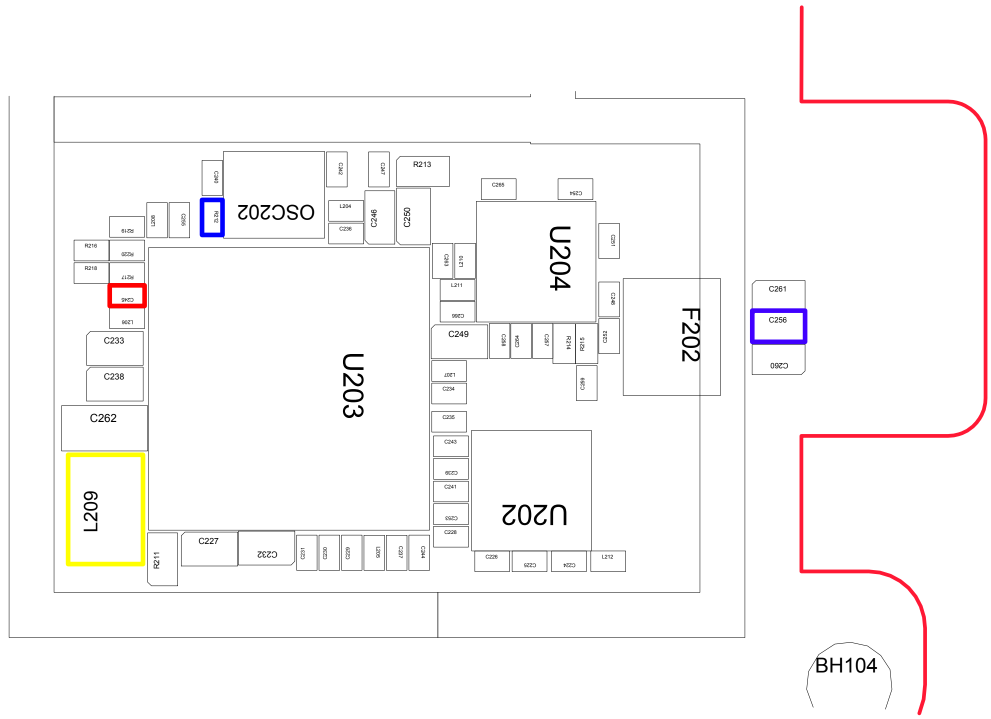
8-3-6 BT/WIFI

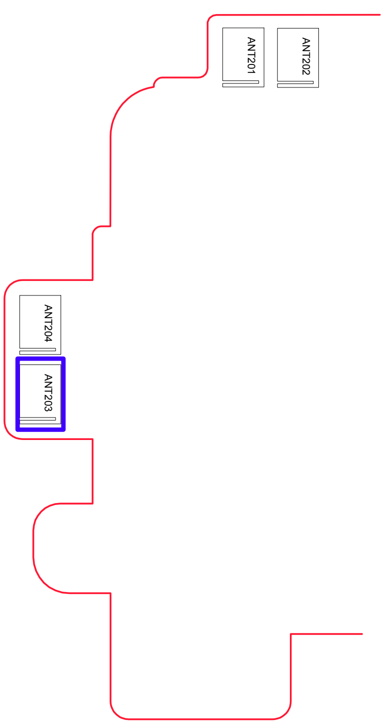

BT/WIFI 不能运作
是
检查 BT 或 WiFi 功能是否开启
否
启用 BT 或 WiFi 功能
是
检查 C245 上的电压是否为 1.8V
否
检查 AP PMIC(U702)
是
检查 R212 上的时钟脉冲是否为 37.4MHz
否
检查 OSC202
是
检查电压 L209 是否为 1.5V
否
重焊或更换 L209
是
检查 C256、ANT203 的状态
否
重焊或更换 C256、ANT203
是
重焊或更换 U203
是
结束


V_BATT
R211
VDD_LN_1.2V
VDD_CORE_1.2V
VOUT_3P3_3.3V
VOUT_2P5_3.1V
VOUT_2P5_3.1V
VOUT_3P3_3.3V
VDD_CORE_1.2V
VDD_LN_1.2V
VAP_IO_1.8V
VIN_LDO
VDD_BT_1.2V
VAP_IO_1.8V
VOUT_3P3_3.3V
VDD_CORE_1.2V
L206
C227
C229
C230
C231
C232
C233
C238
C243
C244
C245
VDD_LN_1.2V
VDD_WRF_1.2V
VDD_BT_1.2V
L204
L205
C236
C237
BT_WAKE
BT_HOST_WAKE
TP_BT_WAKE
TP_BT_HOST_WAKE
WRF_5G_TSSI
VDD_WRF_1.2V
V_BATT
C246
R213
WRF_IN_2G
WRF_OUT_2G
VDD_LN_1.2V
V_BATT
C247
VIN_LDO
Don't tie A4,A3
C249
C250
VDD_WRF_1.2V
BT_PCM_DIN
BT_PCM_DOUT
BT_PCM_CLK
BT_PCM_SYNC
TP212
TP213
TP214
TP215
BT_UART_RTS
BT_UART_CTS
TP216
TP217
BT_UART_TXD
BT_UART_RXD
TP218
TP219
BT_RF_TX
C253
BT_EN
BT_nRST
TP_BT_EN
TP_BT_nRST
U203
XTAL_37400_OUT
XTAL_37400_IN
L208
SW_BT_TX
SW_WLAN_BT_RX
SW_WLAN_TX
SW_2.4G_LNA_EN
SW_5G_TX
SW_5G_RX
LNA_5G_EN
PA_5G_EN
C255
TP_WLAN_HOST_WAKE
WLAN_HOST_WAKE
VAP_IO_1.8V
VIN_LDO
Do not Route under VIN_LDO line
L209
C262
TP_WLAN_EN
WLAN_EN
R216
R217
R218
R219
R220
WLAN_SD_D(0)
WLAN_SD_D(1)
WLAN_SD_D(2)
WLAN_SD_D(3)
WLAN_SD_CLK
WLAN_SD_CMD
TP_SD_CLK
BT_WIFI_CLK32K
WLAN_TX_5G
WLAN_RX_5G

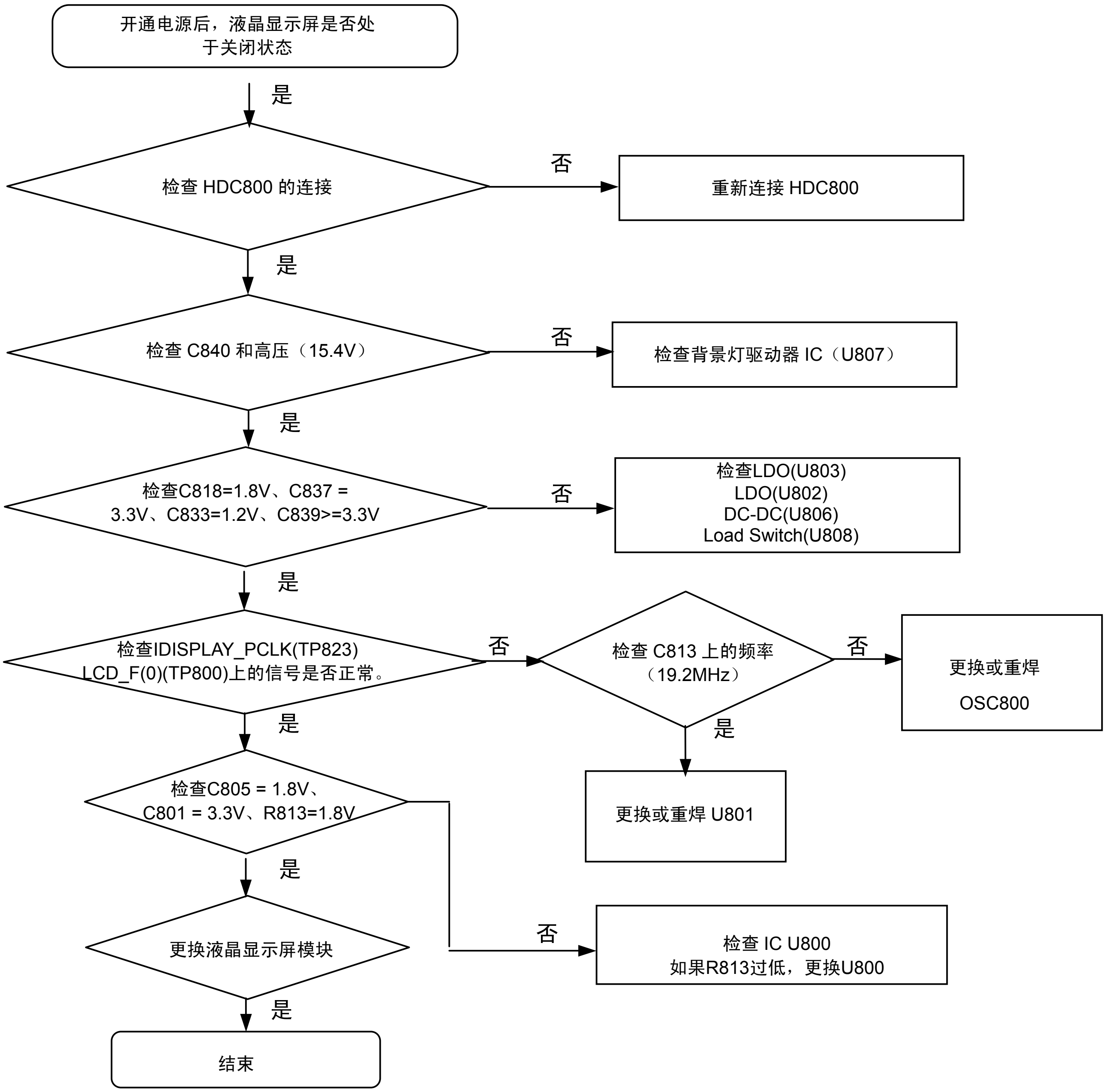

OSC202
XTAL_37400_OUT
R212
XTAL_37400_IN
C240
C242

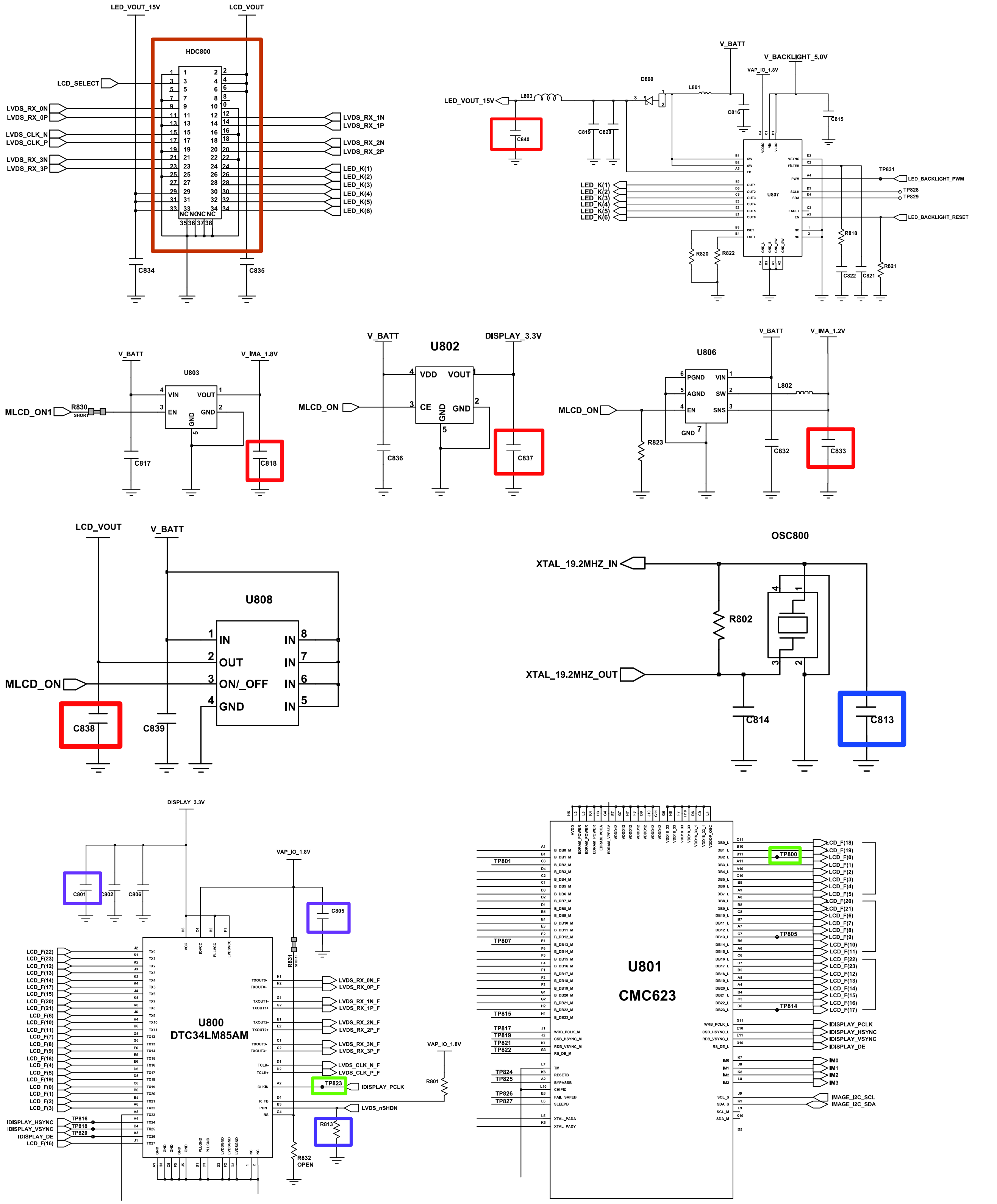

F202
LNA_ANT_5G
LNA_ANT_2G
HB
LB
ANT
C256
C260
C261
ANT203
ANT204


OSC202
U204
U203
F202
U202
L209
C262
C233
C238
C227
C232
C249
C250
C246
R213
R211
C261
C256
C260
BH104

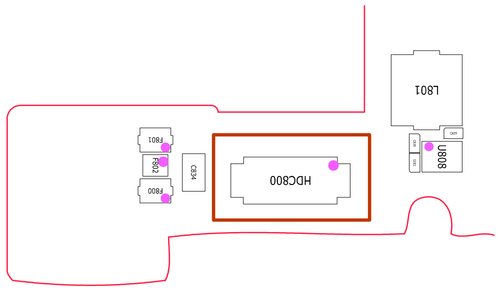

ANT201
ANT202
ANT204
ANT203

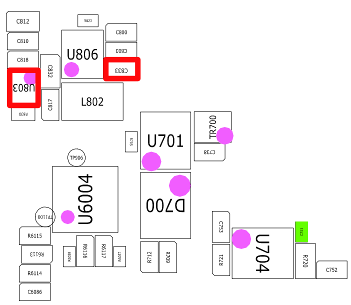
8-3-7.液晶显示屏

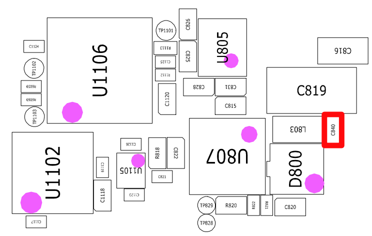

开通电源后，液晶显示屏是否处于关闭状态
是
检查 HDC800 的连接
否
重新连接 HDC800
是
检查 C840 和高压（15.4V）
否
检查背景灯驱动器 IC（U807）
是
检查C818=1.8V、C837 = 3.3V、C833=1.2V、C839>=3.3V
否
检查LDO(U803)
LDO(U802)
DC-DC(U806)
Load Switch(U808)
是
检查IDISPLAY_PCLK(TP823) LCD_F(0)(TP800)上的信号是否正常。
否
检查 C813 上的频率（19.2MHz）
否
更换或重焊 OSC800
是
更换或重焊 U801
是
检查C805 = 1.8V、C801 = 3.3V、R813=1.8V
否
检查 IC U800 如果R813过低，更换U800
是
更换液晶显示屏模块
是
结束



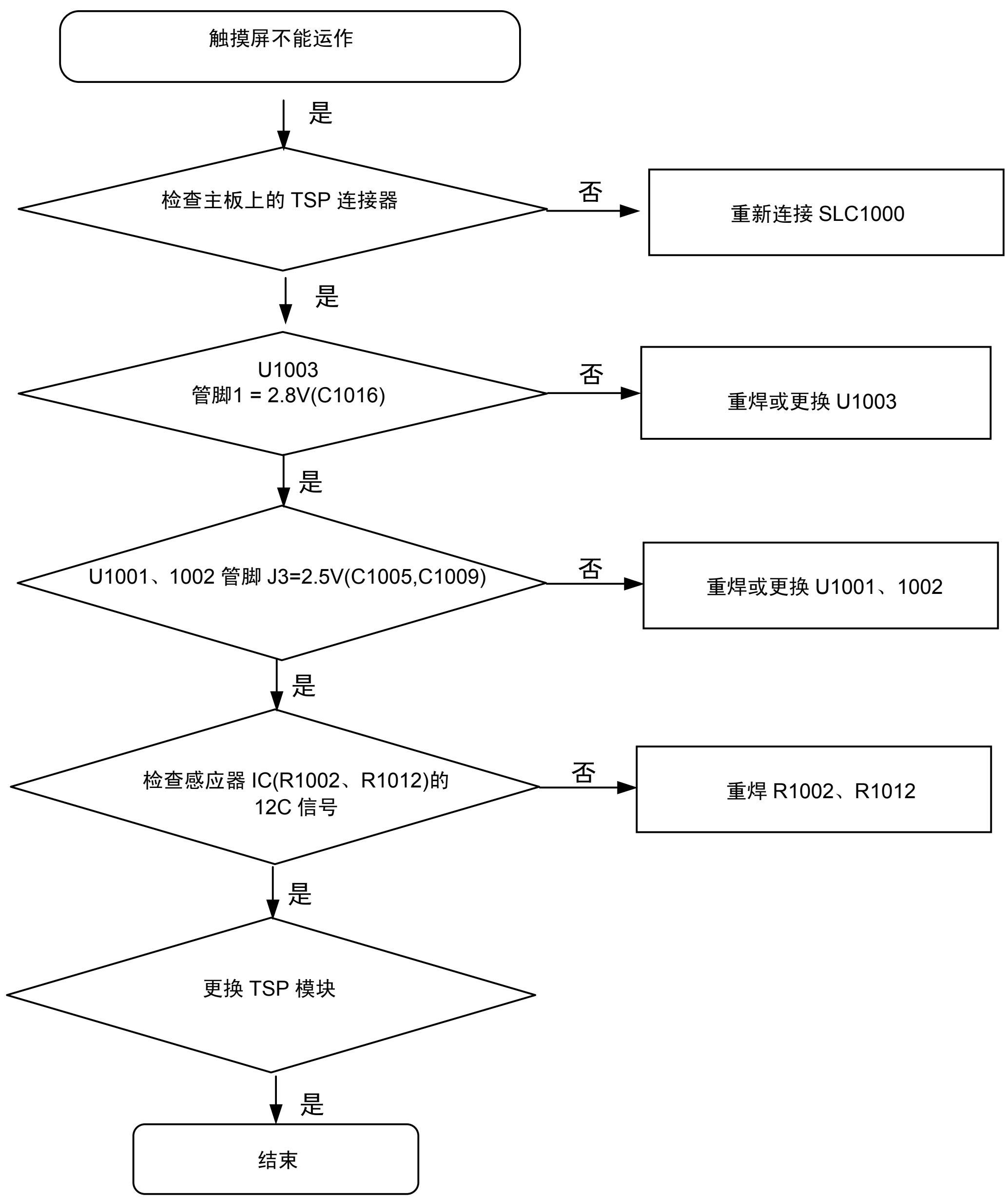
LED_VOUT_15V
LCD_VOUT
HDC800
LCD_SELECT
LVDS_RX_0N
LVDS_RX_0P
LVDS_CLK_N
LVDS_CLK_P
LVDS_RX_3N
LVDS_RX_3P
LVDS_RX_1N
LVDS_RX_1P
LVDS_RX_2N
LVDS_RX_2P
LED_K(1)
LED_K(2)
LED_K(3)
LED_K(4)
LED_K(5)
LED_K(6)
C834
C835
V_BATT
V_BACKLIGHT_5.0V
VAP_IO_1.8V
D800
L801
L803
C840
C816
C815
C819
C820
U807
TP831
LED_BACKLIGHT_PWM
TP828
TP829
LED_BACKLIGHT_RESET
R818
R820
R822
C822
C821
R821
U803
V_IMA_1.8V
MLCD_ON1
R830
C817
C818
U802
DISPLAY_3.3V
MLCD_ON
C836
C837
U806
V_IMA_1.2V
L802
R823
C832
C833
LCD_VOUT
U808
C838
C839
OSC800
XTAL_19.2MHZ_IN
XTAL_19.2MHZ_OUT
R802
C814
C813
U800
DTC34LM85AM
C801
C802
C806
C805
R831
LVDS_RX_0N_F
LVDS_RX_0P_F
LVDS_RX_1N_F
LVDS_RX_1P_F
LVDS_RX_2N_F
LVDS_RX_2P_F
LVDS_RX_3N_F
LVDS_RX_3P_F
LVDS_CLK_N_F
LVDS_CLK_P_F
TP823
IDISPLAY_PCLK
LVDS_nSHDN
R801
R813
R832
OPEN
IDISPLAY_HSYNC
IDISPLAY_VSYNC
IDISPLAY_DE
U801
CMC623
TP800
TP805
TP814
IMAGE_I2C_SCL
IMAGE_I2C_SDA

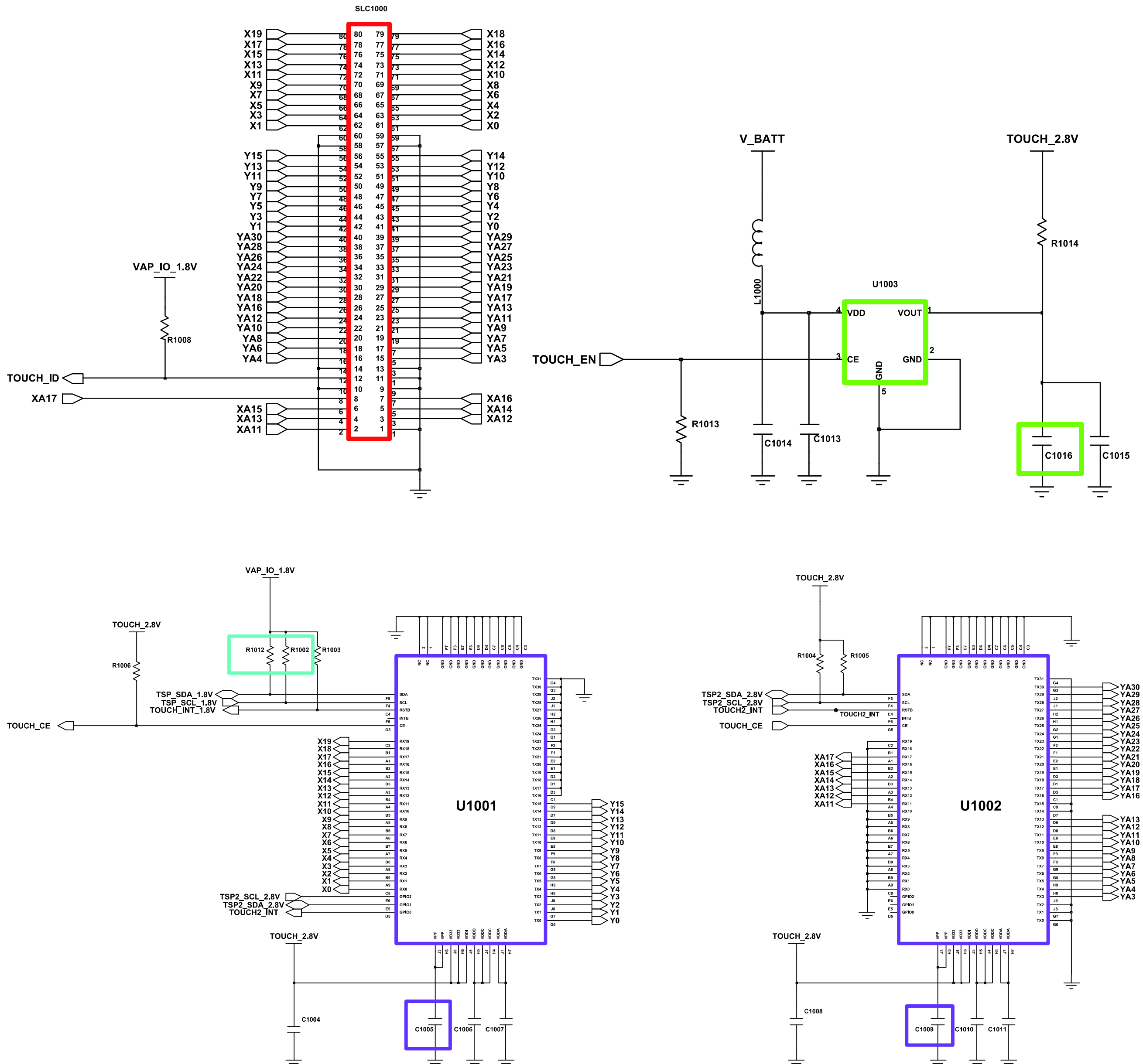


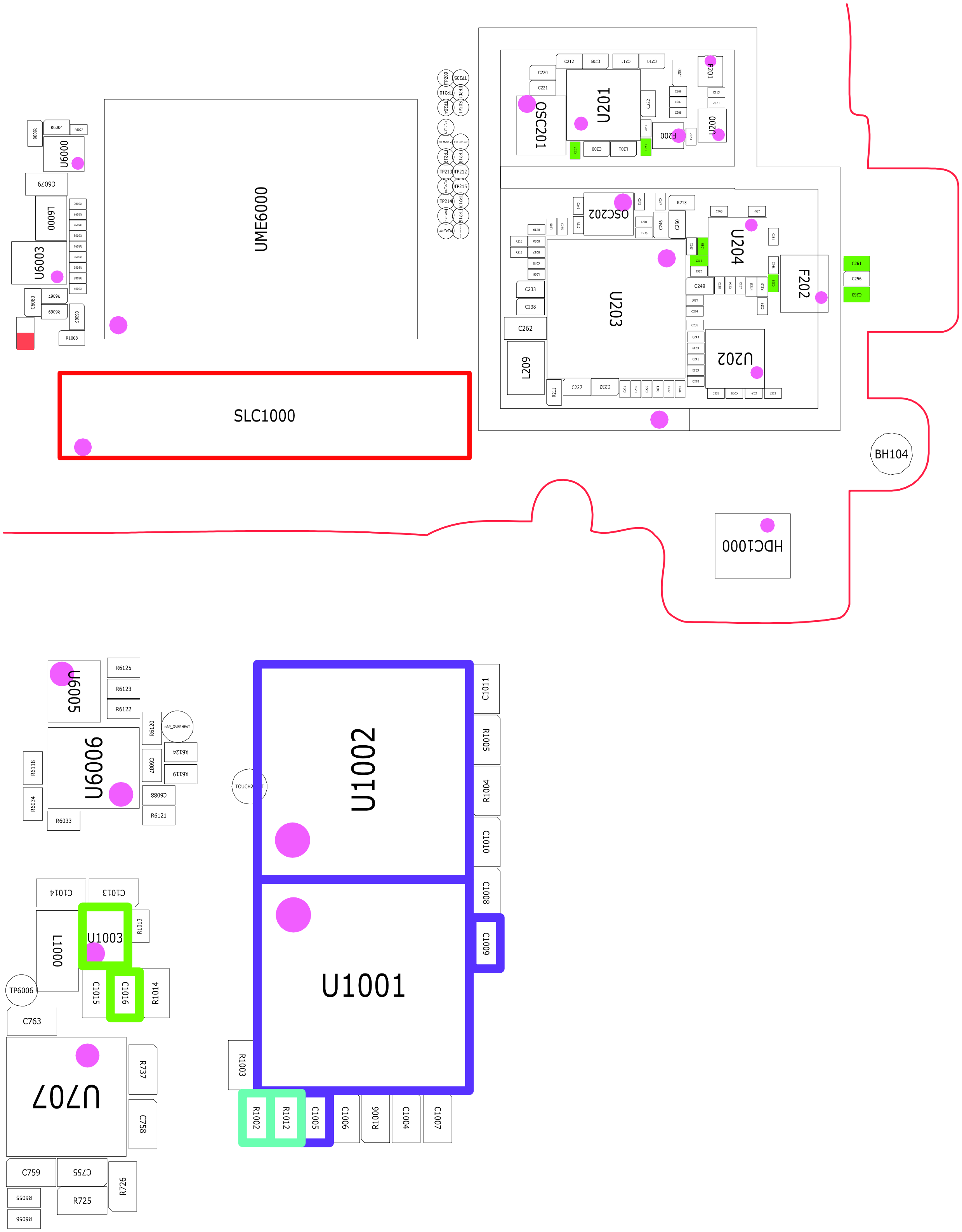




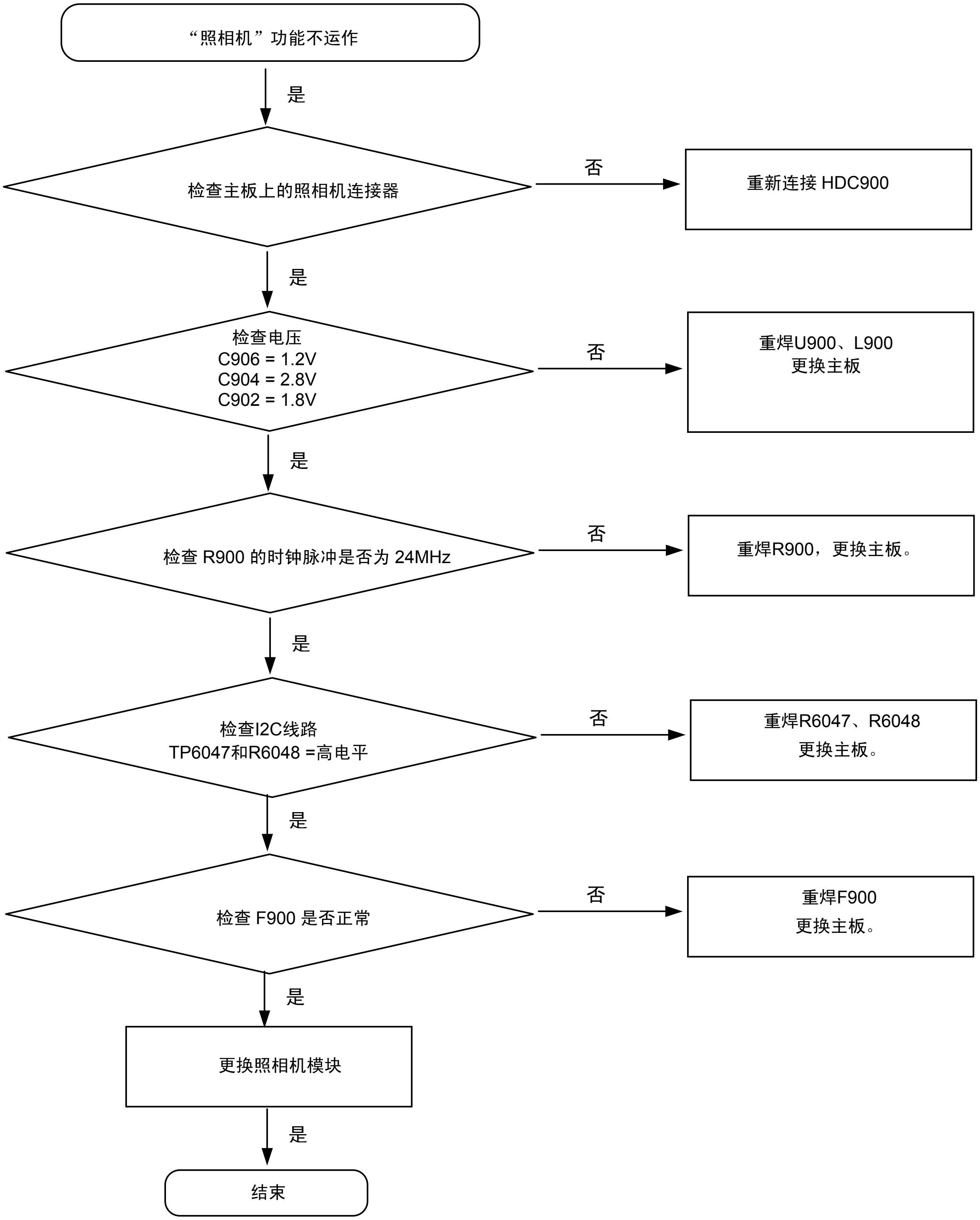
8-3-8.TSP

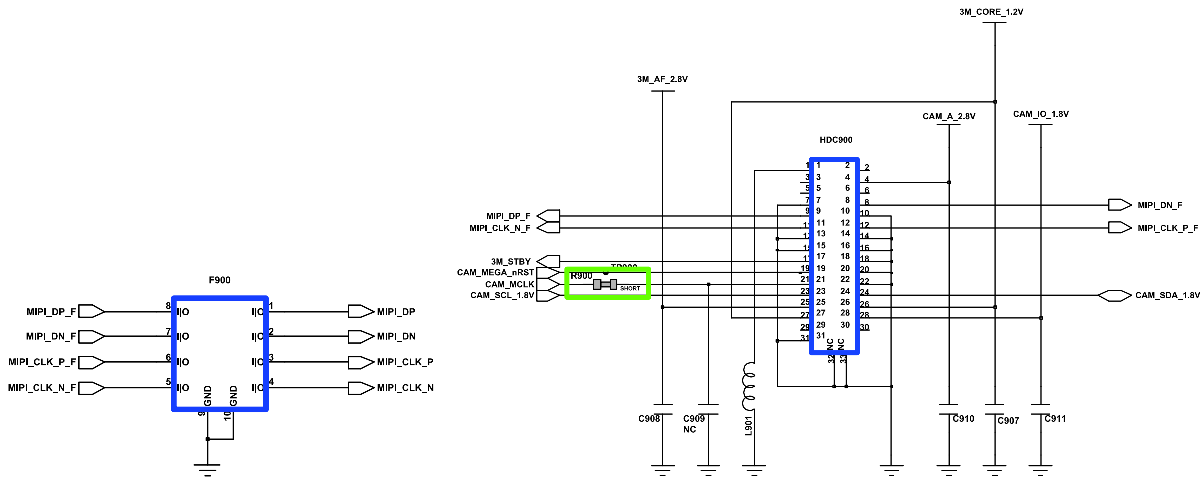

触摸屏不能运作
是
检查主板上的 TSP 连接器
否
重新连接 SLC1000
是
U1003
管脚1 = 2.8V(C1016)
否
重焊或更换 U1003
是
U1001、1002 管脚 J3=2.5V(C1005,C1009)
否
重焊或更换 U1001、1002
是
检查感应器 IC(R1002、R1012)的
12C 信号
否
重焊 R1002、R1012
是
更换 TSP 模块
是
结束

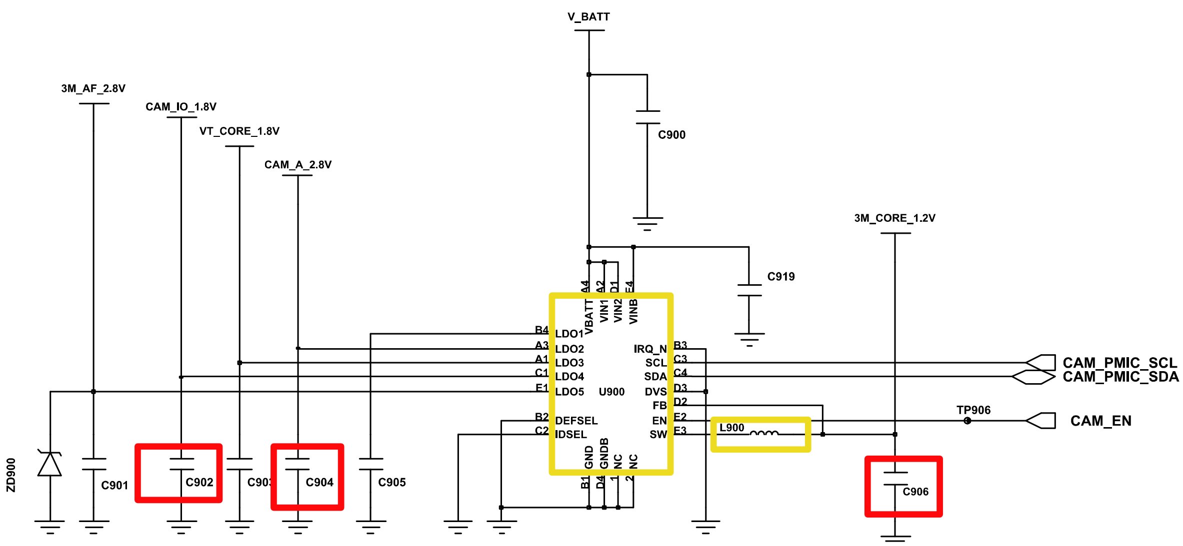

SLC1000
X19
X17
X15
X13
X11
X9
X7
X5
X3
X1
X18
X16
X14
X12
X10
X8
X6
X4
X2
X0
Y15
Y13
Y11
Y9
Y7
Y5
Y3
Y1
YA30
YA28
YA26
YA24
YA22
YA20
YA18
YA16
YA12
YA10
YA8
YA6
YA4
Y14
Y12
Y10
Y8
Y6
Y4
Y2
Y0
YA29
YA27
YA25
YA23
YA21
YA19
YA17
YA13
YA11
YA9
YA7
YA5
YA3
VAP_IO_1.8V
R1008
TOUCH_ID
XA17
XA15
XA13
XA11
XA16
XA14
XA12
V_BATT
TOUCH_2.8V
L1000
U1003
VDD
VOUT
CE
GND
GND
TOUCH_EN
R1013
C1014
C1013
R1014
C1016
C1015
VAP_IO_1.8V
TOUCH_2.8V
R1006
R1012
R1002
R1003
TSP_SDA_1.8V
TSP_SCL_1.8V
TOUCH_INT_1.8V
TOUCH_CE
U1001
TSP2_SCL_2.8V
TSP2_SDA_2.8V
TOUCH2_INT
TOUCH_2.8V
C1004
C1005
C1006
C1007
TOUCH_2.8V
R1004
R1005
TSP2_SDA_2.8V
TSP2_SCL_2.8V
TOUCH2_INT
TOUCH_CE
U1002
C1008
C1009
C1010
C1011

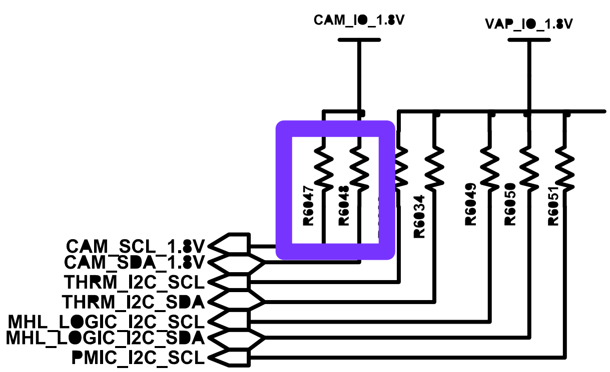

UME6000
U6000
U6003
L6000
C6079
R1008
SLC1000
U201
OSC201
U200
F201
F200
OSC202
U204
U203
U202
F202
L209
C262
C233
C238
C249
C227
C232
R211
R213
C212
C209
C211
C210
C220
C221
C200
L201
C261
C256
C260
BH104
HDC1000
U6005
U6006
R6125
R6123
R6122
R6120
R6124
R6119
C6087
C6088
R6121
R6118
R6034
R6033
U1002
U1001
C1011
R1005
R1004
C1010
C1008
C1009
TOUCH
C1014
C1013
U1003
L1000
R1013
C1015
C1016
R1014
TP6006
C763
U707
R737
C758
C759
C755
R726
R725
R6055
R6056
R1003
R1002
R1012
C1005
C1006
R1006
C1004
C1007

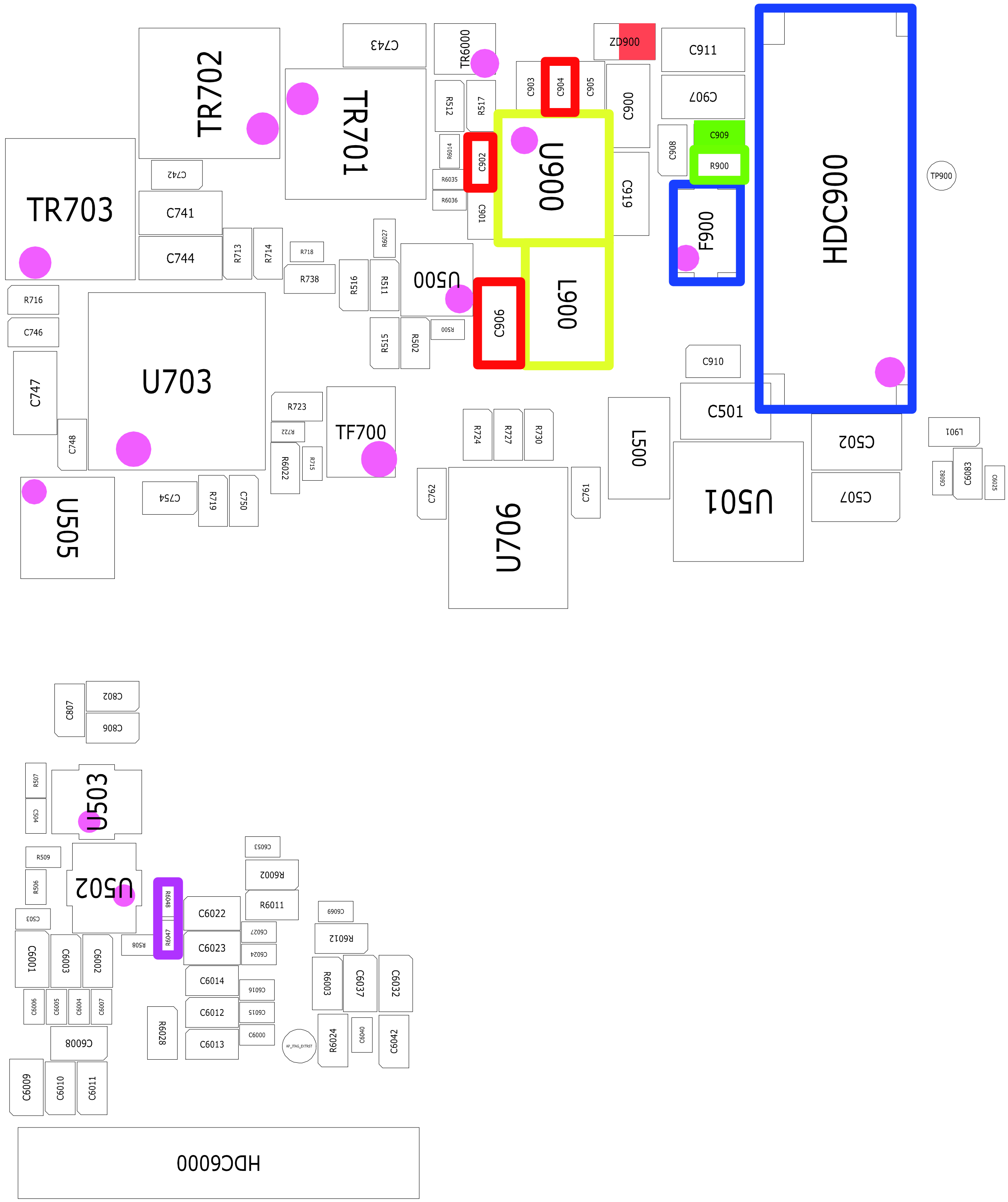
8-3-9.3M 照相机


“照相机”功能不运作
是
检查主板上的照相机连接器
否
重新连接 HDC900
是
检查电压
C906 = 1.2V
C904 = 2.8V
C902 = 1.8V
否
重焊U900、L900
更换主板
是
检查 R900 的时钟脉冲是否为 24MHz
否
重焊R900，更换主板。
是
检查I2C线路
TP6047和R6048 =高电平
否
重焊R6047、R6048
更换主板。
是
检查 F900 是否正常
否
重焊F900
更换主板。
是
更换照相机模块
是
结束

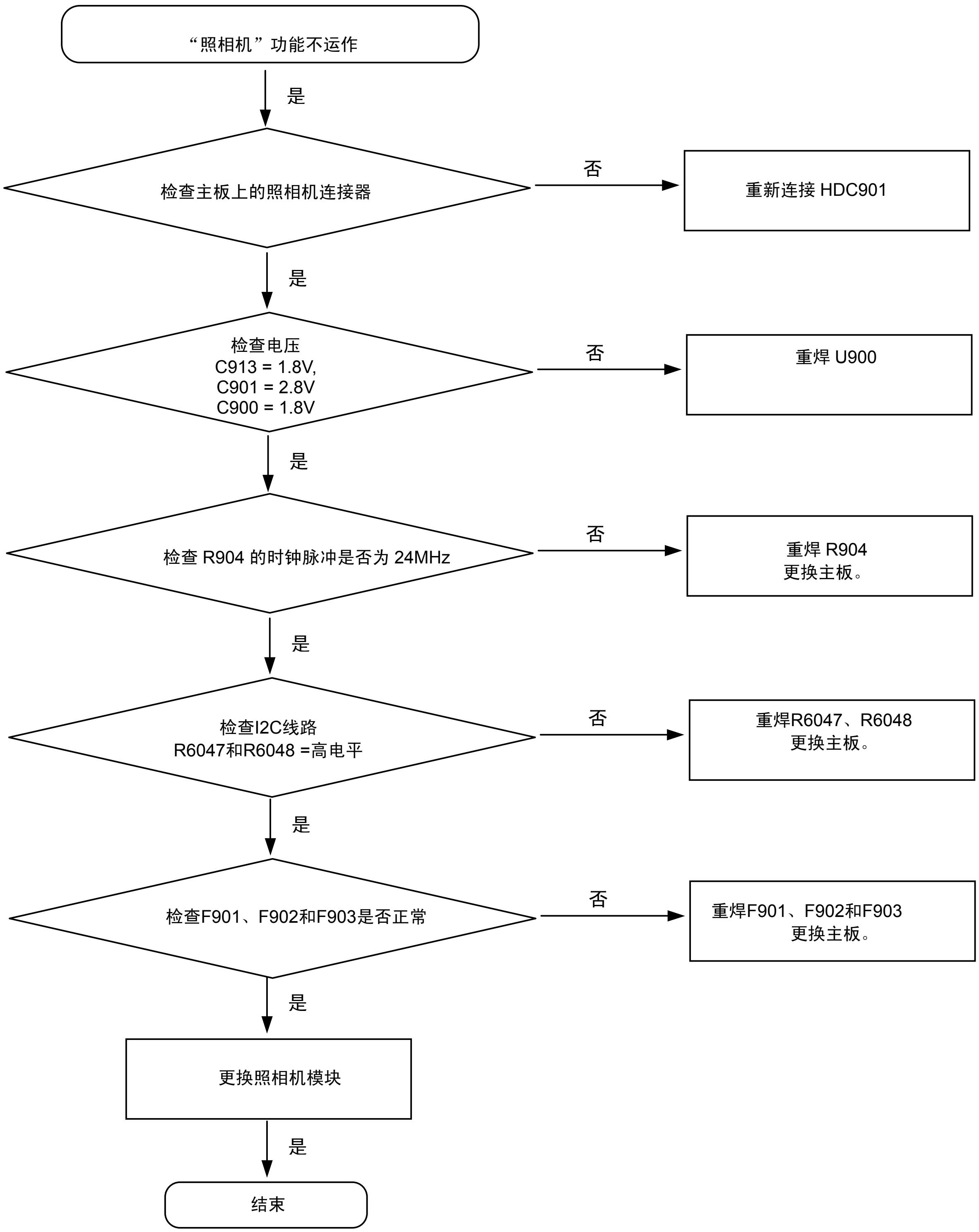


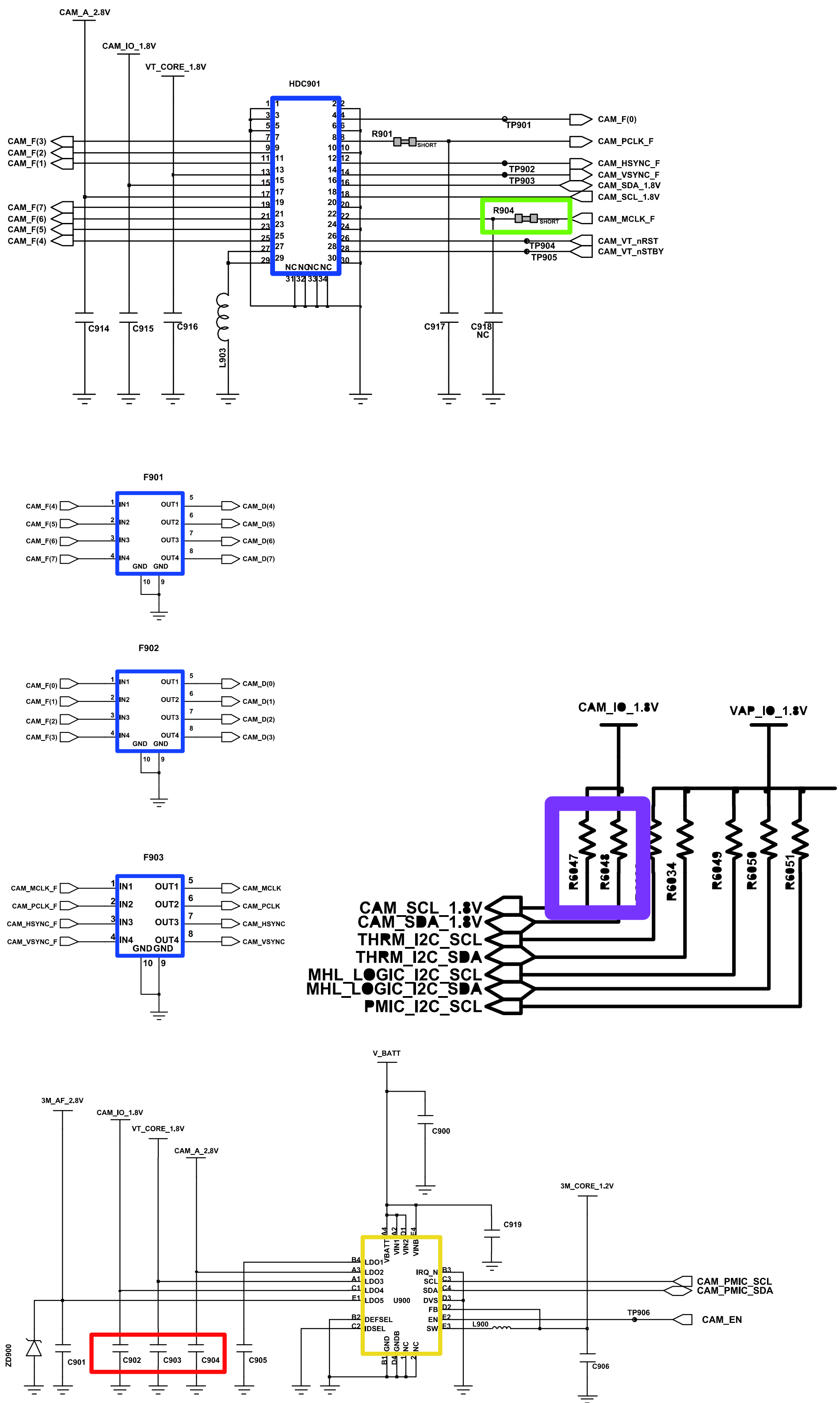


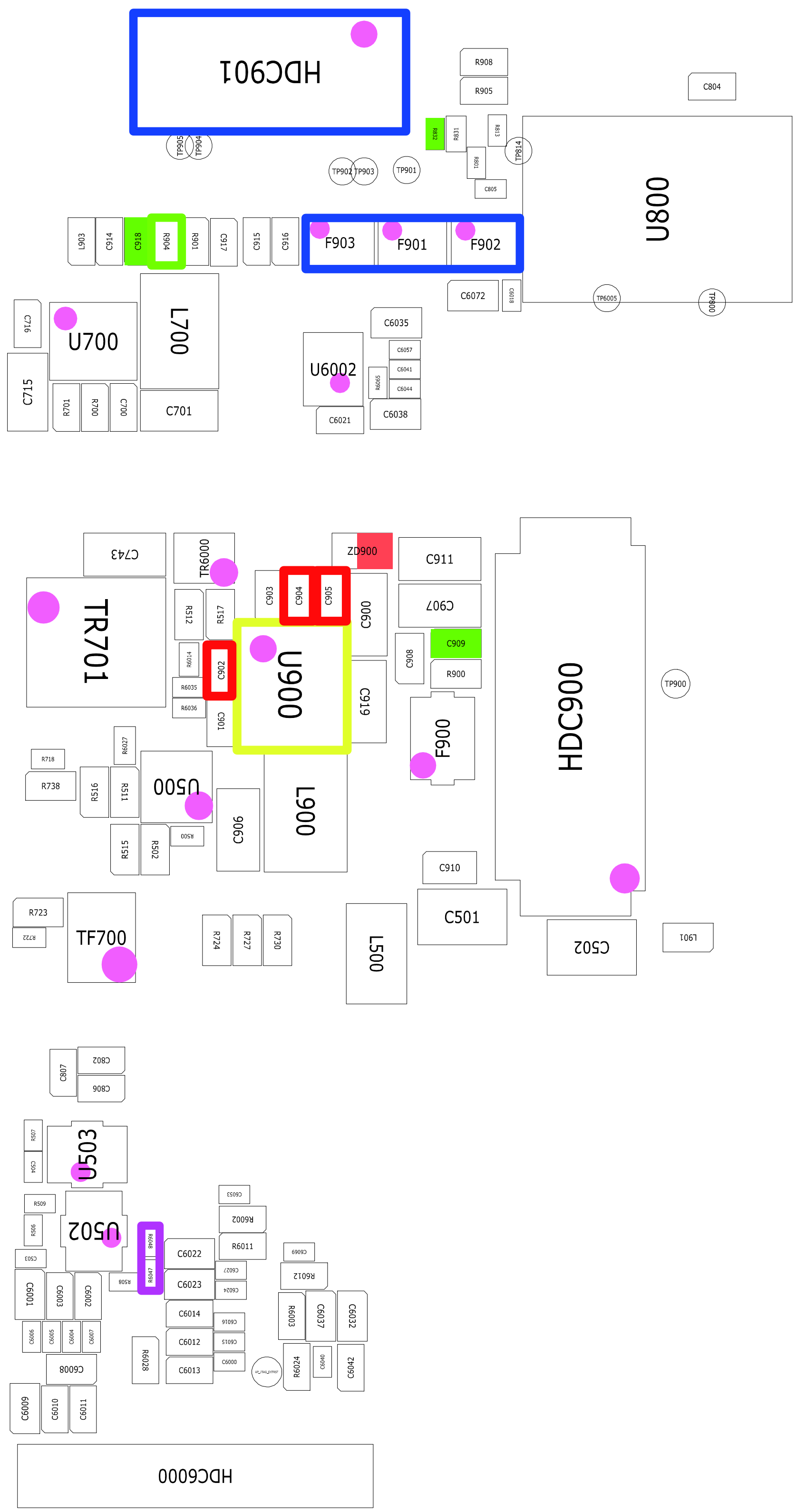

CAM_IO_1.8V
VAP_IO_1.8V
R6047
R6048
R6034
R6049
R6050
R6051
CAM_SCL_1.8V
CAM_SDA_1.8V
THRM_I2C_SCL
THRM_I2C_SDA
MHL_LOGIC_I2C_SCL
MHL_LOGIC_I2C_SDA
PMIC_I2C_SCL



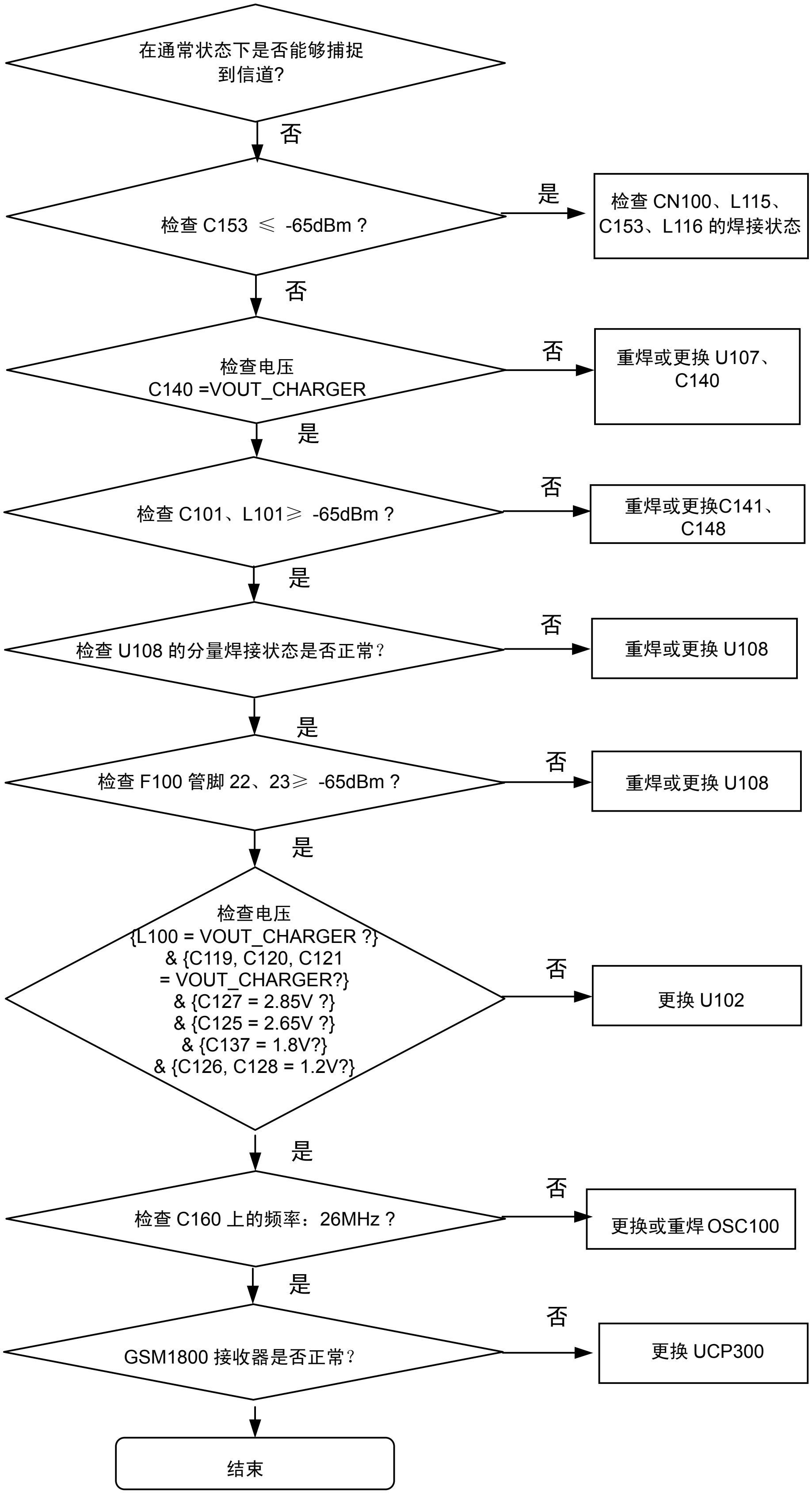
8-3-10. 2M 照相机

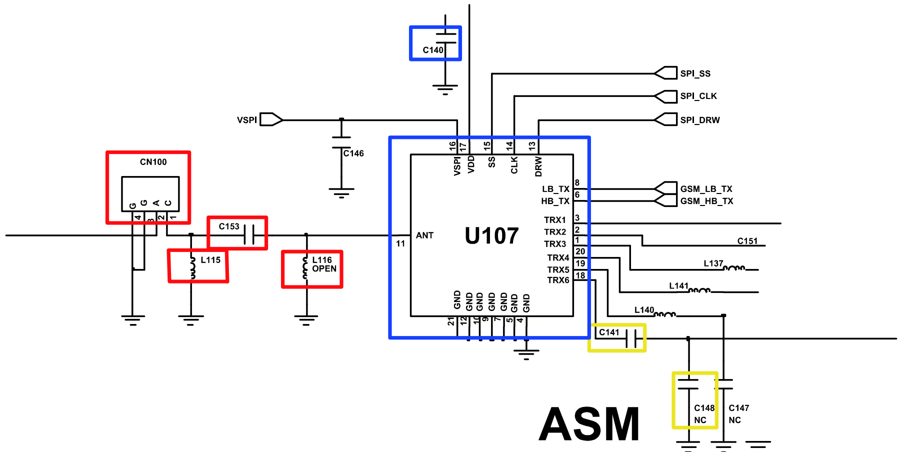

“照相机”功能不运作
是
检查主板上的照相机连接器
否
重新连接 HDC901
是
检查电压
C913 = 1.8V,
C901 = 2.8V
C900 = 1.8V
否
重焊 U900
是
检查 R904 的时钟脉冲是否为 24MHz
否
重焊 R904
更换主板。
是
检查I2C线路
R6047和R6048 =高电平
否
重焊R6047、R6048
更换主板。
是
检查F901、F902和F903是否正常
否
重焊F901、F902和F903
更换主板。
是
更换照相机模块
是
结束

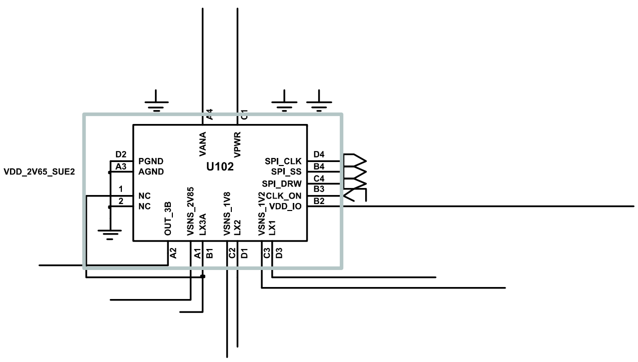

CAM_A_2.8V
CAM_IO_1.8V
VT_CORE_1.8V
HDC901
CAM_F(0)
CAM_PCLK_F
CAM_HSYNC_F
CAM_VSYNC_F
CAM_SDA_1.8V
CAM_SCL_1.8V
CAM_MCLK_F
CAM_VT_nRST
CAM_VT_nSTBY
R901
R904
TP901
TP902
TP903
TP904
TP905
C914
C915
C916
C917
C918 NC
L903
F901
F902
F903
CAM_IO_1.8V
VAP_IO_1.8V
R6047
R6048
R6034
R6049
R6050
R6051
CAM_SCL_1.8V
CAM_SDA_1.8V
THRM_I2C_SCL
THRM_I2C_SDA
MHL_LOGIC_I2C_SCL
MHL_LOGIC_I2C_SDA
PMIC_I2C_SCL
V_BATT
3M_AF_2.8V
3M_CORE_1.2V
U900
C900
C901
C902
C903
C904
C905
C906
C919
ZD900
L900
TP906
CAM_PMIC_SCL
CAM_PMIC_SDA
CAM_EN

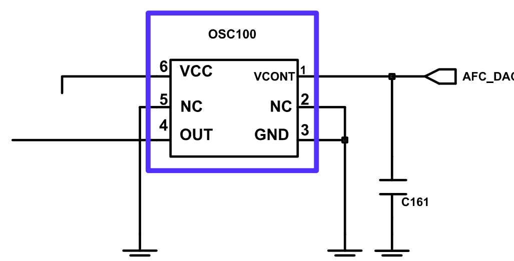

HDC901
R908
R905
C804
U800
F903
F901
F902
C6072
L903
C914
C918
R904
R901
C917
C915
C916
C716
U700
L700
C715
R701
R700
C700
C701
U6002
C6035
C6021
C6038
C743
TR6000
ZD900
C911
C907
TR701
R512
R517
C903
C904
C905
C900
C909
R900
C902
U900
C919
F900
HDC900
TP900
R718
R738
R516
R511
U500
L900
C906
R515
R502
R723
R722
TF700
R724
R727
R730
C910
C501
L500
C502
L901
C807
C802
C806
U503
R509
U502
C6022
C6023
C6014
C6012
C6013
R6002
R6011
R6012
R6003
C6037
C6032
R6024
C6042
R6028
C6001
C6003
C6002
C6008
C6009
C6010
C6011
HDC6000

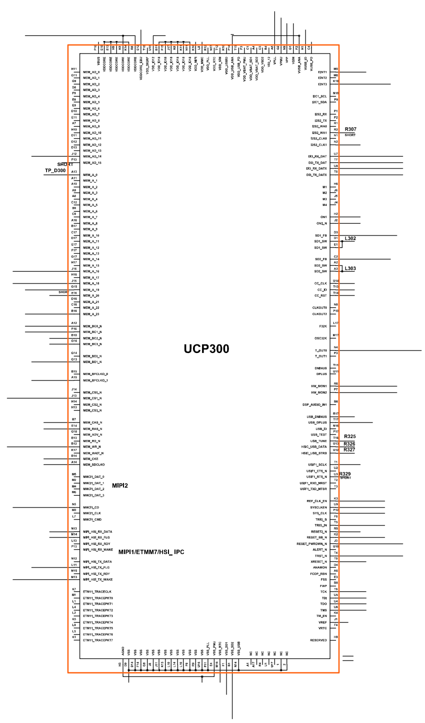
8-3-12.GSM1800 接收器

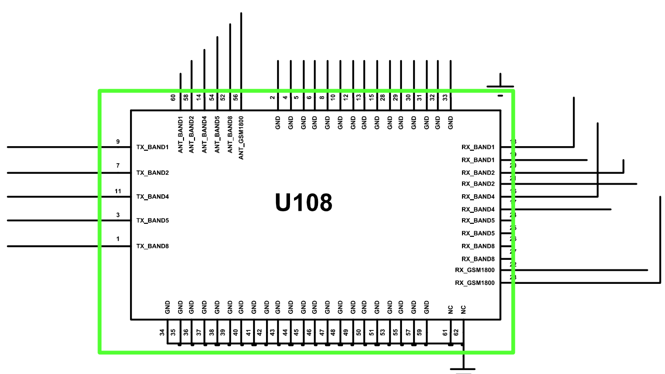


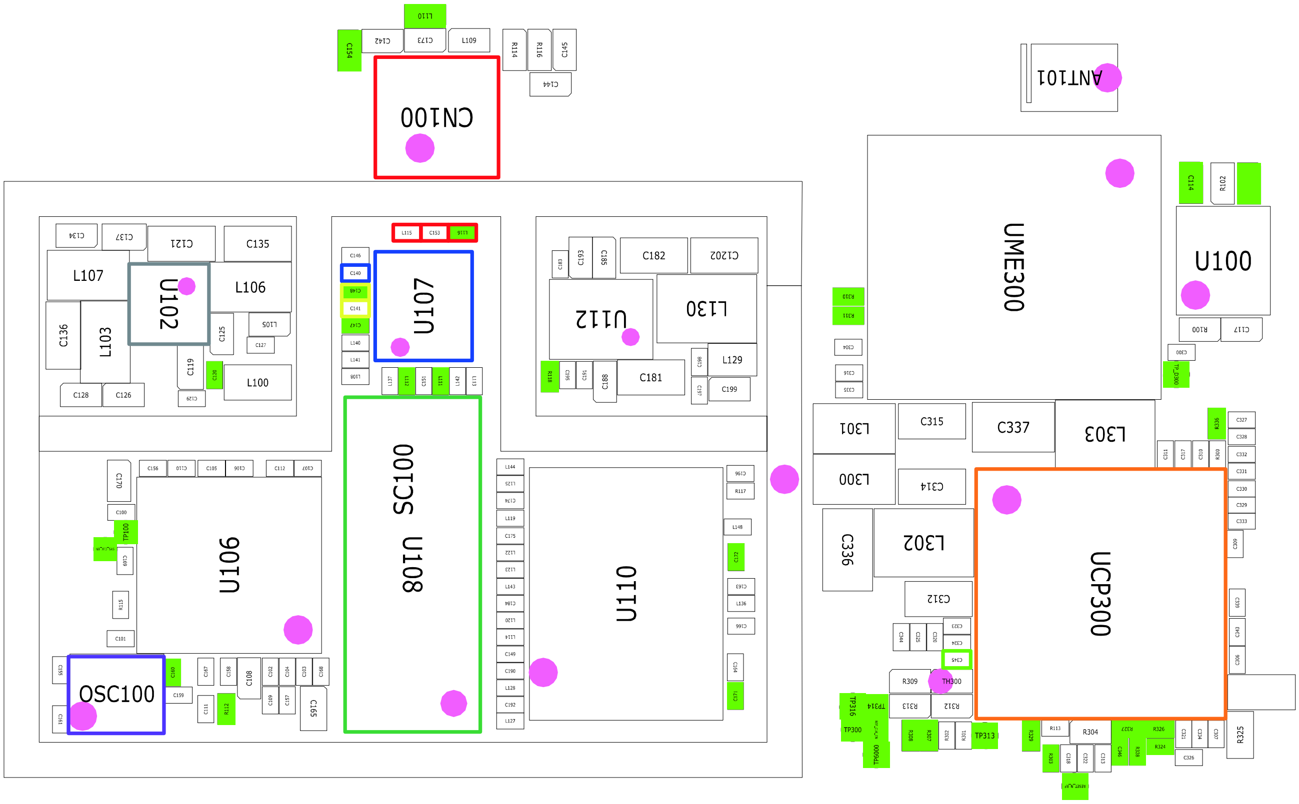

C140
SPI_SS
SPI_CLK
SPI_DRW
VSPI
C146
CN100
C153
L115
L116
OPEN
U107
ANT
VSPI
VDD
SS
CLK
DRW
LB_TX
HB_TX
GSM_LB_TX
GSM_HB_TX
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
C151
L137
L141
L140
C141
C148
NC
C147
NC
ASM


VDD_2V65_SUE2
U102
PGND
AGND
NC
NC
VANA
VPWR
SPI_CLK
SPI_SS
SPI_DRW
CLK_ON
VDD_IO
OUT_3B
VSNS_2V85
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1

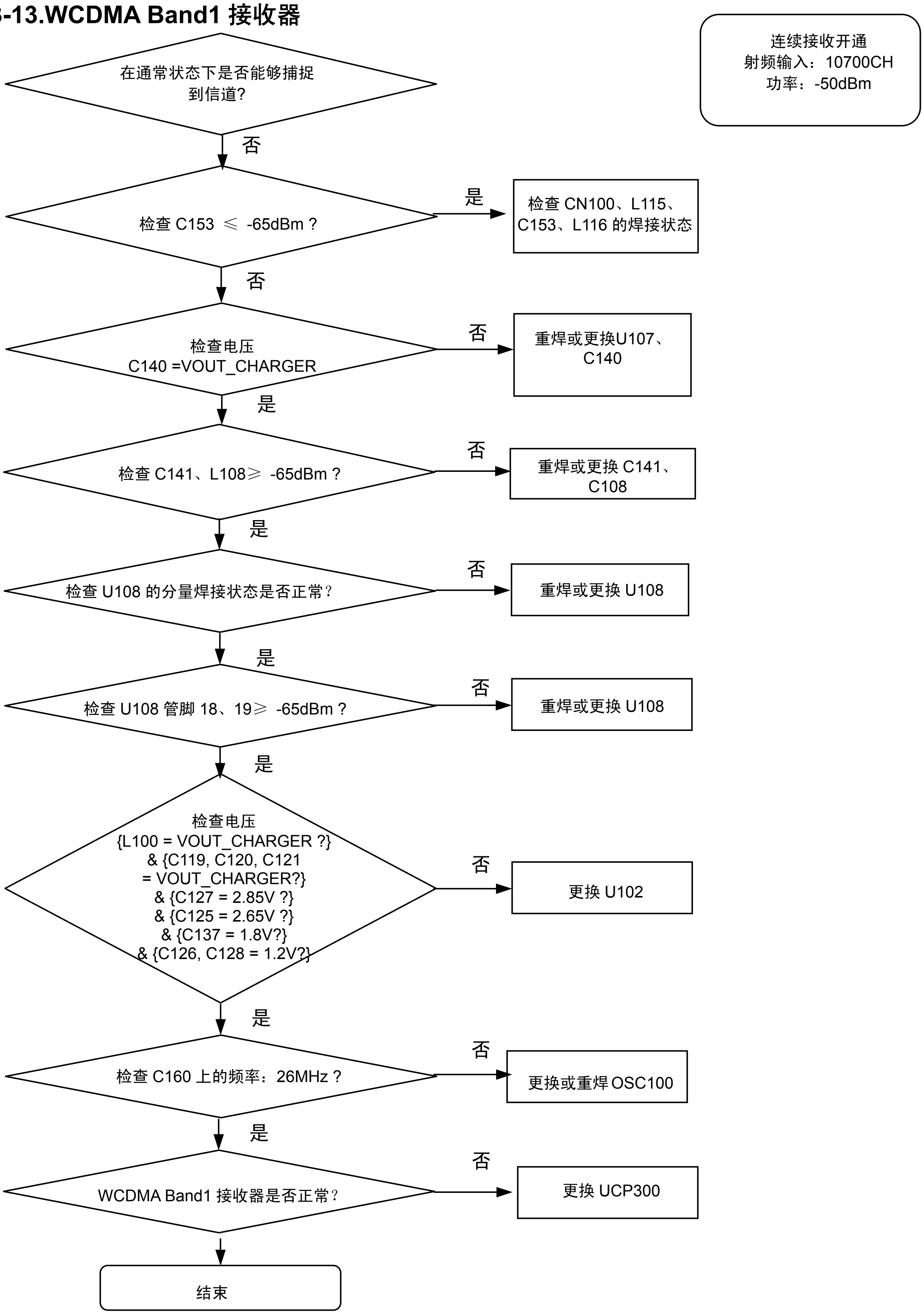

OSC100
VCC
VCONT
NC
NC
OUT
GND
AFC_DAC
C161

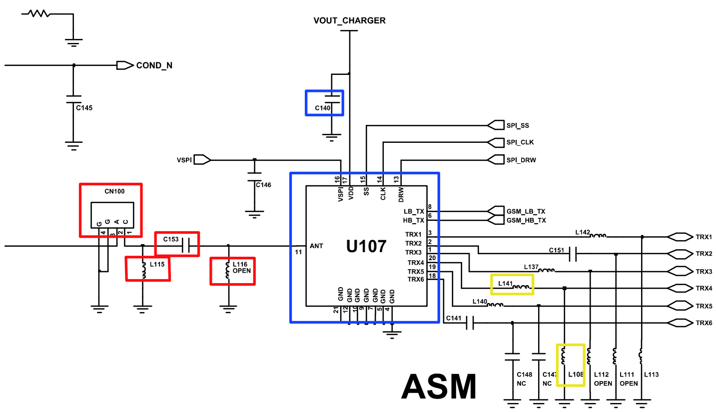

UCP300
MIPI2
MIPI1/ETMM7/HSI_IPC


U108
TX_BAND1
TX_BAND2
TX_BAND4
TX_BAND5
TX_BAND8
ANT_BAND1
ANT_BAND2
ANT_BAND4
ANT_BAND5
ANT_BAND8
ANT_GSM1800
RX_BAND1
RX_BAND1
RX_BAND2
RX_BAND2
RX_BAND4
RX_BAND4
RX_BAND5
RX_BAND5
RX_BAND8
RX_BAND8
RX_GSM1800
RX_GSM1800

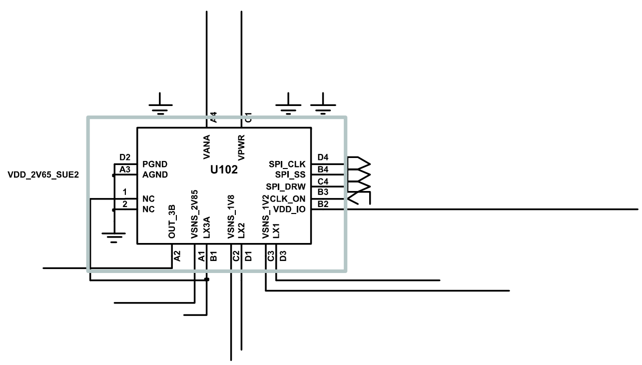

CN100
U102
U107
U112
L130
UME300
U100
ANT101
U106
SC100
U108
U110
UCP300
OSC100
L107
L106
L103
C136
C135
C121
L100
C182
C1202
C181
L129
C199
L301
C315
C337
L303
L300
C314
C336
L302
C312
R309
R313
R312
R304
R325
C165
C108
L109
C173
C142
C154
R114
R116
C145
C144
R102
C114
R100
C117
C128
C126
C119
C125
C170
TP100
C188
C185
C193
TH300
TP300
TP316
TP314
TP313

8-3-13.WCDMA Band1 接收器

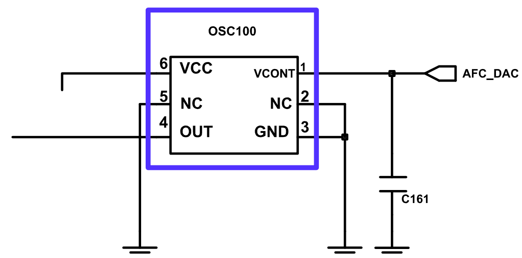




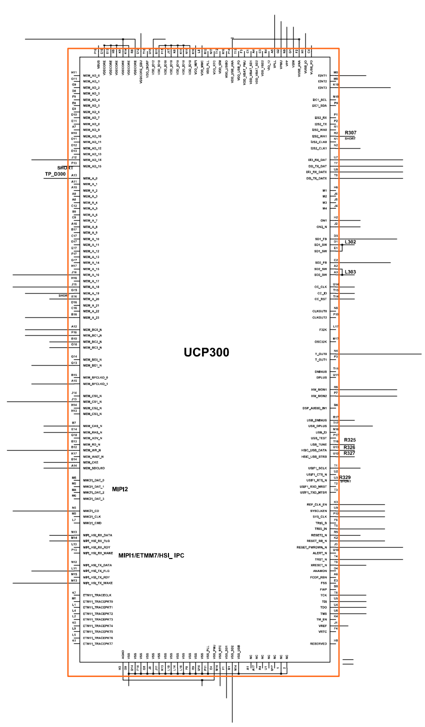
VOUT_CHARGER
COND_N
C145
C140
SPI_SS
SPI_CLK
SPI_DRW
VSPI
C146
CN100
C153
L115
L116
OPEN
U107
ANT
LB_TX
HB_TX
GSM_LB_TX
GSM_HB_TX
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
L142
C151
L137
L141
L140
C141
C148
NC
C147
NC
L108
L112
OPEN
L111
OPEN
L113
ASM



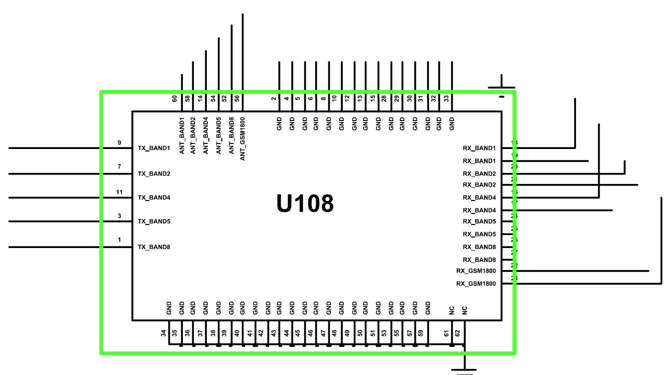
VDD_2V65_SUE2
U102
PGND
AGND
NC
NC
VANA
VPWR
SPI_CLK
SPI_SS
SPI_DRW
CLK_ON
VDD_IO
OUT_3B
VSNS_2V85
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1



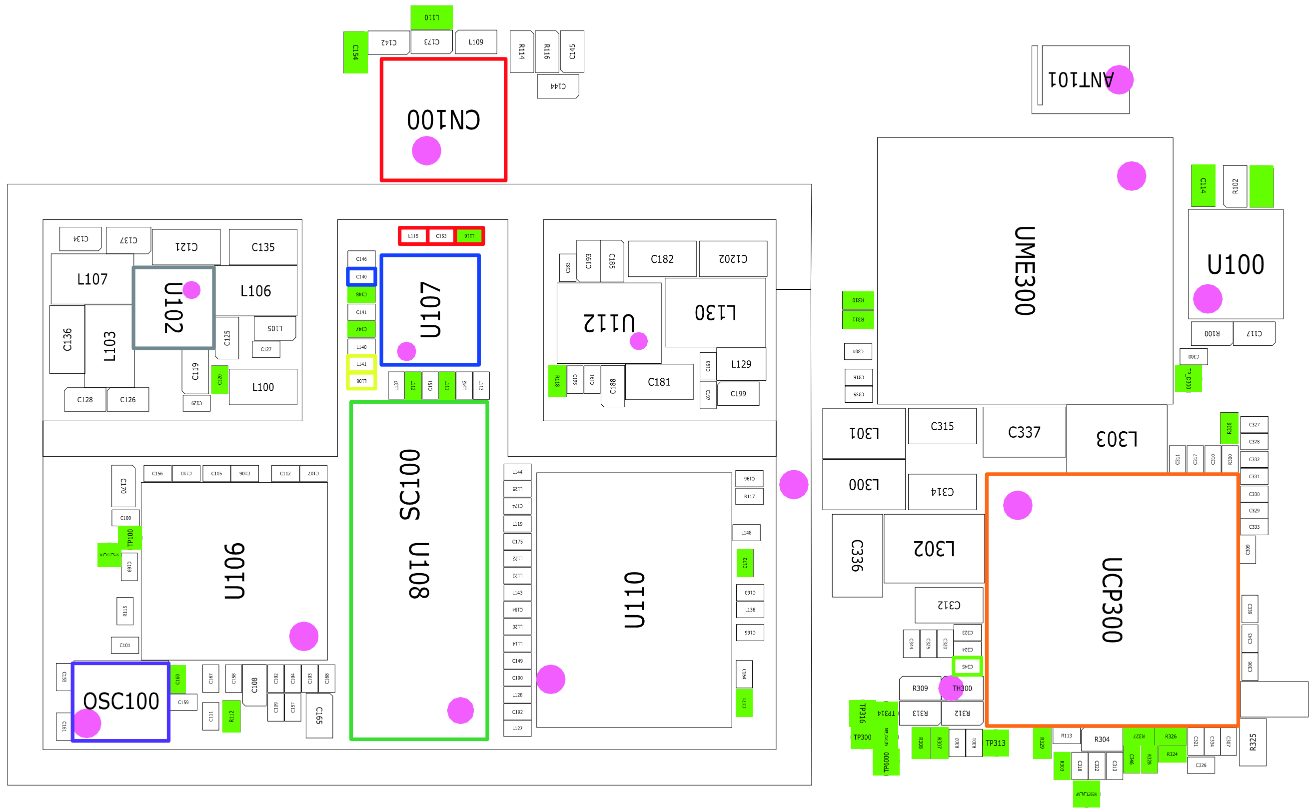
OSC100
VCC
VCONT
NC
NC
OUT
GND
AFC_DAC
C161



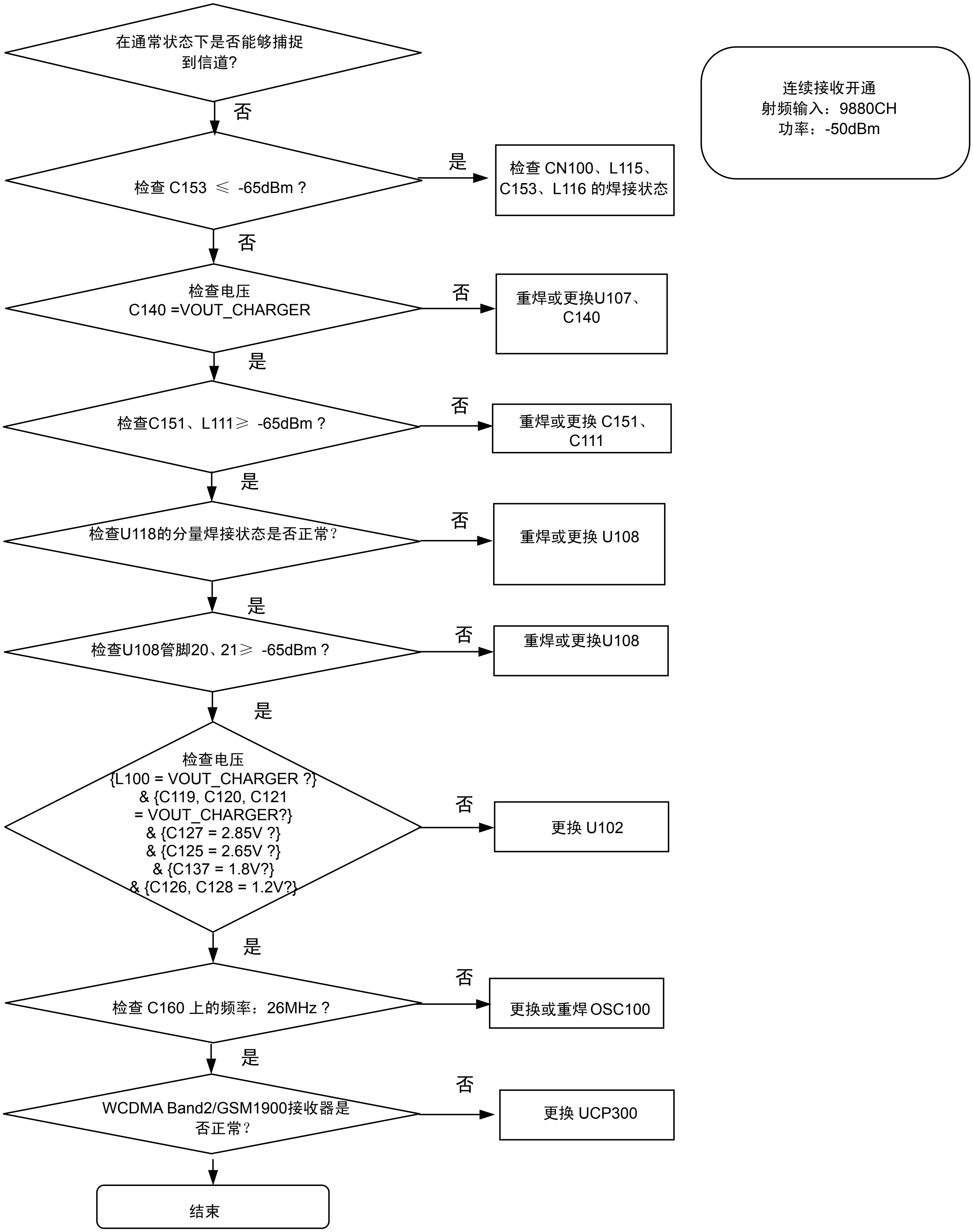
UCP300

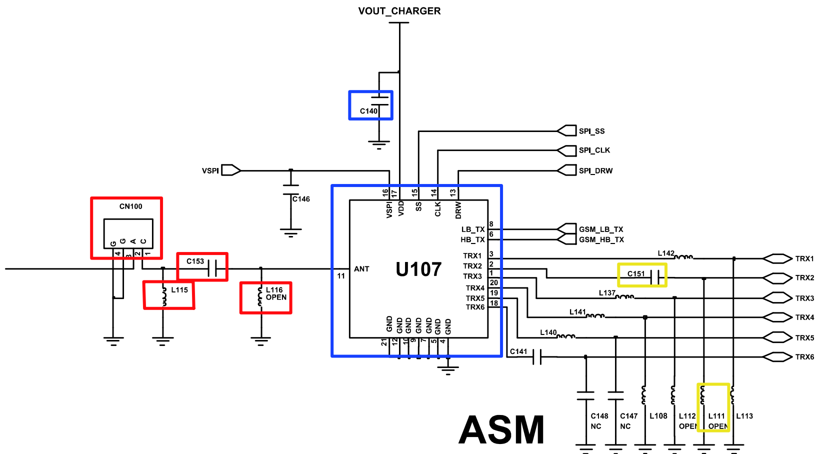


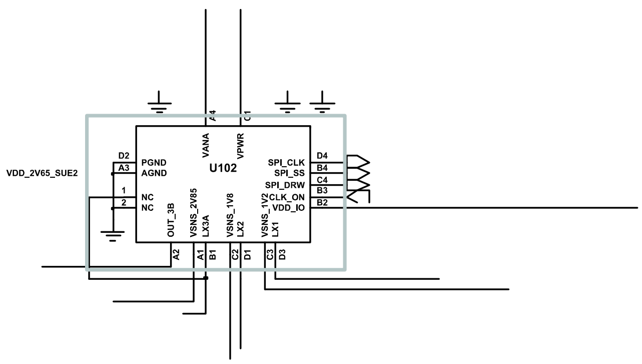
U108
TX_BAND1
TX_BAND2
TX_BAND4
TX_BAND5
TX_BAND8
ANT_BAND1
ANT_BAND2
ANT_BAND4
ANT_BAND5
ANT_BAND8
ANT_GSM1800
RX_BAND1
RX_BAND2
RX_BAND4
RX_BAND5
RX_BAND8
RX_GSM1800
GND
NC

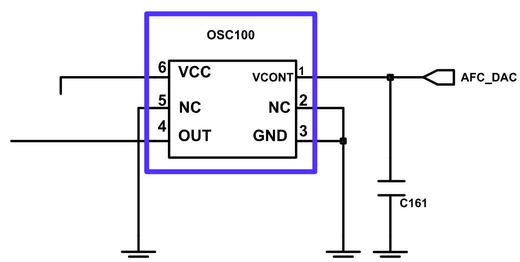

CN100
ANT101
UME300
U100
U102
L107
L106
L103
L100
U107
SC100
U108
U106
U112
L130
U110
OSC100
L301
C315
C337
L303
L300
C336
L302
UCP300

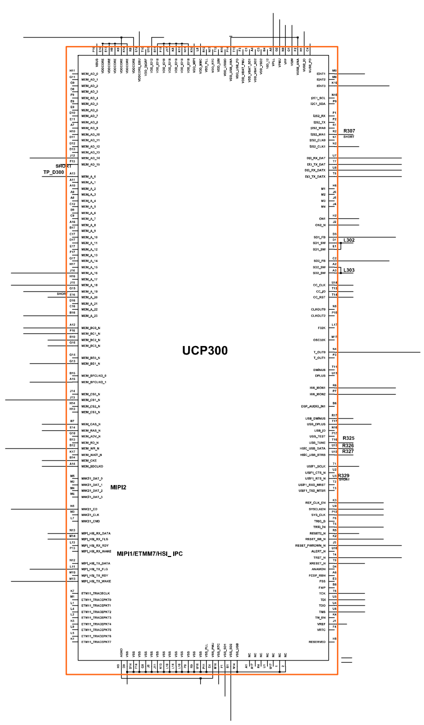
8-3-14.WCDMA Band2/GSM1900 接收器

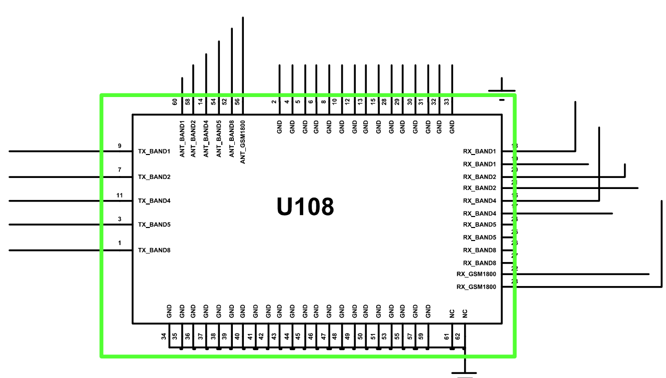


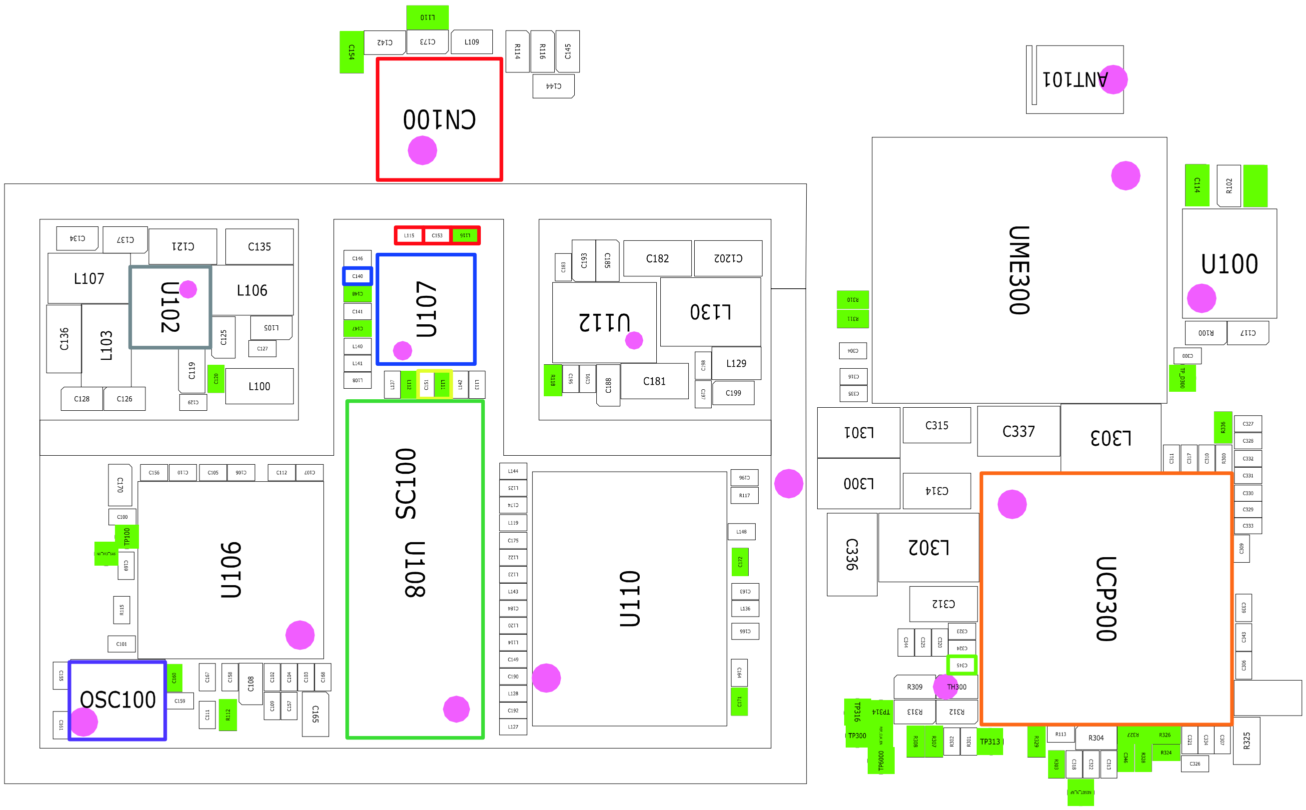

VOUT_CHARGER
C140
SPI_SS
SPI_CLK
SPI_DRW
VSPI
C146
CN100
C153
L115
L116
OPEN
U107
VSPI VDD SS CLK DRW
ANT
LB_TX
HB_TX
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
GND
GSM_LB_TX
GSM_HB_TX
L142
C151
L137
L141
L140
C141
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
C148
NC
C147
NC
L108
L112
L111
OPEN
L113
ASM


VDD_2V65_SUE2
U102
PGND
AGND
NC
NC
VANA
VPWR
SPI_CLK
SPI_SS
SPI_DRW
V2CLK_ON
VDD_IO
OUT_3B
VSNS_2V85
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1

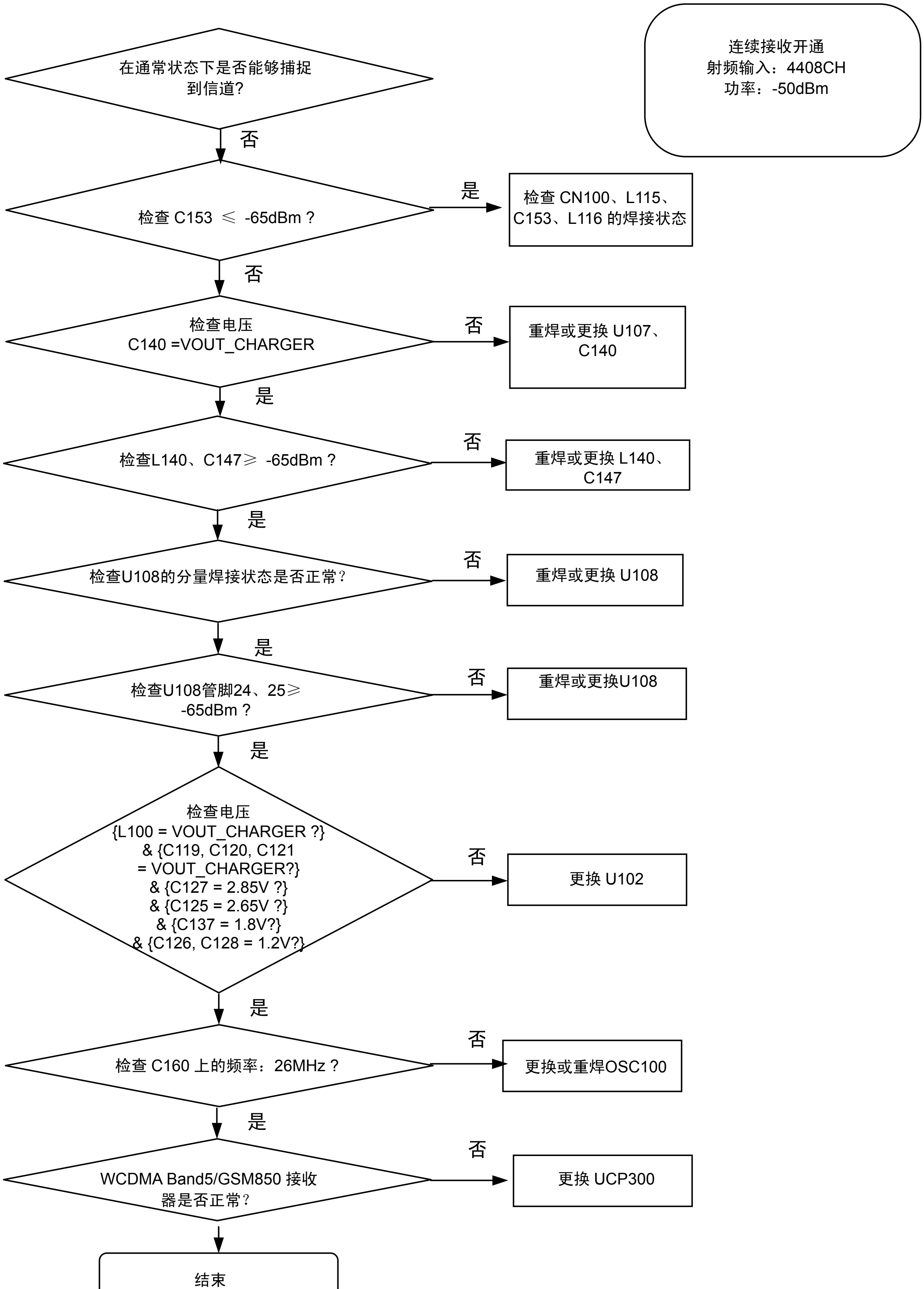

OSC100
VCC
NC
OUT
VCONT
NC
GND
AFC_DAC
C161

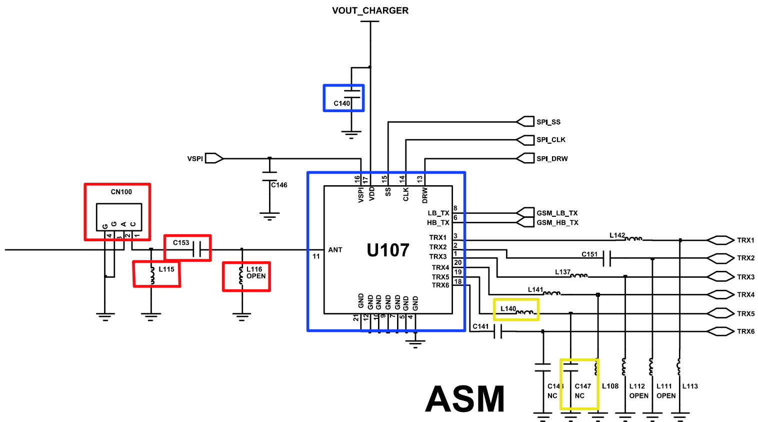

UCP300
MIPI2
MIPI1/ETMM7/HSI_ IPC

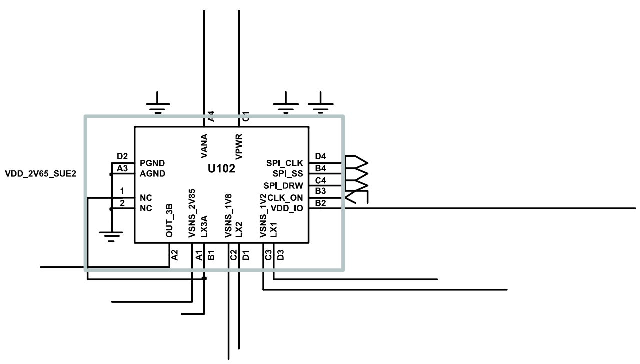

U108
TX_BAND1
TX_BAND2
TX_BAND4
TX_BAND5
TX_BAND8
RX_BAND1
RX_BAND2
RX_BAND4
RX_BAND5
RX_BAND8
RX_GSM1800
ANT_BAND1
ANT_BAND2
ANT_BAND4
ANT_BAND5
ANT_BAND8
ANT_GSM1800
GND
NC

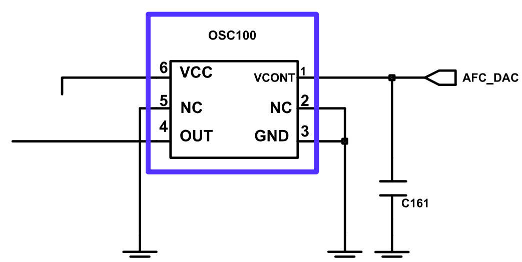

CN100
ANT101
UME300
U100
U102
L107
L106
L103
C136
L100
U107
SC100
U108
U112
L130
C182
C181
U106
U110
OSC100
L301
L300
C315
C337
L303
L302
C336
UCP300
R325

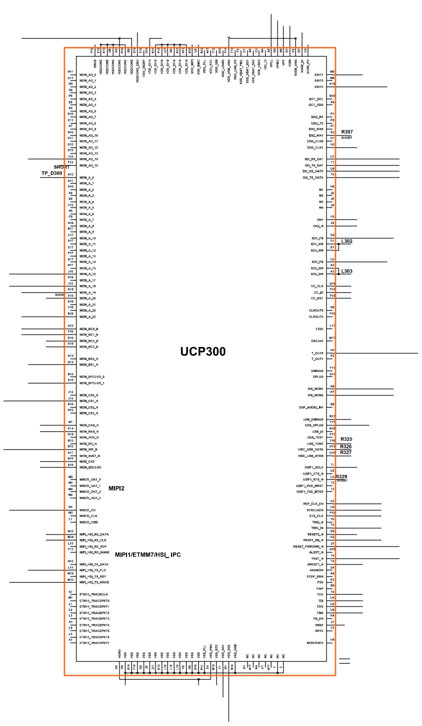
8-3-15.WCDMA Band5/GSM850 接收器

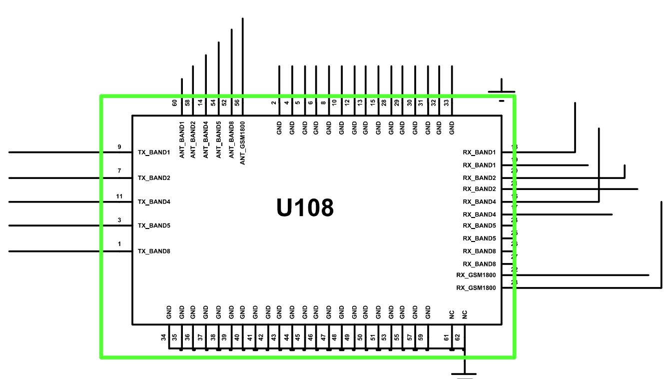


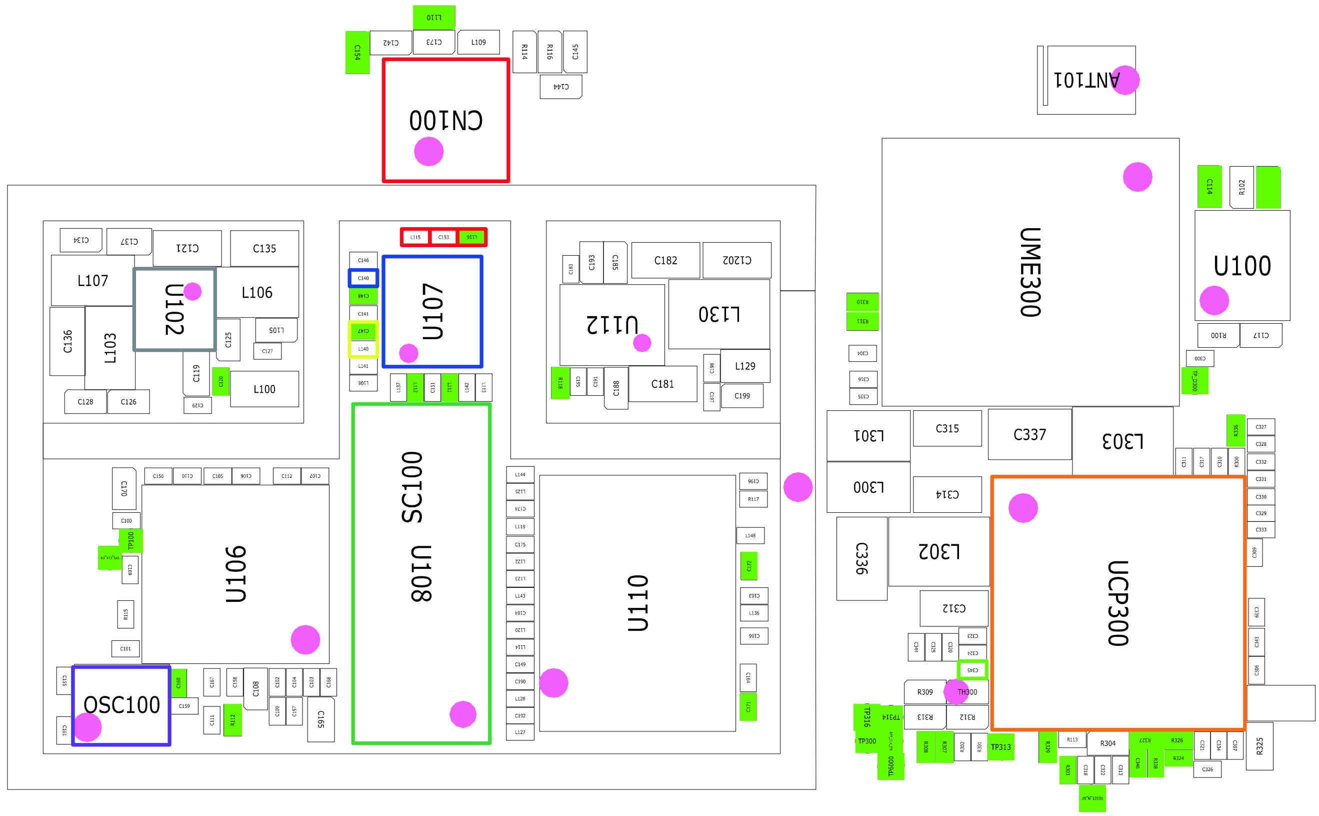

VOUT_CHARGER
C140
SPI_SS
SPI_CLK
SPI_DRW
VSPI
C146
CN100
C153
L115
L116
OPEN
U107
ANT
VSPI
VDD
SS
CLK
DRW
LB_TX
HB_TX
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
GSM_LB_TX
GSM_HB_TX
L142
C151
L137
L141
L140
C141
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
C144
NC
C147
NC
L108
L112
OPEN
L111
OPEN
L113
ASM


VDD_2V65_SUE2
U102
PGND
AGND
NC
NC
VANA
VPWR
SPI_CLK
SPI_SS
SPI_DRW
CLK_ON
VDD_IO
OUT_3B
VSNS_2V85
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1

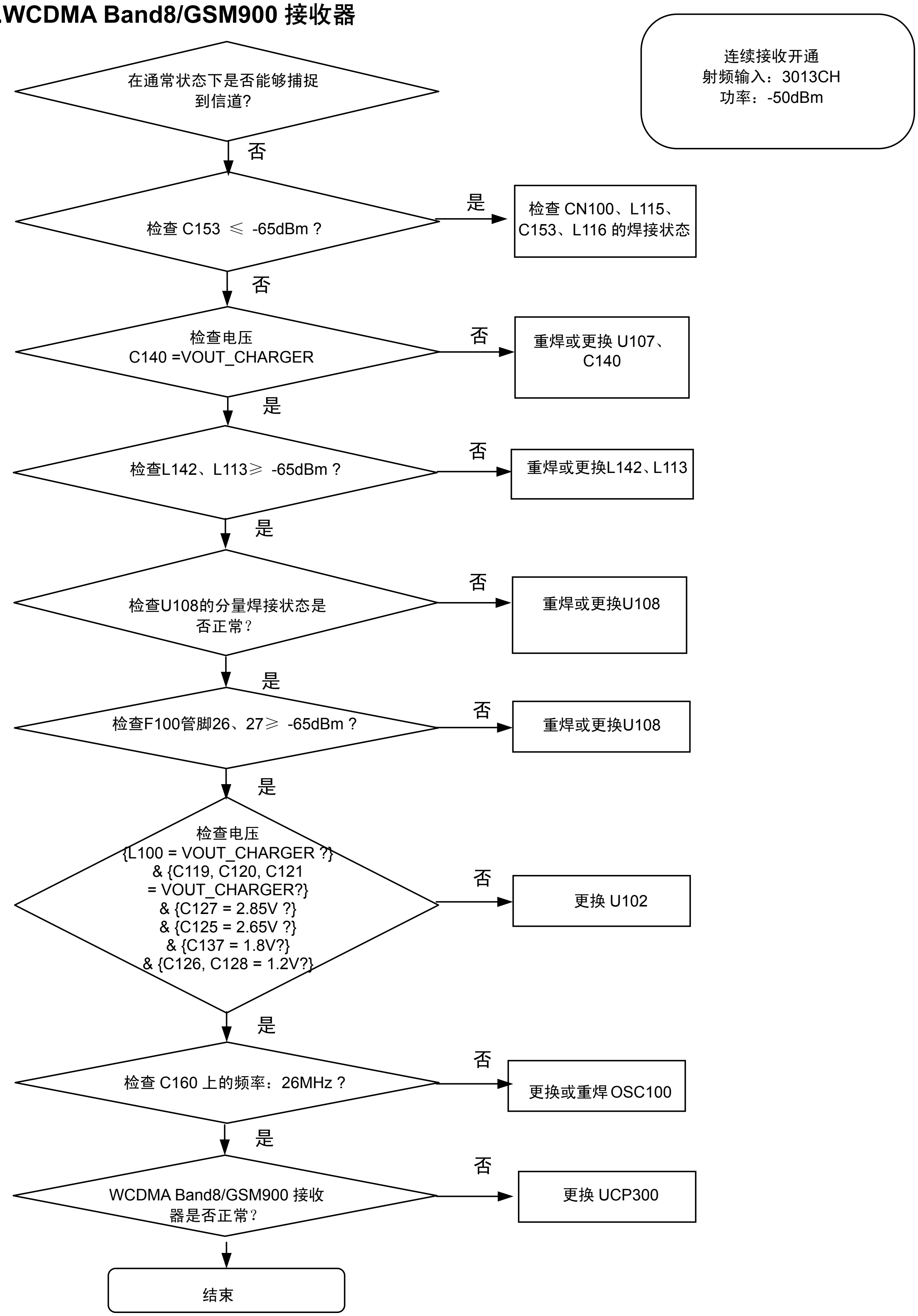

OSC100
VCC
VCONT
NC
NC
OUT
GND
AFC_DAC
C161

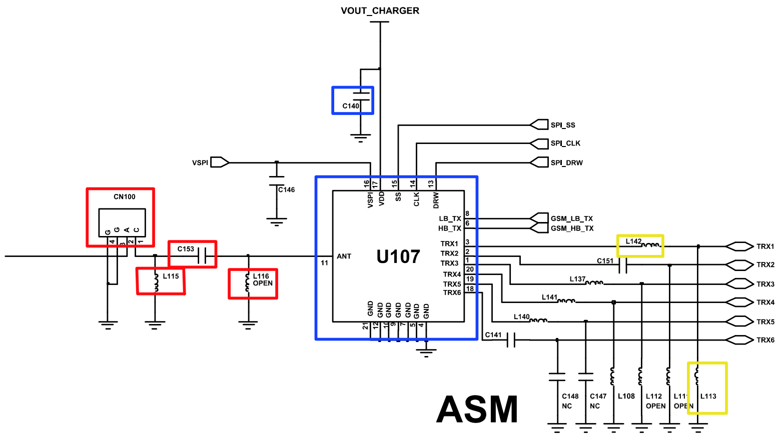

UCP300

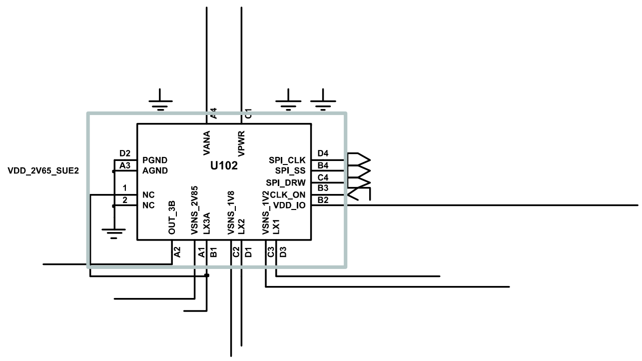

U108
TX_BAND1
TX_BAND2
TX_BAND4
TX_BAND5
TX_BAND8
ANT_BAND1
ANT_BAND2
ANT_BAND4
ANT_BAND5
ANT_BAND8
ANT_GSM1800
RX_BAND1
RX_BAND1
RX_BAND2
RX_BAND2
RX_BAND4
RX_BAND4
RX_BAND5
RX_BAND5
RX_BAND8
RX_BAND8
RX_GSM1800
RX_GSM1800

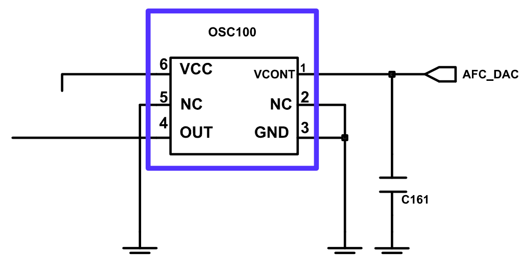

CN100
ANT101
UME300
U100
U102
L107
L106
C121
C135
C136
L103
L100
U107
U112
L130
C182
C1202
L129
C181
L301
C315
C337
L303
L300
C314
C336
L302
C312
UCP300
SC100
U108
U106
U110
OSC100
C165
C108
R309
R313
R312
R304
R325

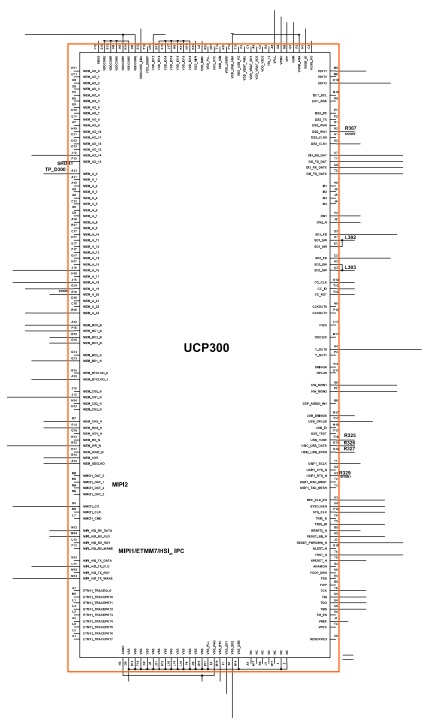
8-3-16.WCDMA Band8/GSM900 接收器

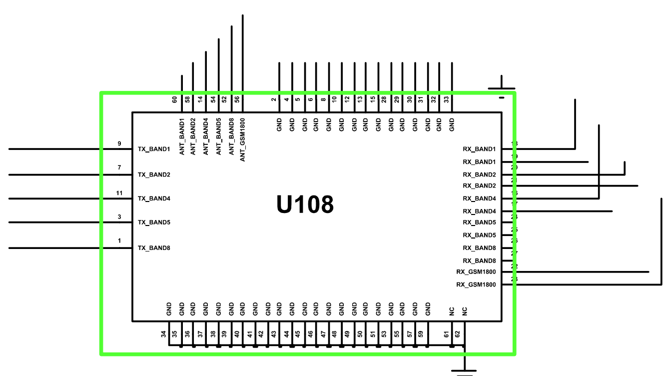


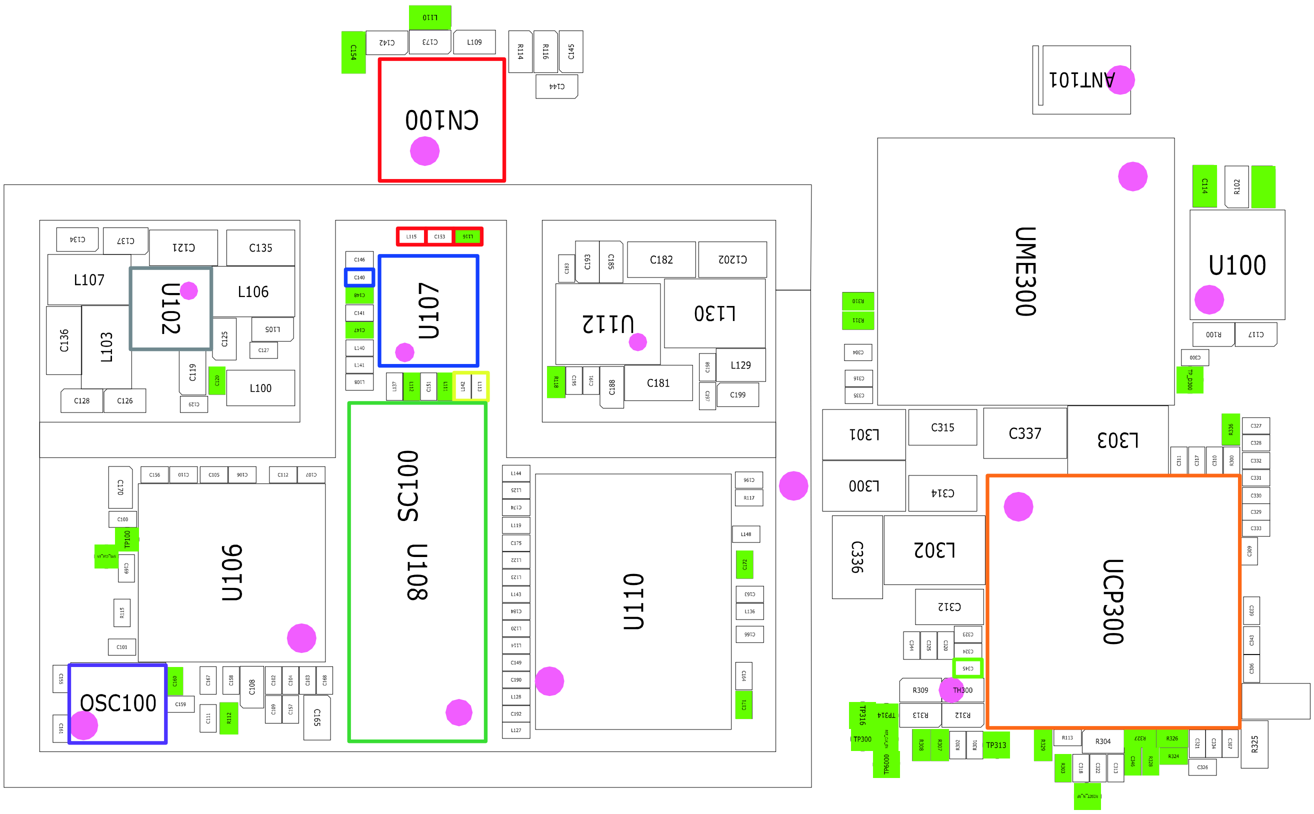

VOUT_CHARGER
C140
SPI_SS
SPI_CLK
SPI_DRW
VSPI
C146
CN100
C153
L115
L116
OPEN
ANT
U107
VSPI
VDD
SS
CLK
DRW
LB_TX
HB_TX
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
GSM_LB_TX
GSM_HB_TX
L142
C151
L137
L141
L140
C141
C148
NC
C147
NC
L108
L112
OPEN
OPEN
L113
ASM


VDD_2V65_SUE2
U102
PGND
AGND
NC
NC
VANA
VPWR
SPI_CLK
SPI_SS
SPI_DRW
CLK_ON
VDD_IO
OUT_3B
VSNS_2V85
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1

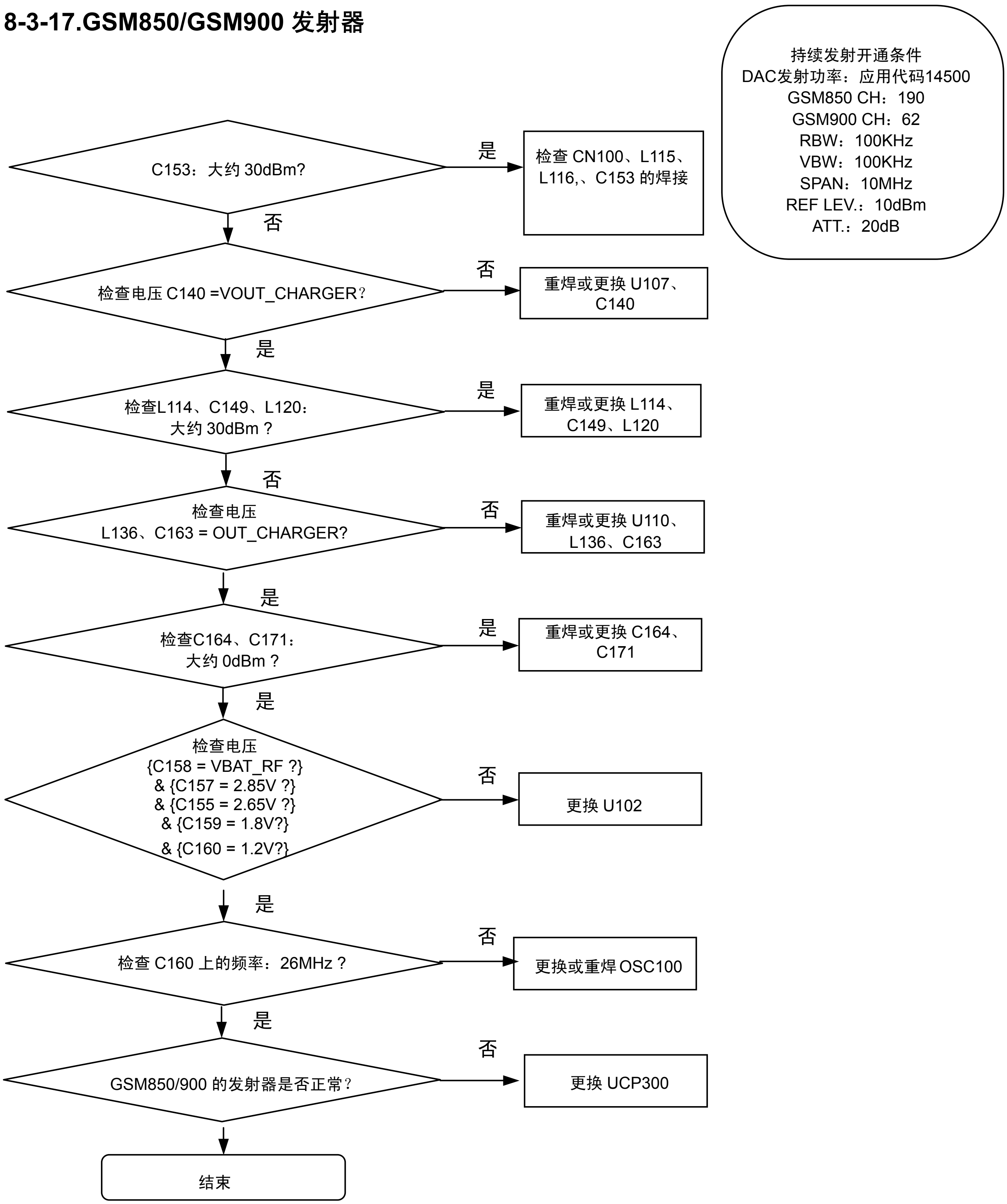

OSC100
VCC
NC
OUT
VCONT
NC
GND
AFC_DAC
C161

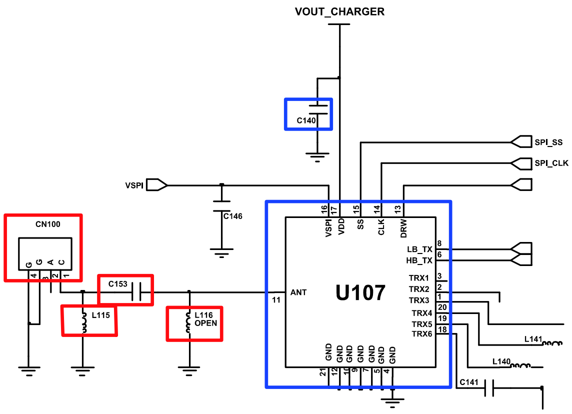

UCP300
MIPI2
MIPI1/ETMM7/HSI_IPC

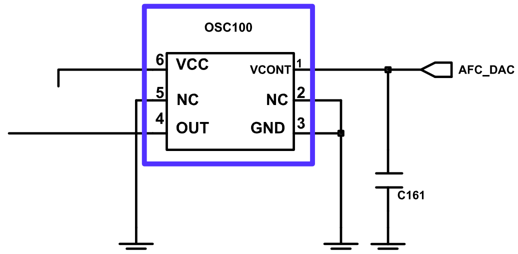

U108
TX_BAND1
TX_BAND2
TX_BAND4
TX_BAND5
TX_BAND8
ANT_BAND1
ANT_BAND2
ANT_BAND4
ANT_BAND5
ANT_BAND8
ANT_GSM1800
RX_BAND1
RX_BAND2
RX_BAND4
RX_BAND5
RX_BAND8
RX_GSM1800
GND
NC

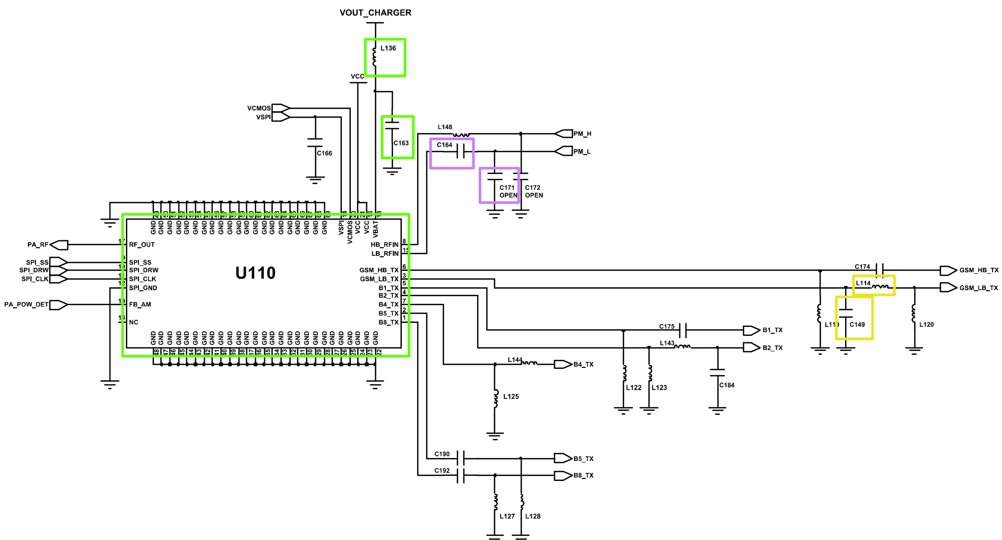

CN100
ANT101
UME300
U100
U102
L107
L106
C135
C121
L103
C136
L100
U107
U112
L130
C182
C1202
C181
L129
SC100
U108
U106
U110
OSC100
UCP300
L301
L300
C315
C337
L303
C314
C336
L302
C312
R309
TH300
TP300
TP313
R304
R325

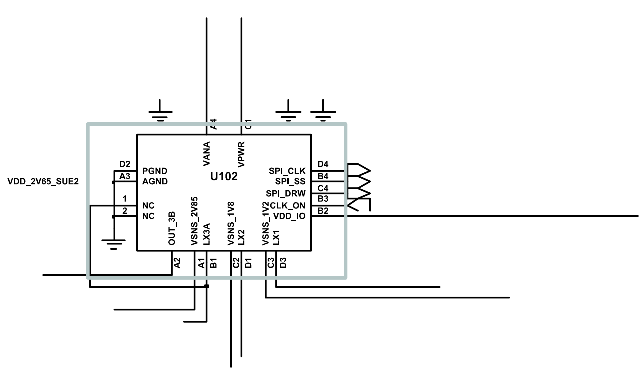
8-3-17.GSM850/GSM900 发射器

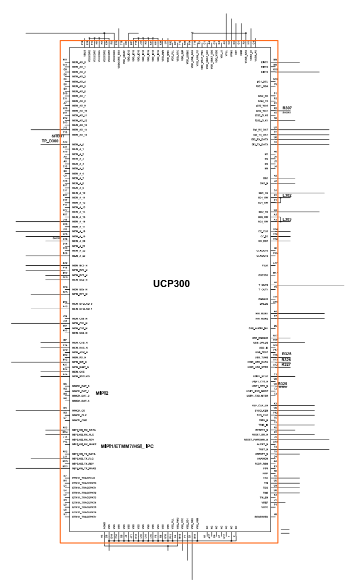


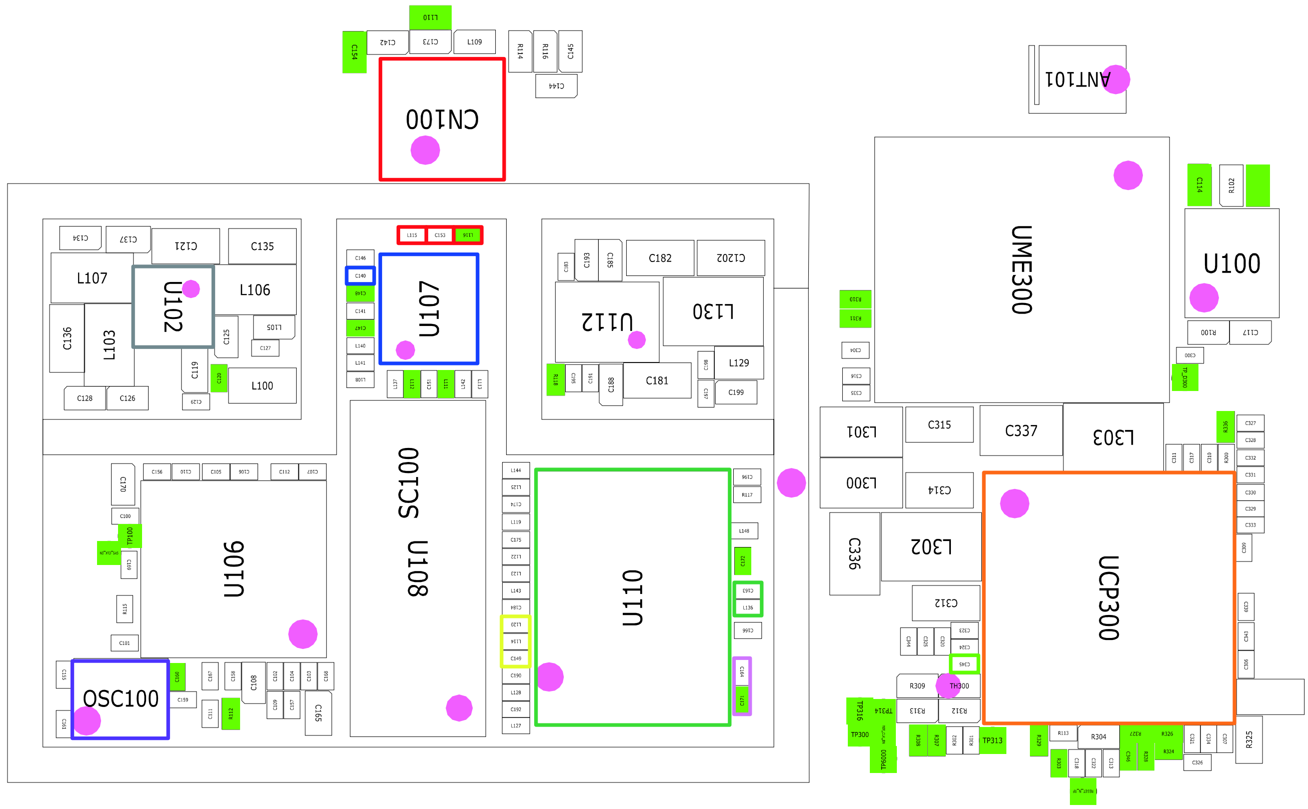

VOUT_CHARGER
C140
SPI_SS
SPI_CLK
VSPI
C146
CN100
G G A C
4 3 2 1
C153
L115
L116
OPEN
VSPI
VDD
SS
CLK
DRW
16 17 15 14 13
ANT
11
U107
LB_TX
HB_TX
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
8
6
3
2
1
20
19
18
GND
L141
L140
C141


OSC100
6 VCC
5 NC
4 OUT
VCONT 1
NC 2
GND 3
AFC_DAC
C161

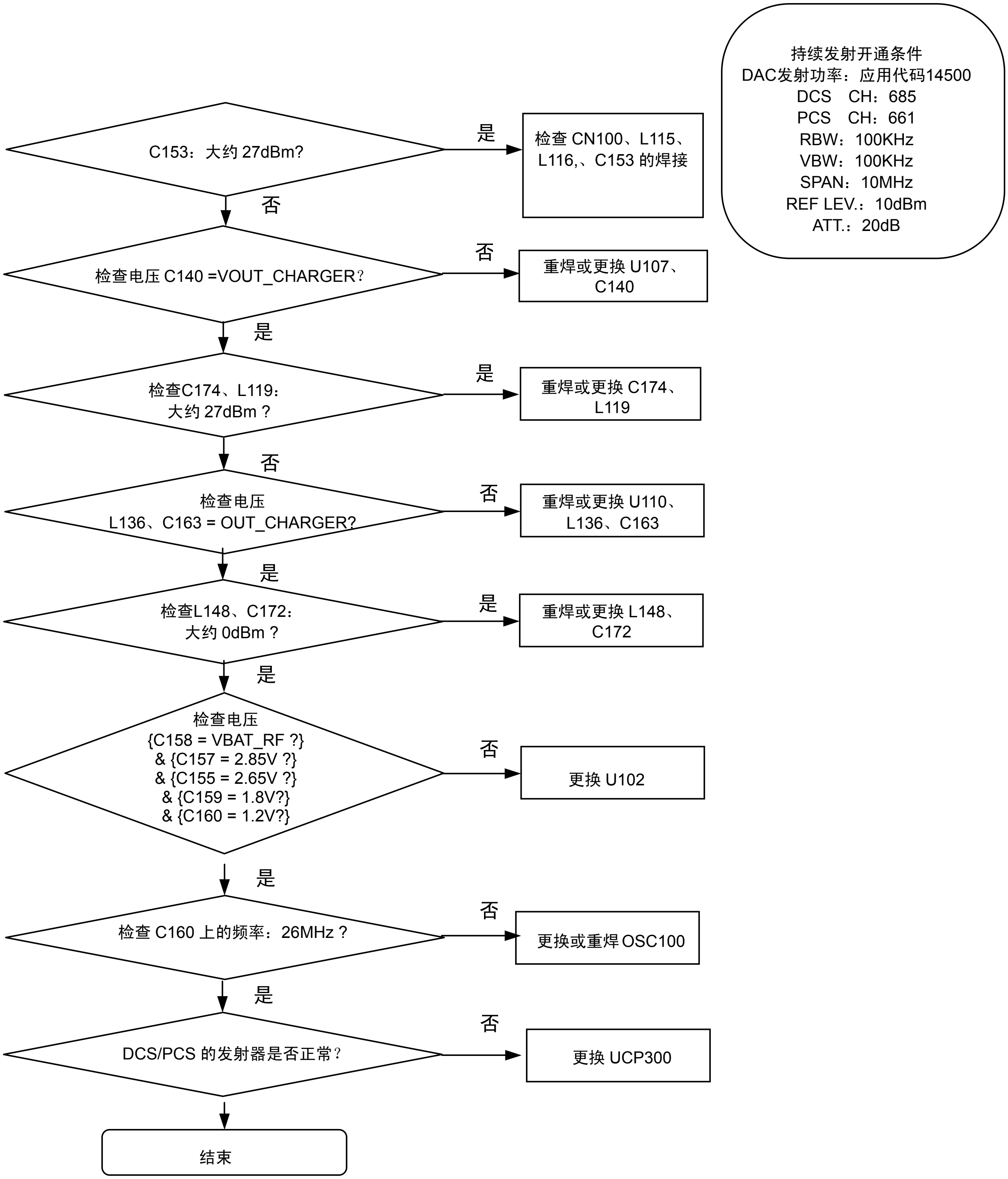

VOUT_CHARGER
L136
VCC
VCMOS
VSPI
C166
C163
L148
C164
PM_H
PM_L
C171
OPEN
C172
OPEN
PA_RF
SPI_SS
SPI_DRW
SPI_CLK
PA_POW_DET
RF_OUT
SPI_SS
SPI_DRW
SPI_CLK
SPI_GND
FB_AM
NC
U110
HB_RFIN
LB_RFIN
GSM_HB_TX
GSM_LB_TX
B1_TX
B2_TX
B4_TX
B5_TX
B8_TX
VSPI
VCMOS
VCC
VBAT
GND
C174
GSM_HB_TX
L114
GSM_LB_TX
C149
L120
C175
B1_TX
L143
B2_TX
L144
B4_TX
L125
L122
L123
C184
C190
B5_TX
C192
B8_TX
L127
L128

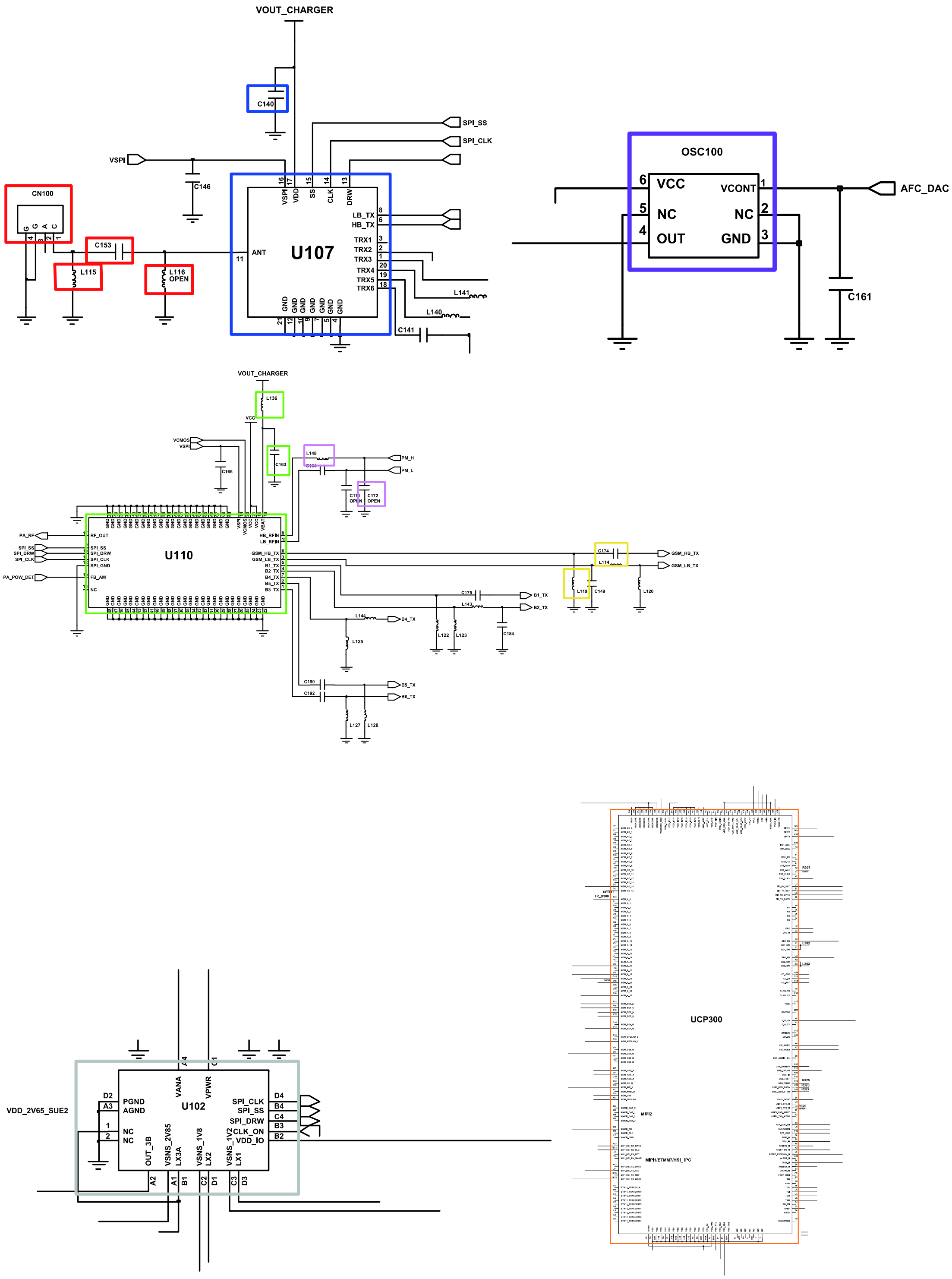

VDD_2V65_SUE2
D2
A3
PGND
AGND
1
2
NC
NC
U102
VANA
VPWR
A4
C1
SPI_CLK
SPI_SS
SPI_DRW
CLK_ON
VDD_IO
D4
B4
C4
B3
B2
OUT_3B
VSNS_2V85
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1
A2
A1
B1
C2
D1
C3
D3


UCP300

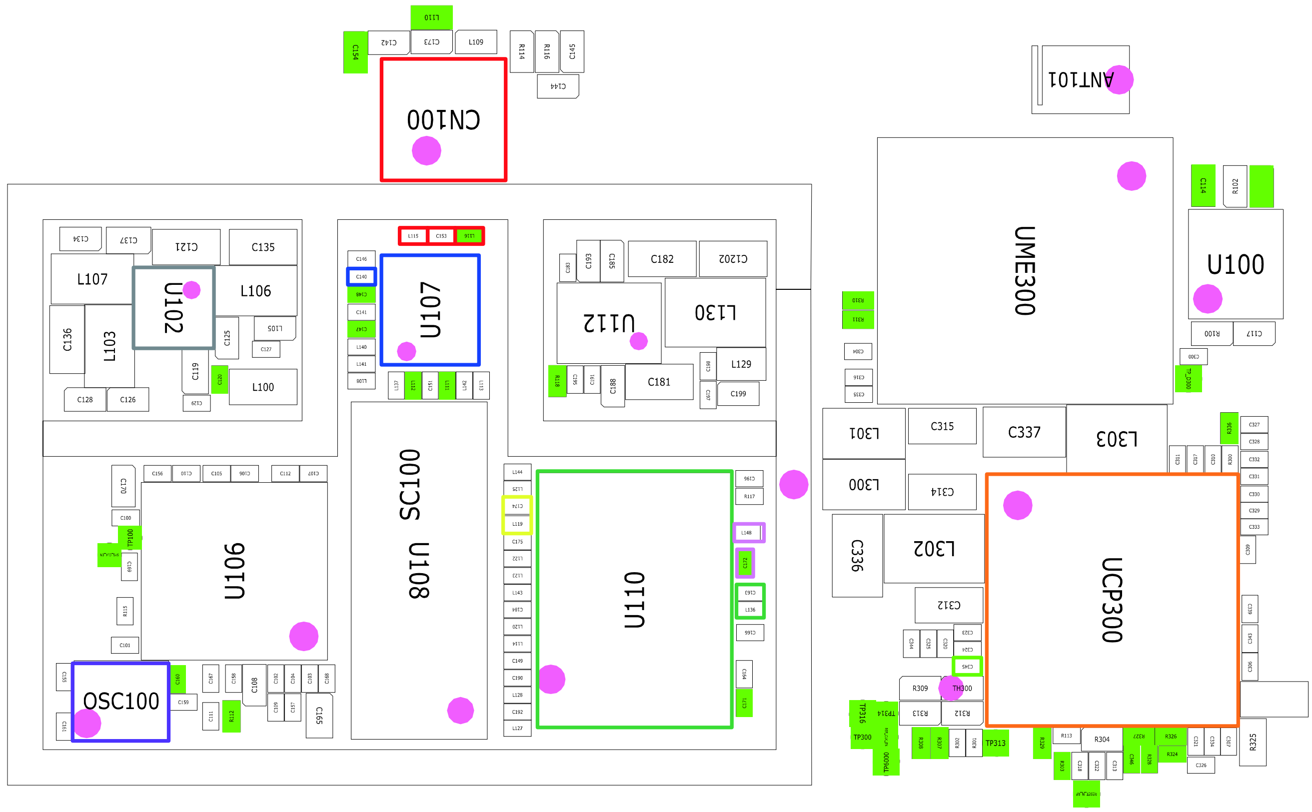

CN100
ANT101
UME300
U100
U102
U107
U112
L130
L107
L106
L103
C136
L100
C121
C135
C182
C1202
C181
L129
SC100
U108
U106
U110
OSC100
UCP300
L301
L300
L303
L302
C315
C337
C314
C336
C312
R325
R304
R309
TH300
TP313
TP300

8-3-18.DCS/PCS 发射器

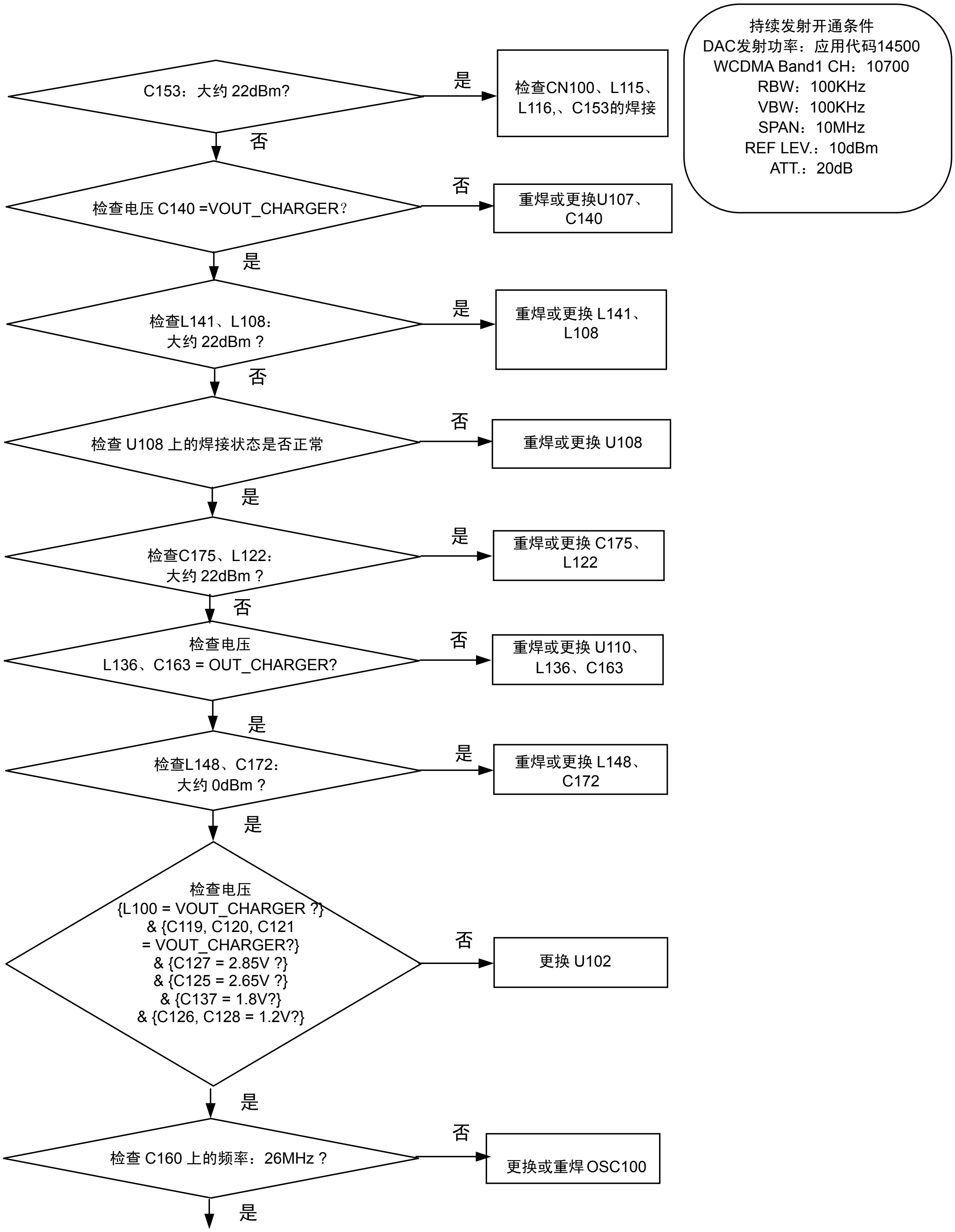


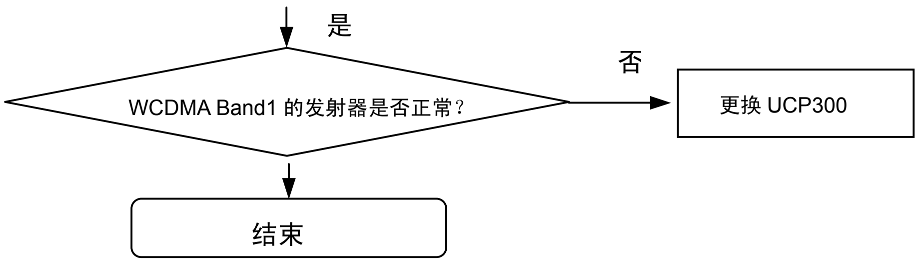


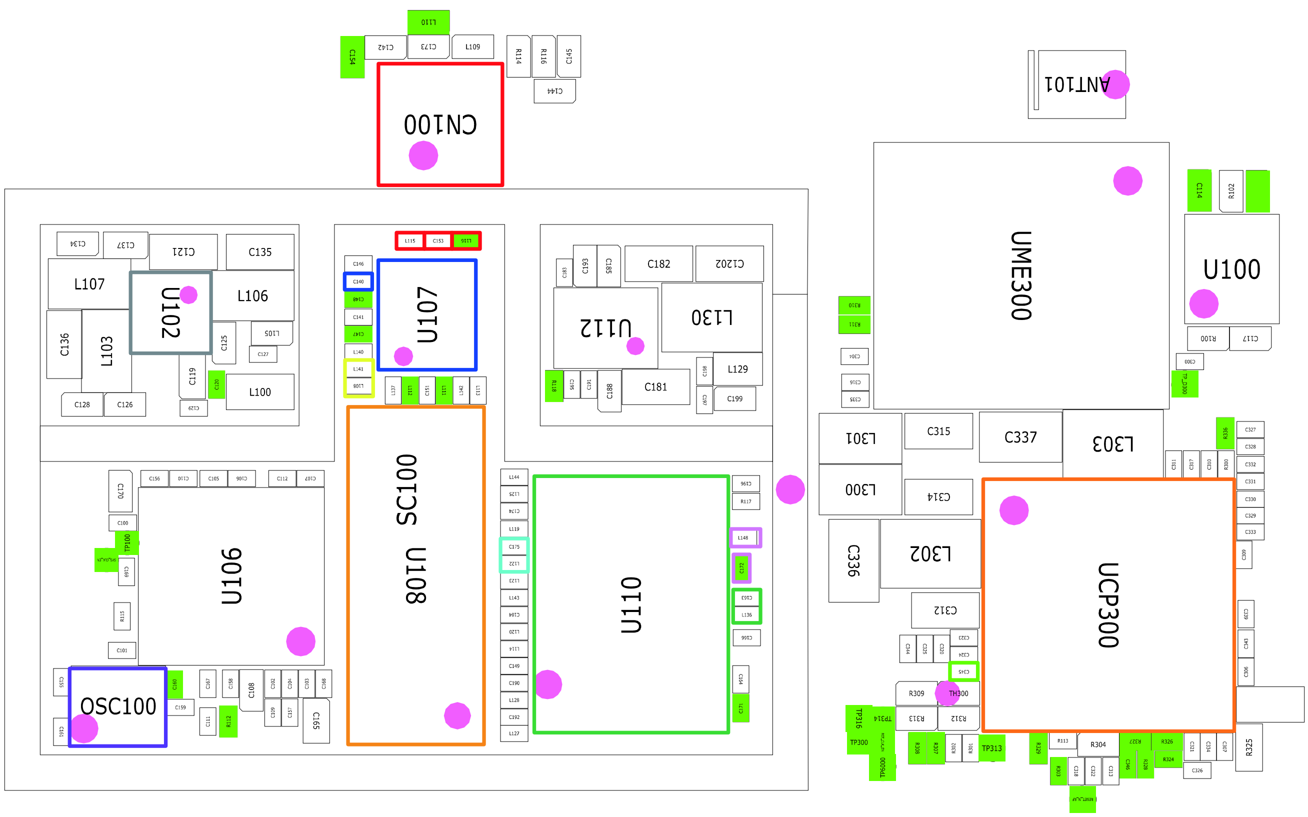
VOUT_CHARGER
C140
SPI_SS
SPI_CLK
VSPI
C146
CN100
C153
L115
L116 OPEN
ANT
U107
VSPI
VDD
SS
CLK
DRW
LB_TX
HB_TX
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
GND
L141
L140
C141
OSC100
VCC
NC
OUT
VCONT
NC
GND
AFC_DAC
C161
VOUT_CHARGER
L136
VCC
VCMOS
VSPI
C166
C163
L148
PM_H
PM_L
C171 OPEN
C172 OPEN
U110
PA_RF
RF_OUT
SPI_SS
SPI_DRW
SPI_CLK
SPI_GND
PA_POW_DET
FB_AM
NC
HB_RFIN
LB_RFIN
GSM_HB_TX
GSM_LB_TX
B1_TX
B2_TX
B4_TX
B5_TX
B8_TX
C174
L114
GSM_HB_TX
GSM_LB_TX
L119
C149
L120
C175
B1_TX
L143
B2_TX
L144
B4_TX
L125
L122
L123
C184
C190
B5_TX
C192
B8_TX
L127
L128
UCP300
VDD_2V65_SUE2
U102
PGND
AGND
NC
NC
VANA
VPWR
SPI_CLK
SPI_SS
SPI_DRW
CLK_ON
VDD_IO
OUT_3B
VSNS_2V85
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1

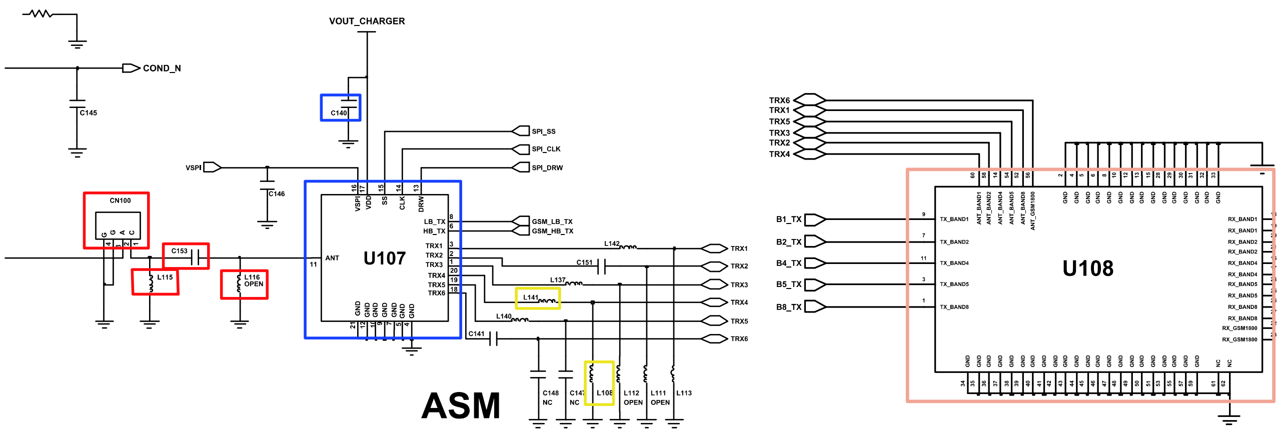

CN100
ANT101
UME300
U100
U102
L107
L106
C135
C136
L103
L100
U107
U112
L130
C182
C181
L129
SC100
U108
U106
U110
OSC100
L301
C315
C337
L300
C314
L303
C336
L302
C312
UCP300
R325

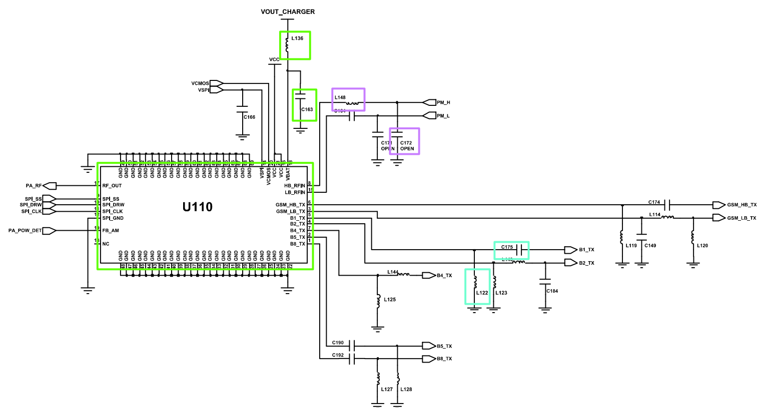
8-3-19.WCDMA BAND1 发射器

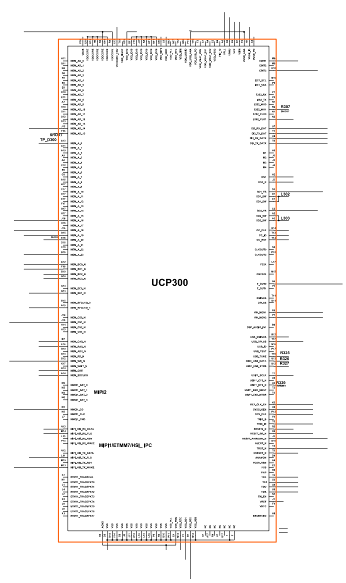


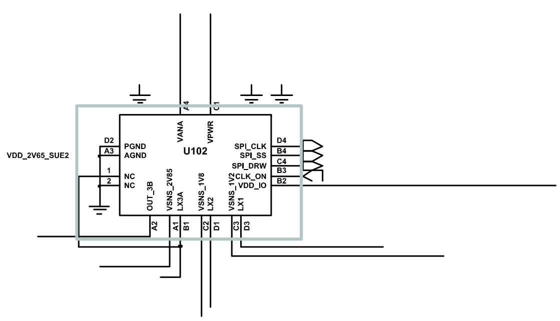

是
WCDMA Band1 的发射器是否正常？
否
更换 UCP300
结束

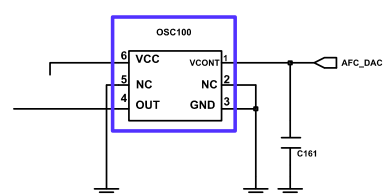

CN100
ANT101
UME300
U100
U102
U107
U112
L130
L107
L106
L103
C136
L100
C135
C182
C181
L129
C199
L301
L300
L303
C315
C337
C336
L302
C312
C314
U106
SC100
U108
U110
UCP300
OSC100
C165
R309
TH300
R313
R312
TP300
R304
R325


VOUT_CHARGER
COND_N
C145
C140
SPI_SS
SPI_CLK
SPI_DRW
VSPI
C146
CN100
C153
L115
L116
OPEN
U107
ANT
LB_TX
HB_TX
GSM_LB_TX
GSM_HB_TX
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
L142
C151
L137
L141
L140
C141
C148
NC
C147
NC
L108
L112
OPEN
L111
OPEN
L113
ASM
TRX6
TRX1
TRX5
TRX3
TRX2
TRX4
B1_TX
B2_TX
B4_TX
B5_TX
B8_TX
U108

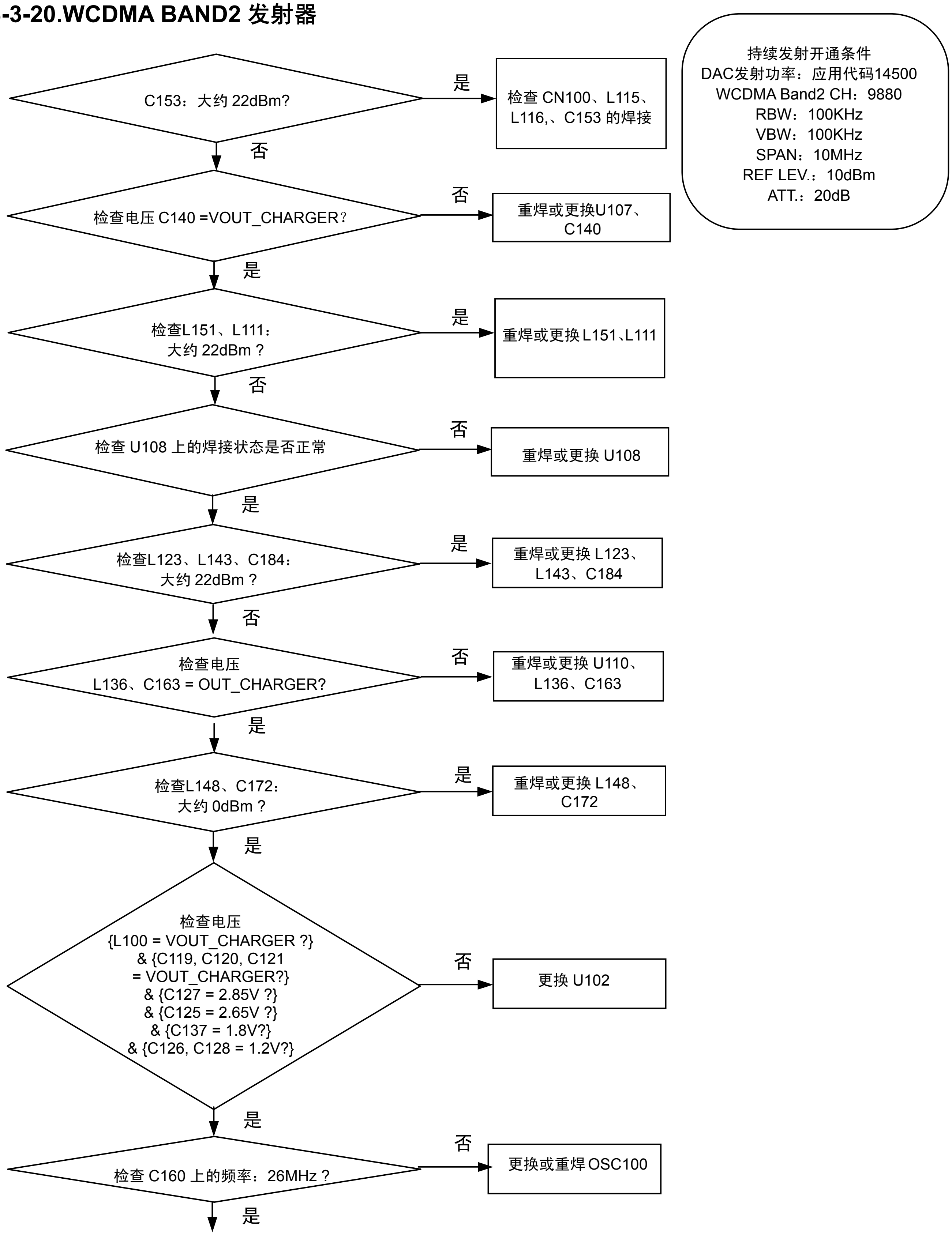

VOUT_CHARGER
L136
U110
PA_RF
PA_POW_DET
GSM_HB_TX
GSM_LB_TX
B1_TX
B2_TX
B4_TX
B5_TX
B8_TX

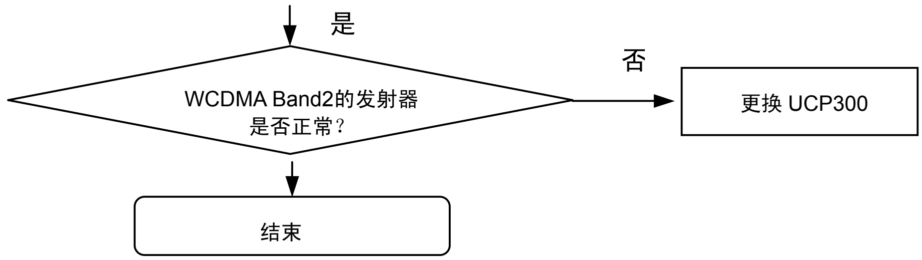

UCP300

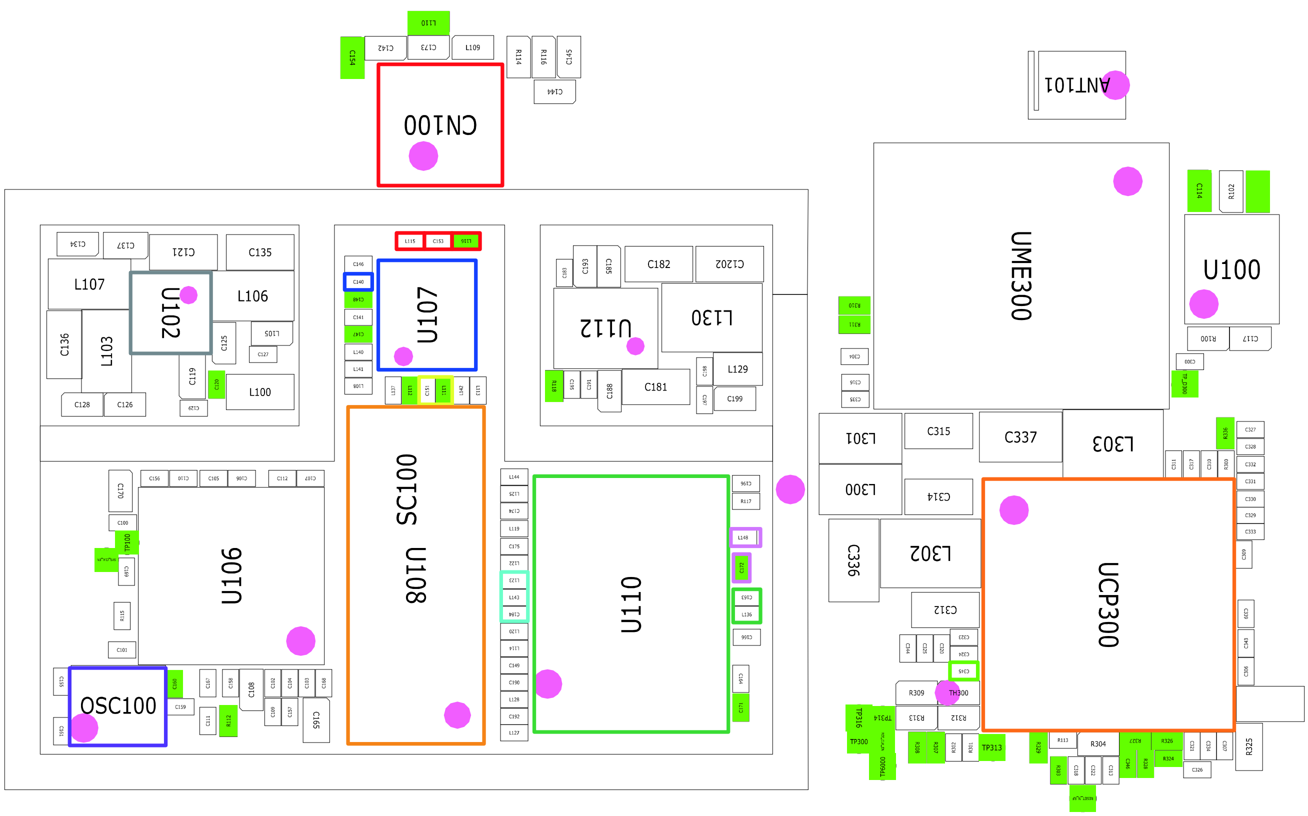

VDD_2V65_SUE2
U102
PGND
AGND
NC
NC
VANA
VPWR
SPI_CLK
SPI_SS
SPI_DRW
CLK_ON
VDD_IO
OUT_3B
VSNS_2V85
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1

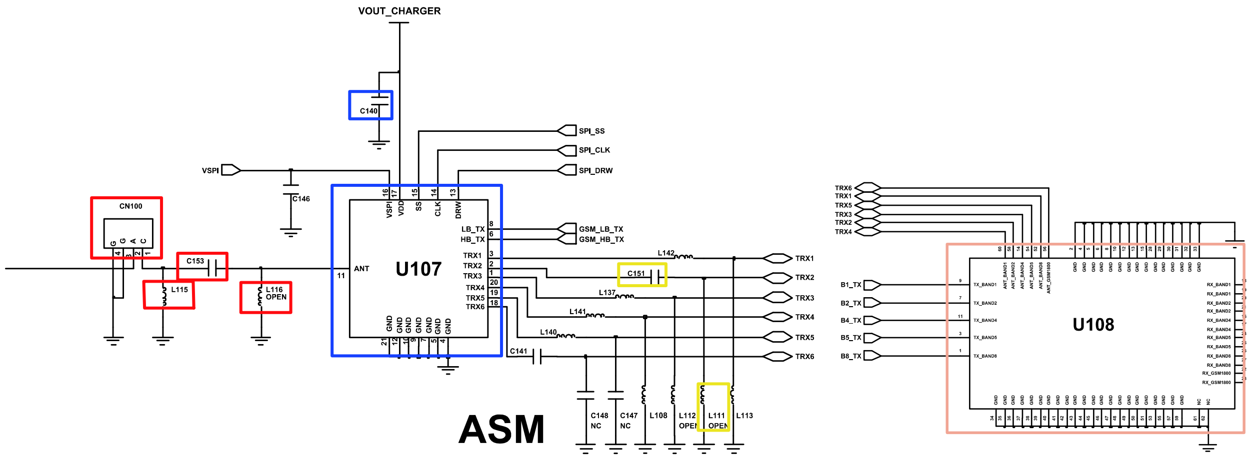

OSC100
VCC
NC
OUT
VCONT
NC
GND
AFC_DAC
C161

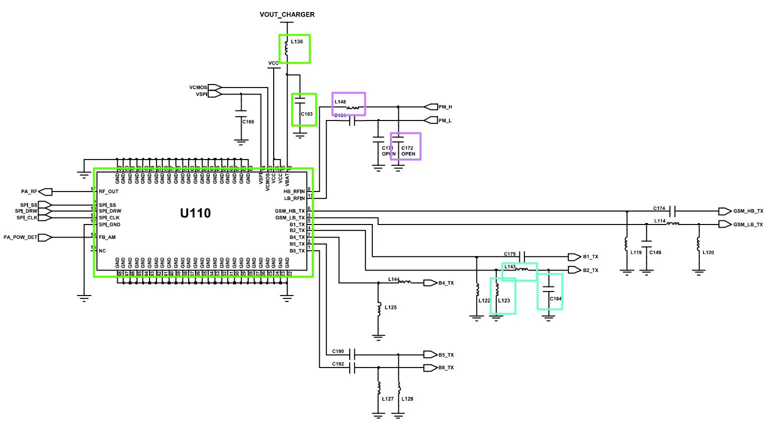
8-3-20.WCDMA BAND2 发射器

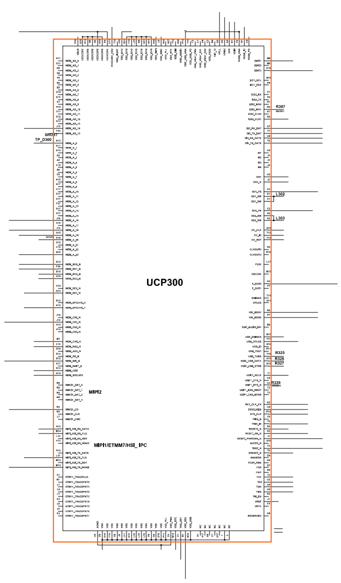


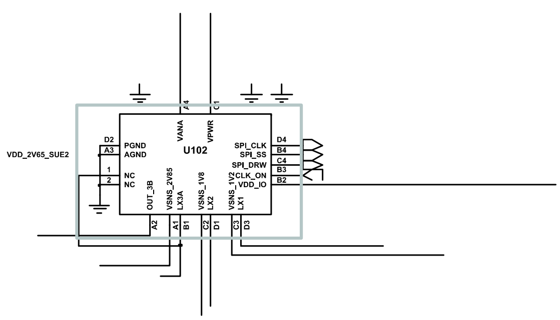

是
WCDMA Band2的发射器
是否正常？
否
更换 UCP300
结束

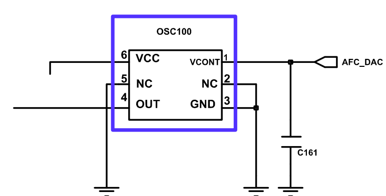

CN100
ANT101
UME300
U100
U102
L107
L106
C135
L103
C136
L100
U107
U112
L130
C182
C181
L129
SC100
U108
U106
U110
OSC100
L301
L300
C315
C337
L303
C314
C336
L302
UCP300
C165
C108
R325


VOUT_CHARGER
C140
SPI_SS
SPI_CLK
SPI_DRW
VSPI
C146
CN100
C153
L115
L116
OPEN
U107
ANT
LB_TX
HB_TX
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
GSM_LB_TX
GSM_HB_TX
L142
C151
L137
L141
L140
C141
C148
NC
C147
NC
L108
L112
L111
OPEN
L113
ASM
B1_TX
B2_TX
B4_TX
B5_TX
B8_TX
U108

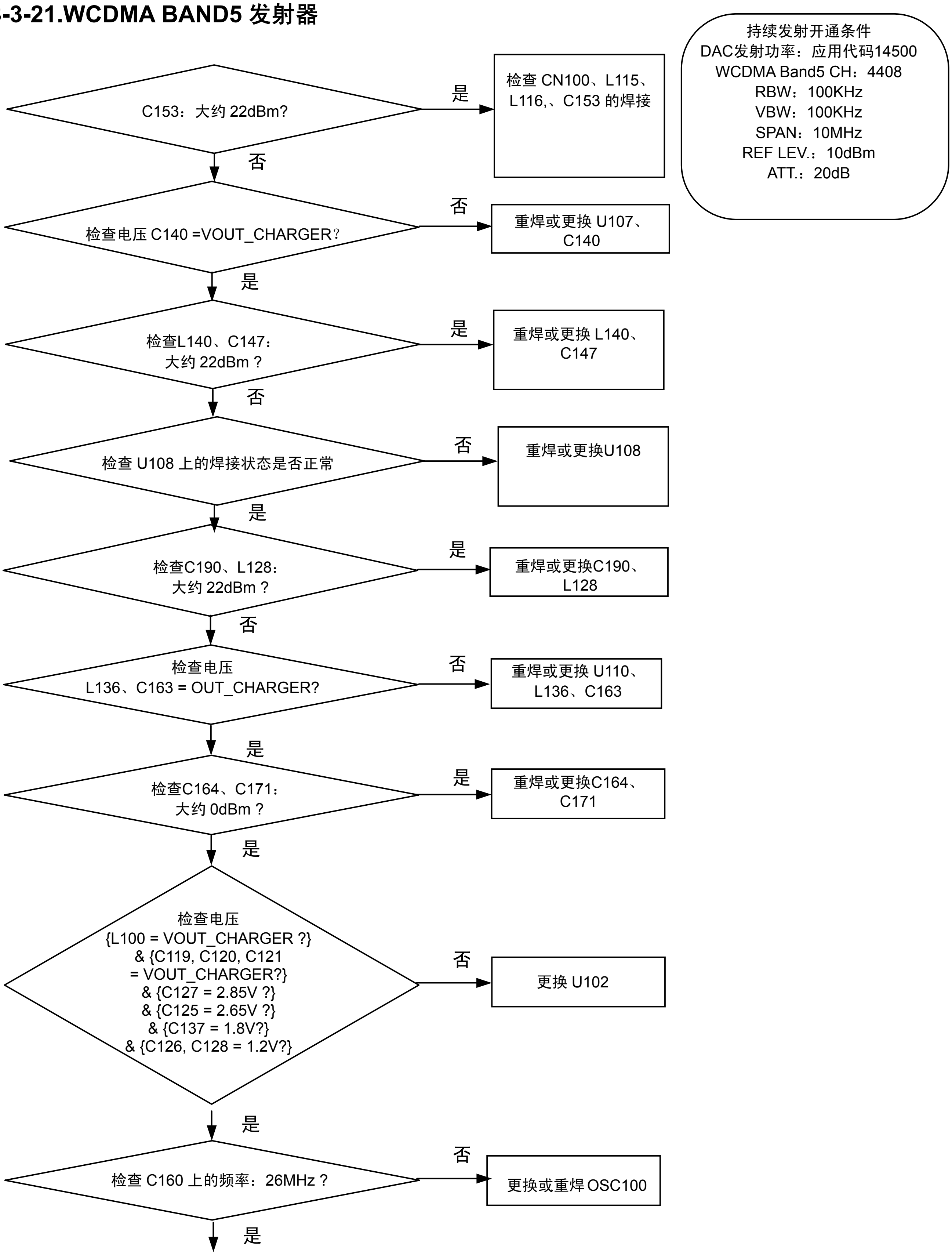

VOUT_CHARGER
L136
VCC
VCMOS
VSPI
C166
C163
L148
PM_H
PM_L
C171
OPEN
C172
OPEN
PA_RF
SPE_SS
SPE_DRW
SPE_CLK
PA_POW_DET
U110
GSM_HB_TX
GSM_LB_TX
C174
L114
L119
C149
L120
C175
L143
B1_TX
B2_TX
L144
B4_TX
L125
L122
L123
C184
C190
C192
B5_TX
B8_TX
L127
L128

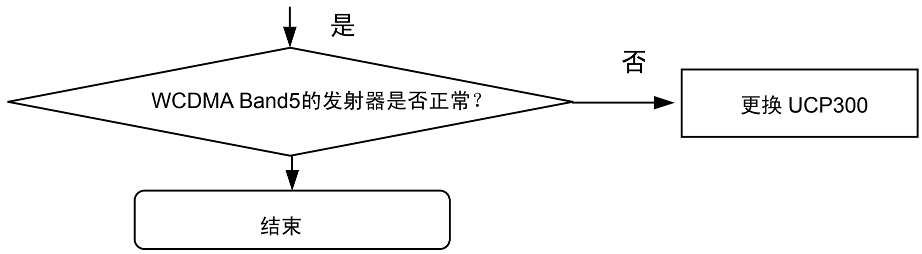

UCP300

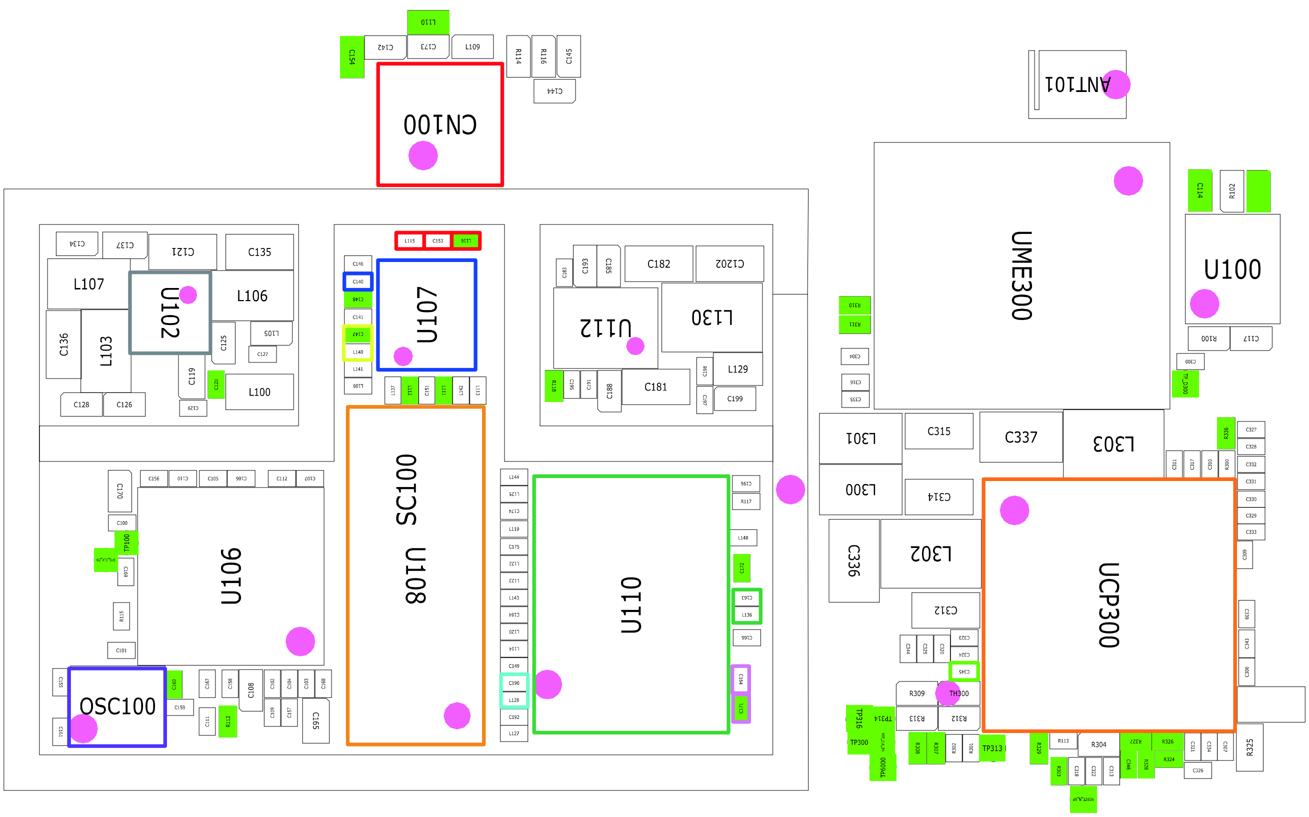

VDD_2V65_SUE2
U102
PGND
AGND
NC
NC
VANA
VPWR
SPI_CLK
SPI_SS
SPI_DRW
CLK_ON
VDD_IO
OUT_3B
VSNS_2V85
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1

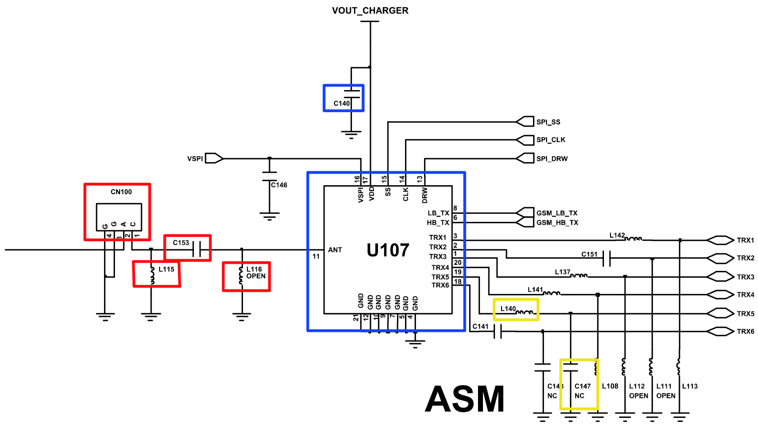

OSC100
VCC
VCONT
NC
NC
OUT
GND
AFC_DAC
C161

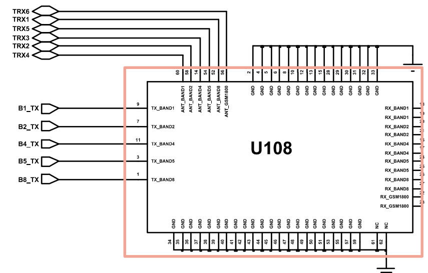
8-3-21.WCDMA BAND5 发射器

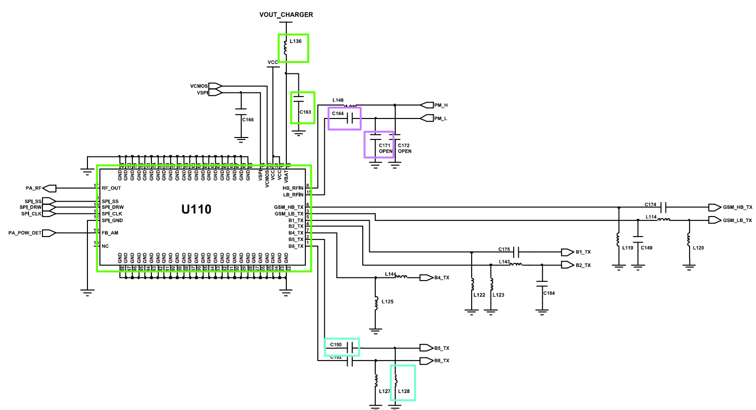


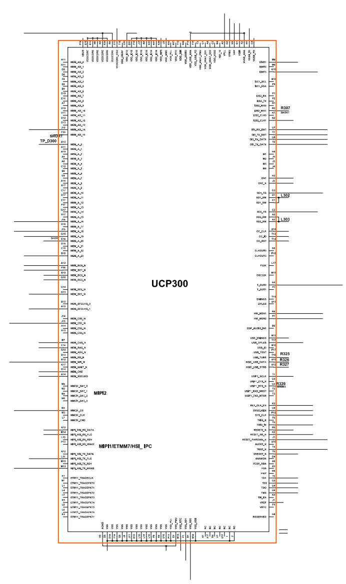

是
WCDMA Band5的发射器是否正常？
否
更换 UCP300
结束

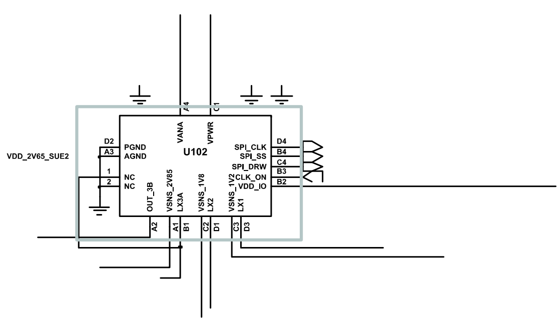

CN100
ANT101
UME300
U100
U102
L107
L106
L103
C136
C135
C121
L100
U107
SC100
U108
U106
OSC100
U112
L130
C182
C181
L129
U110
UCP300
L301
L300
L303
L302
C315
C337
C336
C314
C312
C165
R309
R313
R312
R304
R325

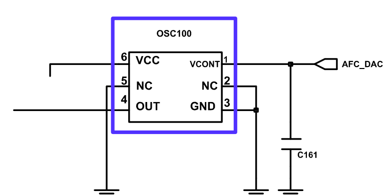

VOUT_CHARGER
C140
SPI_SS
SPI_CLK
SPI_DRW
VSPI
C146
CN100
C153
L115
L116
OPEN
U107
ANT
VSPI
VDD
SS
CLK
DRW
LB_TX
HB_TX
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
GSM_LB_TX
GSM_HB_TX
L142
C151
L137
L141
L140
C141
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
C143
NC
C147
NC
L108
L112
OPEN
L111
OPEN
L113
ASM


TRX6
TRX1
TRX5
TRX3
TRX2
TRX4
B1_TX
B2_TX
B4_TX
B5_TX
B8_TX
U108

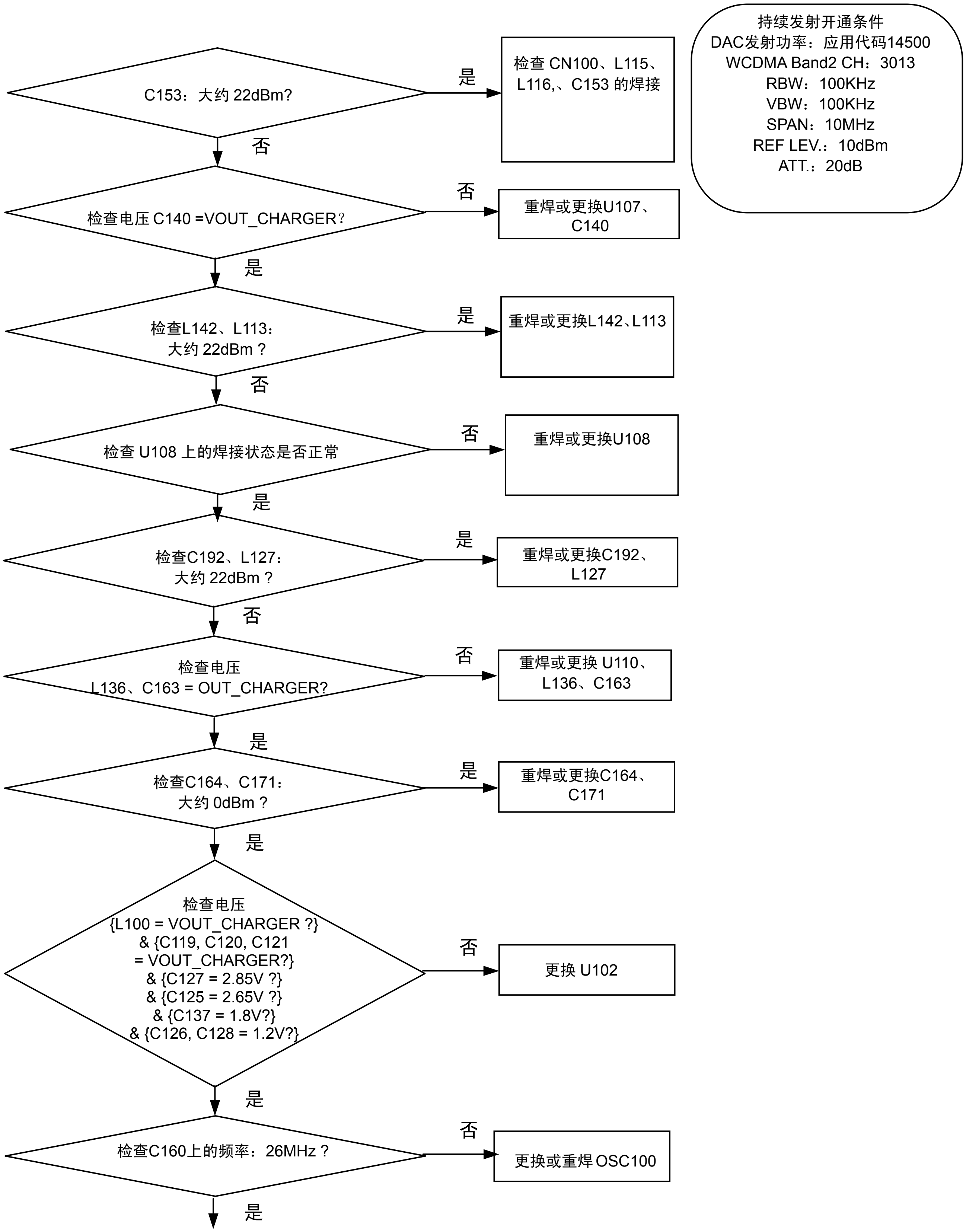

VOUT_CHARGER
L136
C163
U110
PA_RF
PA_POW_DET
C174
GSM_HB_TX
GSM_LB_TX
B1_TX
B2_TX
B4_TX
B5_TX
B8_TX


UCP300


VDD_2V65_SUE2
U102
PGND
AGND
NC
NC
VANA
VPWR
SPI_CLK
SPI_SS
SPI_DRW
CLK_ON
VDD_IO
OUT_3B
VSNS_2V85
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1


OSC100
VCC
NC
OUT
VCONT
NC
GND
AFC_DAC
C161

8-3-22.WCDMA BAND8 发射器



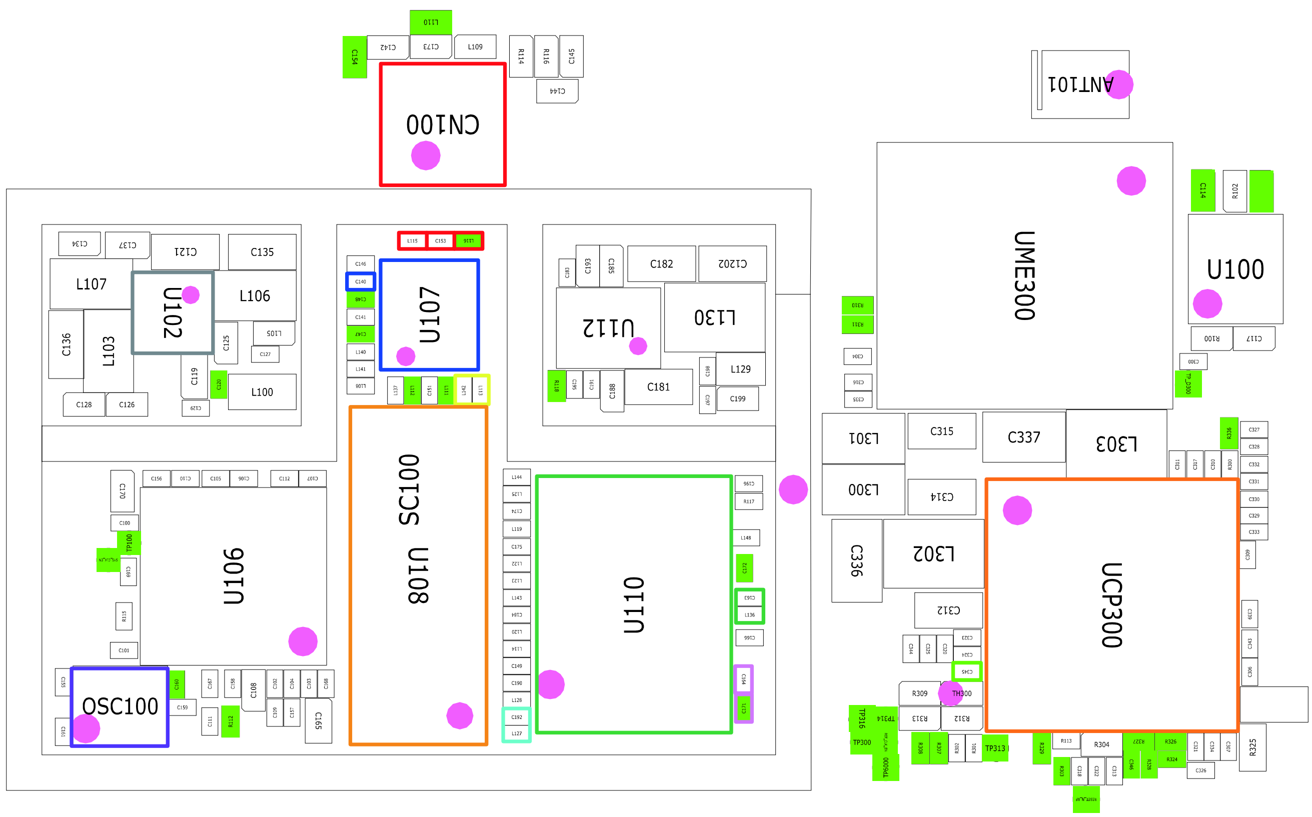
是

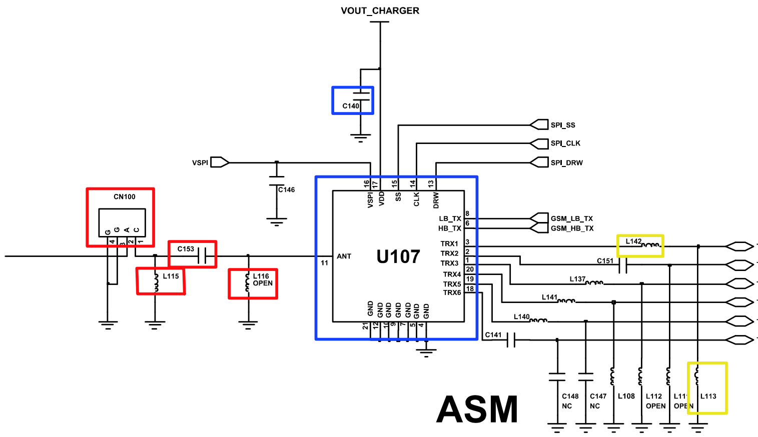
WCDMA Band8 的发射器是否正常？

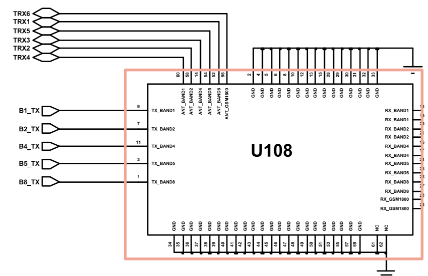
否 → 更换 UCP300

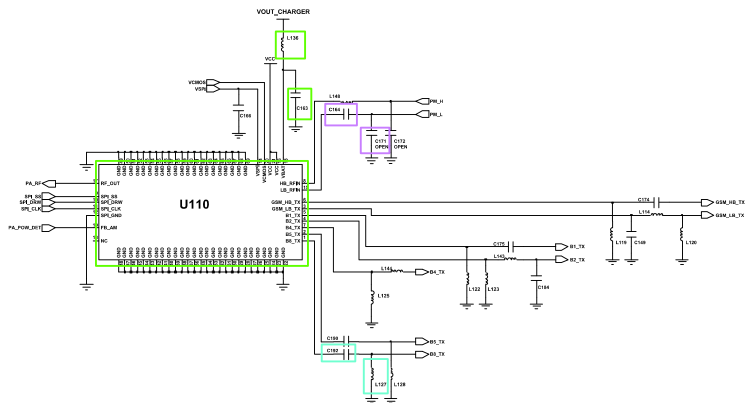
是 ↓

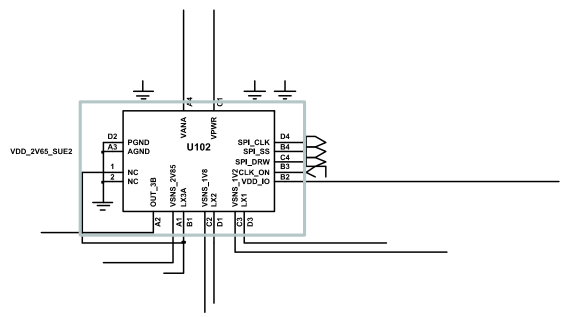
结束

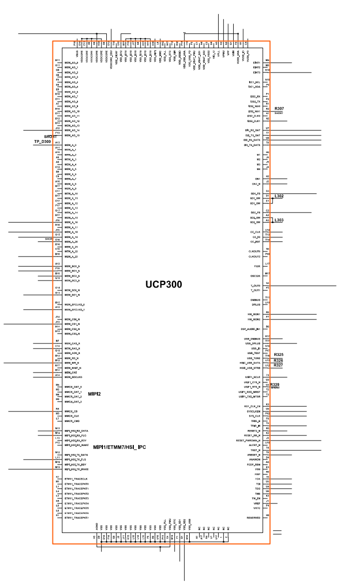


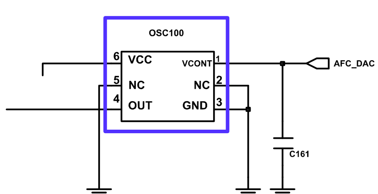

VOUT_CHARGER
C140
SPI_SS
SPI_CLK
SPI_DRW
VSPI
C146
CN100
C153
L115
L116
OPEN
U107
ANT
GSM_LB_TX
GSM_HB_TX
L142
C151
L137
L141
L140
C141
TRX1
TRX2
TRX3
TRX4
TRX5
TRX6
C148
NC
C147
NC
L108
L112
OPEN
L113
ASM


TRX6
TRX1
TRX5
TRX3
TRX2
TRX4
B1_TX
B2_TX
B4_TX
B5_TX
B8_TX
U108


VOUT_CHARGER
L136
C163
C164
C166
C171
OPEN
C172
OPEN
PM_H
PM_L
PA_RF
SPI_SS
SPI_DRW
SPI_CLK
PA_POW_DET
U110
GSM_HB_TX
GSM_LB_TX
B1_TX
B2_TX
B4_TX
B5_TX
B8_TX
L122
L123
L125
L127
L128
C184


VDD_2V65_SUE2
U102
PGND
AGND
NC
NC
VANA
VPWR
SPI_CLK
SPI_SS
SPI_DRW
VCLK_ON
VDD_IO
OUT_3B
VSNS_2V65
LX3A
VSNS_1V8
LX2
VSNS_1V2
LX1


UCP300


OSC100
VCC
NC
OUT
VCONT
NC
GND
AFC_DAC
C161

8-4.维修原理图

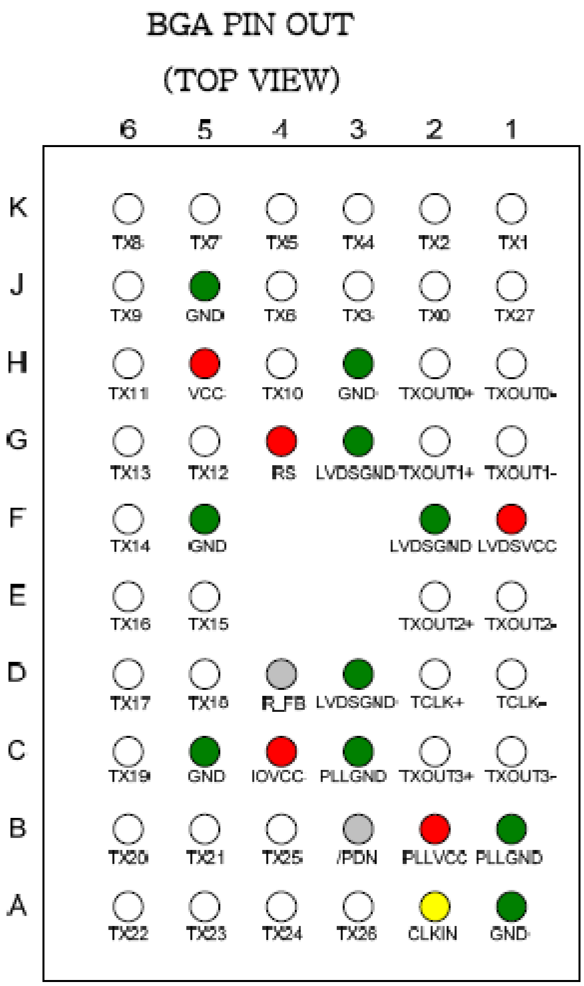
-NC 点（顶部视图）

UCP6000

Table 68 23x23mm FCBGA Ball Map

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	
A	GND	GND	KB_COL6		KB_COL2	KB_COL0		KB_ROW11	KB_ROW9		KB_ROW4	KB_ROW2		TEST_MODE_EN	JTAG_TRST_N		JTAG_RTCK	DDR_QUSE2		DDR_A0	DDR_CKE0		DDR_A9	DDR_BA1		DDR_A8	GND	GND	A
B	GND	GND	KB_COL5	GND	KB_COL3	KB_ROW15	GND	KB_ROW12	KB_ROW8	GND	KB_ROW5	KB_ROW1	GND	CLK_32K_IN	JTAG_TDI	GND	JTAG_TCK	DDR_QUSE3	GND	DDR_A3	DDR_ODT0	GND	DDR_RAS_N	DDR_BA2	GND	DDR_BA0	GND	GND	B
C	PWR_I2C_SCL	PWR_I2C_SDA	KB_COL7		KB_COL4	KB_ROW14		KB_ROW13	KB_ROW7		KB_ROW6	KB_ROW0		CPU_PWR_REQ	JTAG_TDO		JTAG_TMS	DDR_A12		DDR_A11	DDR_CKE1		DDR_CS1_N	DDR_A1		DDR_A5	DDR_A6	DDR_A14	C
D		GND		GND	CLK_32K_OUT	KB_COL1	SYS_CLK_REQ	KB_ROW10	DDR_DM3	DDR_DQS3P	KB_ROW3	DDR_DQ14	DDR_DQ15	PWR_INT_N	DDR_DQ9	DDR_DQ13	SYS_RESET_N	DDR_DQ2	DDR_DQS0N	DDR_A2	DDR_DQ6	DNC	DDR_A10	DDR_DQ21	GND		GND		D
E	GPIO_PU2	XTAL_OUT	XTAL_IN	UART3_CTS_N	GPIO_PU4	THERMD_N	GND	DDR_COMP_PU	DDR_DQ27	GND	DDR_DQS3N	DDR_DQ26	GND	DDR_DQ11	DDR_DM1	GND	DDR_DQ8	DDR_DQ1	GND	DDR_DQS0P	DDR_DQ5	GND	DDR_DQ23	DDR_DQS2N	DDR_CS0_N	DDR_CLK	DDR_CLK_N	DDR_A13	E
F	UART3_RTS_N	UART3_RXD	UART3_TXD	UART2_RXD	UART2_CTS_N	GPIO_PU3	THERMD_P	DDR_COMP_PD	DDR_DQ24	DDR_DQ28	DDR_DQ30	DDR_DQ25	DDR_DQ12	DDR_DQS1N	DDR_DQS1P	DDR_DQ10	DDR_DQ4	DDR_DQ3	DDR_DM0	DDR_DQ0	DDR_DQ7	DDR_DQ20	DDR_DQ16	DDR_DQS2P	DDR_DQ17	DDR_A4	DDR_WE_N	DDR_A7	F
G		GND		GPIO_PU1	GND	UART2_RTS_N		DDR_DQ29	DDR_DQ31		OWR	PLL_S_PLL_LF		CORE_PWR_REQ	DDR_QUSE0		DDR_QUSE1	VDDIO_SYS		DNC	DNC		DDR_DM2	GND	DDR_DQ19		GND		G
H	DAP4_DIN	DAP4_SCLK	DAP4_FS	GPIO_PU5	GEN1_I2C_SDA	GEN1_I2C_SCL	UART2_TXD	GND	VDD_CPU	GND	VDDIO_UART	AVDD_OSC	GND	AVDD_PLLA_P_C	AVDD_PLLM	GND	VDD_DDR_RX	GND	GND	VDDIO_DDR	GND	DDR_DQ18	VI_D0	DDR_DQ22	VI_D1	DDR_CAS_N	CAM_I2C_SDA	CAM_I2C_SCL	H
J	GPIO_PU0	GPIO_PU6	DAP3_FS	DAP4_DOUT	SDIO1_CMD	GPIO_PV3	GPIO_PV1	VDD_CPU	VDD_CPU	VDD_CPU	VDD_CPU	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VDDIO_DDR	VI_GP4	VI_D2	VI_D4	VI_D6	VI_D7	VI_D5	VI_D3	J
K		GND		SDIO1_DAT1	GND	SDIO1_DAT0		GND	VDD_CPU											VDDIO_DDR	GND		VI_D8	GND	VI_VSYNC		GND		K
L	DAP3_DOUT	DAP3_DIN	ULPI_DATA3	ULPI_DATA4	SDIO1_DAT3	ULPI_DATA5	AVDD_PLLX	GND	VDD_CPU		GND	GND	GND	GND	GND	VDDIO_DDR	VDDIO_DDR	GND		VDDIO_DDR	VDDIO_VI	VI_GP0	VI_D10	VI_MCLK	VI_HSYNC	VI_D9	VI_GP5	VI_PCLK	L
M	ULPI_NXT	ULPI_CLK	ULPI_DIR	DAP3_SCLK	SDIO1_DAT2	SDIO1_CLK	DNC	VDDIO_BB	VDD_CPU		VDD_CPU	GND	VDD_CORE	VDD_CORE	VDD_CORE	VDD_CORE	VDDIO_DDR	VDDIO_DDR		VDDIO_DDR	DNC	VI_D11	VI_GP3	VI_GP6	SPI2_MISO	SPI1_MOSI	SPI1_CS0_N	SPI1_SCK	M
N		GND		ULPI_DATA2	GND	ULPI_DATA7		GND	VDD_CPU_SENSE		VDD_CPU	VDD_CPU	GND	GND	VDD_CORE	GND	GND	VDDIO_DDR		VDDIO_DDR	GND		SPDIF_OUT	GND	SPI2_MOSI		GND		N
P	VDDIO_SDIO	GPIO_PV0	ULPI_STP	ULPI_DATA0	ULPI_DATA6	ULPI_DATA1	GPIO_PV4	AVDD_PEX_PLL	GND_CPU_SENSE		VDD_CPU	VDD_CPU	VDD_CPU	VDD_CPU	VDD_CORE	VDD_CORE	GND	GND		VDDIO_DDR	VDDIO_AUDIO	SPI2_CS1_N	SPI2_CS0_N	SPI2_SCK	DAP1_SCLK	SPI1_MISO	DAP_MCLK1	SPI2_CS2_N	P
R	SDIO3_CMD	SDIO3_CLK	SDIO3_DAT0	SDIO3_DAT3	SDIO3_DAT6	GPIO_PV6	GPIO_PV5	AVDD_PEX	AVDD_PEX		GND	VDD_CPU	VDD_CPU	GND	GND	VDD_CORE	GND	GND		VDDIO_DDR	DNC	DAP2_FS	DAP2_SCLK	DAP_MCLK2	SPDIF_IN	DAP1_DIN	DAP1_FS	DAP1_DOUT	R
T		GND		SDIO3_DAT4	GND	SDIO3_DAT5		GND	AVDD_PEX		GND	GND	VDD_CORE	GND	VDD_CORE	VDD_CORE	GND	GND		VDDIO_DDR	GND		DAP2_DOUT	GND	DAP2_DIN		GND		T
U	PEX_REFCLKN	PEX_REFCLKP	SDIO3_DAT1	SDIO3_DAT2	DNC	GPIO_PV2	SDIO3_DAT7	VDD_PEX	VDD_PEX		GND	GND	VDD_CORE	VDD_CORE	GND	VDD_CORE	VDD_CORE	GND		VDDIO_DDR	GND	VDDIO_LCD	LCD_DE	LCD_PWR1	LCD_D12	LCD_D15	LCD_D14	LCD_D13	U
V	PEX_TSTCLKN	PEX_TSTCLKP	PEX_L2_RXP	PEX_L2_RXN	PEX_L3_RXP	PEX_L3_RXN	DNC	VDDIO_PEX_CLK	VDD_PEX		GND	VDD_RTC	VDD_RTC	GND	GND	GND	VDD_CORE	VDD_CORE		VDDIO_DDR	GND	VDDIO_LCD	HDMI_INT_N	LCD_CS0_N	LCD_D22	LCD_D17	LCD_D16	LCD_PCLK	V
W		GND		DNC	GND	PEX_TERMP		GND	VDD_TP											VDD_CORE	GND		DDC_SCL	GND	LCD_SDIN		GND		W
Y	AVDD_PLLE	PEX_L3_TXP	PEX_L3_TXN	PEX_CLK_OUT2_N	PEX_CLK_OUT2_P	DNC	VDDIO_NAND	VDDIO_NAND	VDDIO_NAND	GND_TP	GND	GND	GND	AVDD_USB	AVDD_HDMI	AVDD_DSI_CSI	GND_CORE_SENSE	VDD_CORE_SENSE	VDD_CORE	VDD_CORE	VDD_CORE	CRT_HSYNC	DDC_SDA	LCD_M1	LCD_D9	LCD_D6	LCD_D7	LCD_D8	Y
AA	PEX_L2_TXN	PEX_L2_TXP	DNC	PEX_L0_RXP	PEX_L0_RXN	PEX_L1_RXP	PEX_L1_RXN	GND	GND	GND	VGND_TP	AVDD_HDMI_PLL	GND	AVDD_USB	AVDD_HDMI	GND	AVDD_DSI_CSI	AVDD_PLLU	GND	VDD_CORE	GND	LCD_D21	LCD_D19	LCD_PWR2	LCD_SDOUT	LCD_D0	LCD_D11	LCD_D10	AA
AB		GND		DNC	GND	DNC		VDDIO_HSIC	AVDD_IC_USB		AVDD_USB_PLL	GMI_CS5_N		VPP_KFUSE	VPP_FUSE		VDAC_R	VDAC_G		CSI_CLKBN	LCD_SCK		LCD_D20	GND	LCD_D18		GND		AB
AC	PEX_L1_TXP	PEX_L1_TXN	DNC	PEX_CLK_OUT1_N	GMI_AD22	GMI_CS3_N	GMI_RST_N	GMI_AD27	GMI_DPD	GMI_IORDY	GMI_AD25	GMI_CS4_N	DNC	USB1_DN	HSIC_DATA	USB_REXT	VDAC_VREF	HDMI_TXD1N	GND	CSI_CLKBP	DSI_CLKAP	LCD_D23	CRT_VSYNC	LCD_DC1	LCD_CS1_N	LCD_D1	LCD_D2	LCD_D3	AC
AD	PEX_L0_TXN	PEX_L0_TXP	GMI_AD17	PEX_CLK_OUT1_P	GMI_CS6_N	GEN2_I2C_SDA	GND	GMI_AD20	GMI_CS7_N	GND	GEN2_I2C_SCL	GMI_CS2_N	GND	USB1_DP	HSIC_STROBE	GND	USB1_VBUS	HDMI_TXD1P	GND	CSI_D1AN	DSI_CLKAN	GND	LCD_DC0	LCD_PWR0	LCD_D4	LCD_VSYNC	LCD_HSYNC	LCD_D5	AD
AE		GND		GND	GMI_AD18	GMI_AD8	GMI_AD21	GMI_AD24	GMI_AD1	GMI_AD26	GMI_AD23	GMI_AD9	IC_REXT	ACC1_DETECT	HSIC_REXT	HDMI_TXD0N	HDMI_TXD0P	VDAC_RSET	VDAC_B	CSI_D1AP	DNC	DNC	DNC	LCD_WR_N	GND		GND		AE
AF	GMI_WAIT	GMI_WP_N	GMI_CLK		GMI_CS0_N	GMI_WR_N		GMI_AD7	GMI_AD6		GMI_AD2	GMI_CS1_N		IC_DN	ACC3_DETECT		HDMI_TXCN	HDMI_RSET		DSI_D1AN	DSI_CSI_RUP		DNC	DNC		DNC	DNC	DNC	AF
AG	GND	GND	GMI_AD13	GND	GMI_AD0	GMI_AD3	GND	GMI_AD19	GMI_OE_N	GND	GMI_AD5	GMI_AD16	GND	IC_DP	USB3_DP	GND	HDMI_TXCP	HDMI_TXD2P	GND	DSI_D1AP	DSI_D2AP	GND	CSI_D2AP	CSI_D1BP	GND	CSI_CLKAP	GND	GND	AG
AH	GND	GND	GMI_ADV_N		GMI_AD15	GMI_AD11		GMI_AD4	GMI_AD14		GMI_AD12	GMI_AD10		DNC	USB3_DN		AVDD_VDAC	HDMI_TXD2N		DSI_CSI_RDN	DSI_D2AN		CSI_D2AN	CSI_D1BN		CSI_CLKAN	GND	GND	AH
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	

U800

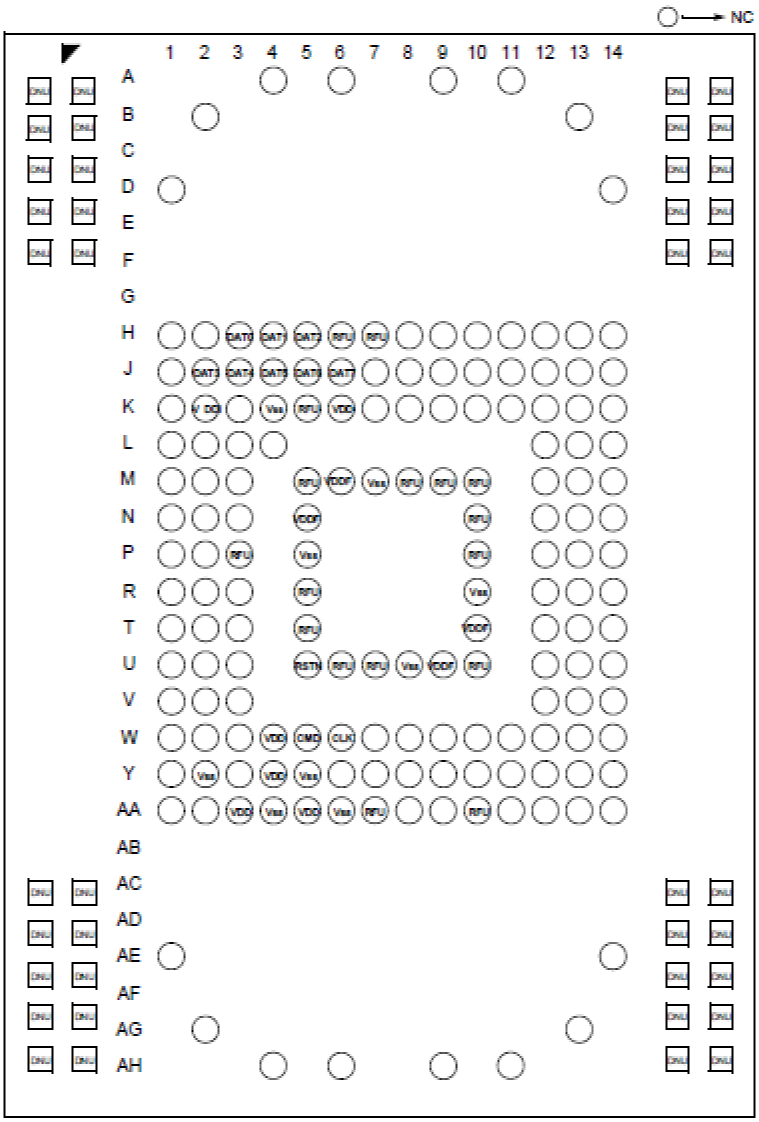

BGA PIN OUT
(TOP VIEW)
6 5 4 3 2 1
K J H G F E D C B A
TX8 TX7 TX5 TX4 TX2 TX1
TX9 GND TX6 TX3 TX0 TX27
TX11 VCC TX10 GND TXOUT0+ TXOUT0-
TX13 TX12 RS LVDSGND TXOUT1+ TXOUT1-
TX14 GND LVDSGND LVDSVCC
TX16 TX15 TXOUT2+ TXOUT2-
TX17 TX18 R_FB LVDSGND TCLK+ TCLK-
TX19 GND IOVCC PLLGND TXOUT3+ TXOUT3-
TX20 TX21 TX25 /PDN PLLVCC PLLGND
TX22 TX23 TX24 TX26 CLKIN GND

U702

2.11.1 ZQZ Package Pinout (Top View)

L11 DIG_PWN	L10 LED_PWM	L9 GREEN1	L8 GREEN2	L7 LDO9	L6 LDO1	L5 XTAL1	L4 GPIO4	L3 GPIO1	L2 ANLG2	L1 ADC_REF
K11 L3	K10 nINT	K9 RED1	K8 RED2	K7 LDO4	K6 XTAL2	K5 LDO0	K4 GPIO2	K3 ANLG3	K2 RSVDC	K1 LDO7
J11 PGND3	J10 SM3	J9 PWM	J8 BLUE1	J7 VIN_LDO9	J6 VIN_LDO01	J5 GPIO3	J4 COMP	J3 AGND2	J2 PSCLK	J1 LDO8
H11 AVDD6	H10 SM3IG	H9 SM3_SW	H8 VTSBIAS	H7 VIN_LDO4	H6 DGND1	H5 ANLG1	H4 LDO4PG	H3 SDAT	H2 LDO6	H1 VIN_LDO678
G11 VREF1V25	G10 V2V2	G9 FB3	G8 nNOPOWER	G7 BLUE2	G6 HSK	G5 PSDAT	G4 SCLK	G3 DIG_PWM2	G2 VIN_SM1	G1 VIN_SM1
F11 RSVDB	F10 RSVDB	F9 RSVDA	F8 AGND1	F7 HSK	F6 HSK	F5 HSK	F4 HSK	F3 HSK	F2 L1	F1 L1
E11 BAT	E10 RTC_OUT	E9 TS	E8 HSK	E7 HSK	E6 HSK	E5 HSK	E4 LDO4EN	E3 HSK	E2 PGND1	E1 PGND1
D11 BAT	D10 FLTDPPM	D9 SM0EN	D8 HSK	D7 LDO5	D6 HSK	D5 HSK	D4 TNOPOWER	D3 nNORTC	D2 PGND0	D1 PGND0
C11 USB	C10 USB	C9 SM2	C8 SM1EN	C7 SYNCEN	C6 OUT32K	C5 DGND2DT	C4 SM1PG		C2 L0	C1 L0
B11 AC	B10 SYS	B9 VIN_SM2	B8 L2	B7 PGND2	B6 V32K	B5 LDO3	B4 SM1	B3 nHOT_RST	B2 VIN_SM0	B1 VIN_SM0
A11 AC	A10 SYS	A9 VIN_SM2	A8 L2	A7 PGND2	A6 LDO2	A5 VIN_LDO23	A4 AGND3	A3 SM0PG	A2 SM0	A1 RESUME

UME6000

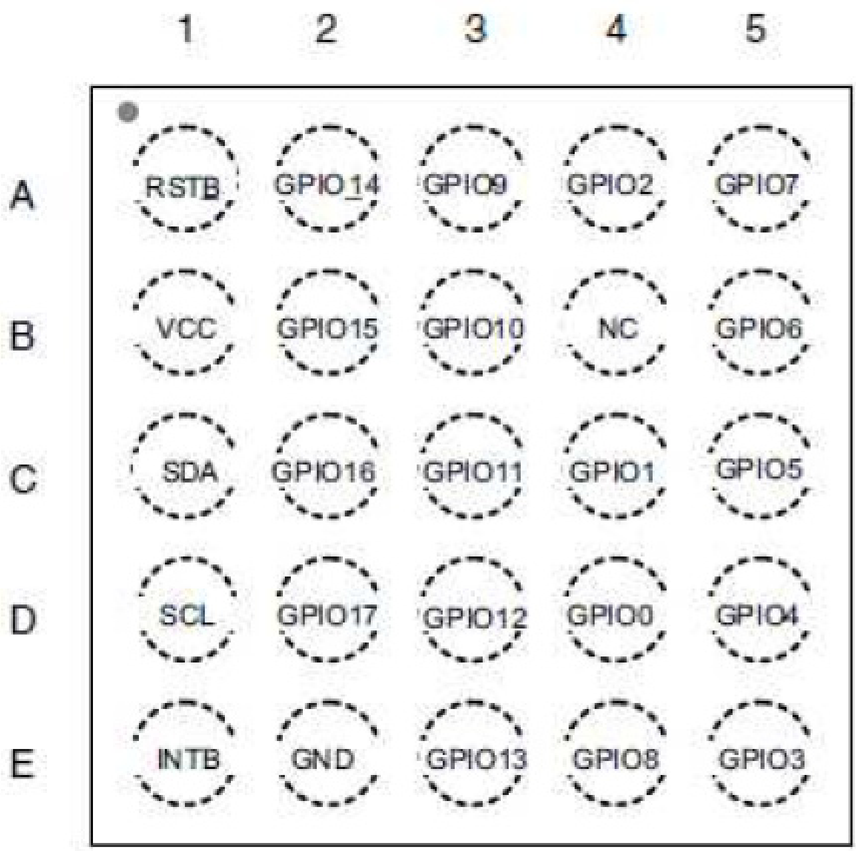


U203

	12	11	10	9	8	7	6	5	4	3	2	1	
A	FM_AOUT1	BT_VCOVDD1p2	BT_FEVSS	BT_RF	BT_PAVDD3p3	WRF_rfin	WRF_rfout		WRF_pa_VDD5p0		WRF_rfout_5G		A
B	FM_AOUT2	BT_PLLVDD1p2	BT_RFVSS	BT_RFVDD1p2	BT_IFVDD1p2	WRF_lna2g_VDD1p2	WRF_pa_GND		WRF_pa_GND		WRF_pa_GND	WRF_rfin_5G	B
C	FM_TX	FM_VSSAUDIO	BT_PLLVSS	BT_VSS	BT_IFVSS	WRF_lna2g_GND		WRF_GND	WRF_padrv_GND	WRF_padrv_VDD5p0		WRF_ana_GND	C
D	FM_RXN	FM_RXP	FM_AUDIOVDD	BT_GPIO_1	WL_GPIO_6	NC (CLK_REQ_IN)	JART_TXD	WL_GPIO_3	WRF_A_TSSI_IN	WRF_LOGEN_A_GND	WRF_LOGEN_A_VDD1P2	WRF_ana_VDD1p2	D
E	FM_RFVDD1p2	FM_RFVSS	FM_PLLVDD	NC (CLK_REQ_MODE)	BT_VDDC_2	WL_VDDC_2	JART_RXD	WL_VSS_2	WRF_gpio_out	WRF_res_ext		WRF_vco_ldo_in_VDD1p8	E
F	FM_VDD2p5	FM_VSSVCO	FM_PLLVSS	BT_GPIO_0	BT_VSSC_1	BT_VDDO_0	UART_RTS_N	JTAG_SEL	WRF_afe_GND	WRF_TCXO_VDD3p3	WRF_VCO_LDO_OUT_VDD1P2	WRF_vco_GND	F
G	CLK_REQ_OUT	TM0(NC for A0)	BT_RST_N		BT_GPIO_7	I2S_DI	UART_CTS_N	WL_GPIO_1	WRF_afe_VDD1p2	WRF_TCXO_in		WRF_xtal_OP	G
H	BT_GPIO_4		ET_GPIO_2	BT_GPIO_3	BT_GPIO_6	I2S_DO	PCM_CLK	WL_GPIO_2	WL_GPIO_0	WRF_xtal_VDD1p2	WRF_xtal_GND	WRF_xtal_ON	H
J	VOUT_3P3	VOUT_3P1	EXT_PWM_REQ	BT_GPIO_5	WL_GPIO_4	LPO	PCM_IN	PCM_OUT	rf_sw_ctrl_5	rf_sw_ctrl_6	rf_sw_ctrl_1	HSIC_RREF	J
K	SR_VDDBAT1	SR_VDDBAT2	BT_REG_ON	EXT_SMPS_REQ	BT_VDDC_0	WL_VDDC_1	PCM_SYNC	rf_sw_ctrl_0	rf_sw_ctrl_7	rf_sw_ctrl_4	WL_VSS_0	WL_VDDC_0	K
L	SR_VDDBAT1	PMU_AVSS	VOUT_LNLDO1	WL_REG_ON	SDIO_DATA3	SDIO_CMD	WL_GPIO_5	WL_VSS_1	rf_sw_ctrl_2	VDDIO_RF_0	HSIC_DATA	HSIC_AVDD12	L
M	SR_VLX	SR_PVSS	VDD_LDO	VOUT_CLDO	SDIO_DATA1	SDIO_CLK	SDIO_DATA_0	SDIO_DATA_2	rf_sw_ctrl_3	WL_VDDO_0	HSIC_STROBE	HSIC_AVSS	M
	12	11	10	9	8	7	6	5	4	3	2	1	

Figure 36: 133-WLBGA Ball Map (bottom view)

U6004



Flip-chip CSP 25